



सत्यमेव जयते
भारत सरकार

Government of India

विद्युत मंत्रालय

Ministry of Power

पूर्वी क्षेत्रीय विद्युत समिति

Eastern Regional Power Committee

14, गोल्फ क्लब रोड, टॉलीगंज, कोलकाता-700033

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सं/NO. ERPC/EE/OPERATION/2025/1185

दिनांक/DATE: 29.09.2025

सेवा में /To

संलग्न सूची के अनुसार /As per list enclosed.

विषय : 22.09.2025 (सोमवार) को ईआरपीसी सचिवालय, कोलकाता में भौतिक रूप से आयोजित 231वीं ओसीसी बैठक के कार्यवृत्त – के संबंध में

Sub: Minutes of 231st OCC Meeting held on 22.09.2025 (Monday) physically at ERPC Secretariat, Kolkata – reg

महोदय/महोदया,

Sir(s)/Madam,

कृपया अपनी जानकारी और आवश्यक कार्रवाई के लिए 22 सितम्बर 2025 (सोमवार) को आयोजित 231वीं ओसीसी बैठक के कार्यवृत्त संलग्न देखें। यह ईआरपीसी वेबसाइट (www.erpc.gov.in) पर भी उपलब्ध है।

Please find enclosed Minutes of 231st OCC Meeting held on 22.09.2025 (Monday) physically at ERPC Secretariat, Kolkata at 10:30 hrs. This is for your kind information and necessary action. The same is also available at ERPC website (www.erpc.gov.in).

टिप्पणियाँ, यदि कोई हों, कृपया यथाशीघ्र इस कार्यालय को अग्रेषित करें।

Observations, if any, may please be forwarded to this office at the earliest.

इसे सदस्य सचिव के अनुमोदन से जारी किया जाता है।

This issues with the approval of Member Secretary.

भवदीय /Yours faithfully.

(S.Kejriwal)

SE(Operation)

एसई (ऑपरेशन)

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ERPC:: Kolkata

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9. महाप्रबंधक, टीटीपीएस, तालचेर, (फैक्स नंबर 06760-243212)
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20. कार्यकारी निदेशक, ईआरएलडीसी, पोसोको, कोलकाता, (फैक्स नंबर 033-2423-5809)
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34. मुख्य विद्युत अभियंता, पूर्वी रेलवे, कोलकाता-700 001 (फैक्स नं.: 033-22300446)
35. मुख्य विद्युत अभियंता, दक्षिण पूर्व रेलवे, कोलकाता-43 (फैक्स: 033-24391566)।
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ईआरपीसी:: कोलकाता



Eastern Regional Power Committee

MINUTES

OF

231st OCC MEETING

Venue: ERPC Secretariat, Kolkata

Date: 22.09.2025

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EASTERN REGIONAL POWER COMMITTEE

MINUTES OF 231st OCC MEETING HELD ON 22.09.2025 (MONDAY) AT 10:30 HRS

At the outset, the Member Secretary, ERPC warmly welcomed all esteemed OCC members and other participants to the 231st OCC Meeting.

He outlined the performance of ER grid during August, 2025 and highlighted the following points:

- ❖ In August-2025, energy consumption of ER was 19.12 BU which is 6.04 % more than August-2024 (18.03 BU).
- ❖ In August-2025, Peak demand met of ER is to be 31331 MW which is 8.79% more than August-2024 (28798 MW).
- ❖ Thermal PLF of ER during August-2025 was 67.97 % w.r.t All India PLF of 60 %.
- ❖ Generating stations whose PLF was more than 90% during August-2025:

Utility	Generating station	PLF (%)
WBPDCL	SANTALDIH TPS	93
IPP	HALDIA TPP	91
	Jindal India power Limited	92

- ❖ During August-2025, **76.22 %** of the time, the ER grid frequency was in IEGC Band (49.90Hz-50.05Hz).
- ❖ Coal stock position (As on 18.09.2025) is as follows:

SL.	Name of States/Power Stns.	% of Actual Stock vis-à-vis Normative Stock
1.	Jharkhand (TVNL)	135%
2.	Odisha/IBTPS	150%
3.	WBPDCL	36% (Min. SAGARDIGHI TPS - 28%, Max.Bandel TPS -63%)
4.	D.P.L. TPS	64%
5.	DVC	79%(Min. Mejia TPP-67%, Max Chandrapur (TPS `A-132%)
6.	NTPC	107% (Max. Darlipalli TPP -167% & Min. North Karapura TPP - 24%)

He further highlighted the following:

- Buxar TPP unit#1(660MW) test synchronized on 21.08.2025.
- 400/220KV,500 MVA ICT#3 at New Dubri, OPTCL Substation first time loaded on 29th August 2025.
- During the **229th OCC (Operational Coordination Committee) Meeting** of ERPC, the matter regarding the **submission of comments by ER Generators on the**

report of the Committee constituted by Chairperson, CEA, on “Technical Minimum Operation of Thermal Power Plants” was deliberated.

It was noted with concern that, despite the report being shared for stakeholder consultation, **no comments have been received from ER Generators till date.**

❖ **ED, ERLDC** at the outset highlighted challenges of grid operation showcasing areas of concern as follows:

- Occurrence of a low frequency event on 8th and 9th September, 2025 where frequency plunged to as low as 49.4 Hz owing to sudden demand surge in Northern and Western regions.
- Low system demand coupled with RE generation is often resulting in high frequency grid operation. Also, due to low requisition by the beneficiaries, generation schedule of most generating units is hovering well below respective technical minimum limits. Discussions have been held with NTPC in ER in this regard.
- ERLDC is set to conduct a workshop on maintaining adequate reserves by ER states.
- At NLDC, modelling of inverter-based sources has already taken a kick start. So, in view of large scale RE integration in the grid in upcoming years, all SLDCs were requested to take initiative in simulation of inverter-based sources.
- Derivatives trading has commenced in Indian Power market but DISCOMs are yet to take part in the Derivative market. It was also proposed that NSE may be invited to enlighten all ER constituents on modalities of participation in Derivative market.

❖ In this regard, WBSEDCL representative submitted that DISCOMs are not yet participating in derivative power market owing to absence of specific guidelines from CERC i.r.o participation in the derivative market and also the corresponding impact on tariff is not clear from the existing CERC tariff regulations.

❖ WBSEDCL further pointed out that up-ancillary services have been observed to be used in high frequency region while ancillary services should be in down mode during high grid frequency.

In response, ERLDC submitted:

- Frequency forecasting is extremely difficult in real time grid operation as backdown of thermal generation practically takes 3 to 4 time blocks owing to constraints of ramping capability.
- It was further assured that, to maintain transparency, reserve utilization in the previous month will be shared with all utilities by ERLDC in monthly OCC meetings.

❖ List of participants is attached at **Annexure-A**.

1. PART-A: CONFIRMATION OF MINUTES

1.1. Confirmation of Minutes of 230th OCC Meeting held physically at Power Training Centre, OPTCL, Bhubaneswar on 22th August 2025

The minutes of 230th Operation Coordination Sub-Committee meeting held on 22.08.2025 was circulated vide letter dated 01.09.2025.

Members may confirm the minutes of 230th OCC meeting.

OCC Decision

Based on the observations received from GMR, following amendments in the minutes of 230th OCC Meeting were approved:

Agenda No. B.2.10(GMRKEL)

➤ **Shutdown of GMR Unit#3(350MW):**

In the minutes of 230th OCC meeting, the shutdown time period was inadvertently recorded from 1st week of September 2025 for a period of 26 days. However, GMRKEL requested to avail the shutdown from 2nd November 2025 to 27th November 2025.

➤ *OCC consented to the proposal of GMR.*

➤ *These modifications shall form part and parcel of MOM of 230th OCC circulated vide letter dated 01.09.2025.*

➤ *Other members confirmed the minutes of 230th OCC meeting.*

2. PART-B: ITEMS FOR DISCUSSION

2.1 Intrastate Transmission Network Assessment & Mitigation across Eastern Region States

Eastern Region has witnessed a significant rise in power demand, recording an annual growth of nearly 7-8% YoY. The regional peak demand met reached 33,450 MW on 23.07.2025, with several states surpassing their previous peak demand records this year.

Constituents	ER Power Position: 19-Aug-2025							
	Daily Power Position			All Time High				
	Max demand	Time	MU/Day	Demand Met	Demand Date	Demand Time	MU/Day	Energy Date
Bihar	7776	22:30	161.36	8752	23-Jul-25	22:02	186.78	24-Jul-25
DVC	3078	19:30	69.85	3674	14-Jun-24	19:30	81.22	22-Apr-22
Jharkhand	2040	19:00	42.13	2406	13-Jun-25	20:00	52.47	14-Jun-25
Odisha	6268	19:14	131.33	7302	12-Aug-25	22:33	148.54	19-Apr-24
Sikkim	79	18:34	1.28	137	11-Jan-24	18:59	2.50	28-Jan-20
W. Bengal	10365	00:38	220.48	13108	14-Jun-25	23:33	268.22	14-Jun-25
ER	28635	23:01	626.61	33452	23-Jul-25	22:56	703.71	23-Jul-25

Sharp increase in demand highlights the rapid growth trajectory of the region. However, intra-state transmission capacity addition/augmentation particularly of new 220 kV and 400 kV corridors has not kept pace. Intra-state network upgradation continues to be a matter

of serious concern due to missed execution timelines and lack of clear visibility at the regional level especially after the dissolution of the Standing Committee.

In this context, an assessment of **Intrastate Transmission Network and Mitigation Measures** across Eastern Region states has been carried out based on operational feedback of ER, as detailed below.

West Bengal:

Nearly 30% of West Bengal's demand is concentrated in Kolkata and adjoining areas, supplied mainly through the **765 kV New Ranchi–Medinipur–New Jeerat D/C**, **400 kV New Ranchi–New PPSP D/C**, and **400 kV Kahalgaon–Farakka D/C** corridors. In the last summer, these corridors carried **~4500 MW (~60–70% of WB's total import)**. Even with all lines in service, voltages in the capital **pocket remain below 385 kV** during peak summer, and the outage of any one corridor leads to severe low-voltage stress. An N-2 contingency of 765 kV Medinipur–New Jeerat D/C may result in complete voltage collapse in Kolkata region.

This issue has been repeatedly deliberated, including in the 226th OCC, where it was recommended to implement dynamic reactive compensation as a long-term measure and UVLS as an immediate defence mechanism. An assessment has been undertaken to examine the existing network issues concerning Inter-state lines, Tie lines and Intra-state networks across the ER states.

A) Inter-State (ISTS) associated network of WB:

Inter State Element(s)	Contingency/Risk in Operation	Actionable points
765kV New Ranchi–Medinipur–New Jeerat D/C	Voltage collapse in case N-2 contingency.	Short term measure (Summer '26): ✓ Implementation of UVLS Scheme Long term measure (2028 onwards): ✓ Installation of STATCOM ✓ Requirement of another 765kV corridor connecting New Jeerat
400kV New Ranchi–New PPSP D/C	N-1 non-compliant.	Short term measure (Summer '26): ✓ Limiting drawl by PPSP during solar hours. (Pump Mode) Long term measure (2028 onwards): LILO of this line agreed at New Jamshedpur with reconductoring of New Jamshedpur to –NPPSP.
400kV Farakka- Kahalgaon-D/C & 400kV Farakka-Sagardighi-D/C	N-1 non-compliant.	Short term measure (Summer '26):

	& low Voltage scenario	✓ Commissioning of Sagardighi Unit 5 Long term measure (2028 onwards): ✓ Reconductoring of Farakka-Kahalgaon-D/C approved. ✓ 765kV corridor in this area connecting Northern Region.
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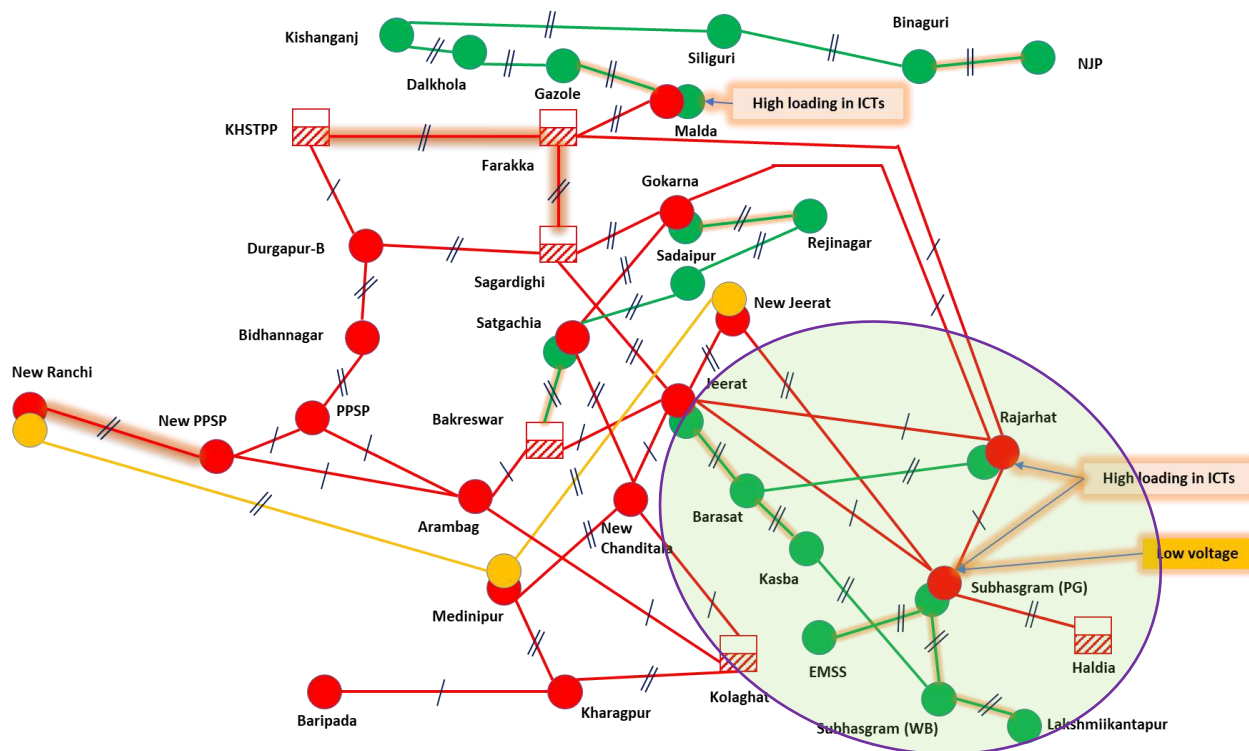
West Bengal -Intra-state system:

220 kV Kolkata ring which is under STU network face high loading during high demand in summer due to high concentration of space cooling load. In the last summer season, several 220 kV lines already exceeded their thermal capacity. The situation could be temporarily managed through load rearrangement, and Load Trimming Schemes but with the prevailing demand growth trend, it is anticipated that all critical 220 kV corridors would be highly loaded in the coming summer.

This scenario of no margin in the system, poses a serious risk to reliable supply in the Kolkata pocket, and proactive measures need to be finalised to ensure continued growth.

Intra State Element(s)	Contingency/Risk in Operation	Actionable points
220kV Kolkata Network: • Jeerat–Barasat D/C • Barasat–Kasba D/C • Subhasgram(PG)–EMSS (CESC) D/C • Subhasgram(PG)–Subhasgram(WB) D/C • Subhasgram(WB)–Lakshmikantpur D/C	▪ N-1 violations ▪ potential thermal limit violation. It will impact to meet load of the capital city in coming summer '26	Short term measure (Summer '26): ✓ SPS/LTS need to be checked and ready. ✓ Load re-arrangement/ Network reconfiguration- 220kV Network, if required. Long term measure (2028 onwards): ✓ Reconductoring of these 220 KV Feeders with HTLS planned. ✓ Upgradation of Lakshmikantapur S/S to 400kV level is approved. <i>Upgradation work needs to be expedited. Upgradation of other 220kV lines to HTLS or 400kV may be considered.</i>
220kV Malda–Gazole D/C	N-1 violation	Short term measure (Summer '26): ✓ Network reconfiguration- 220kV Network.

		Long term measure (2028 onwards): ✓ Reconductoring already approved
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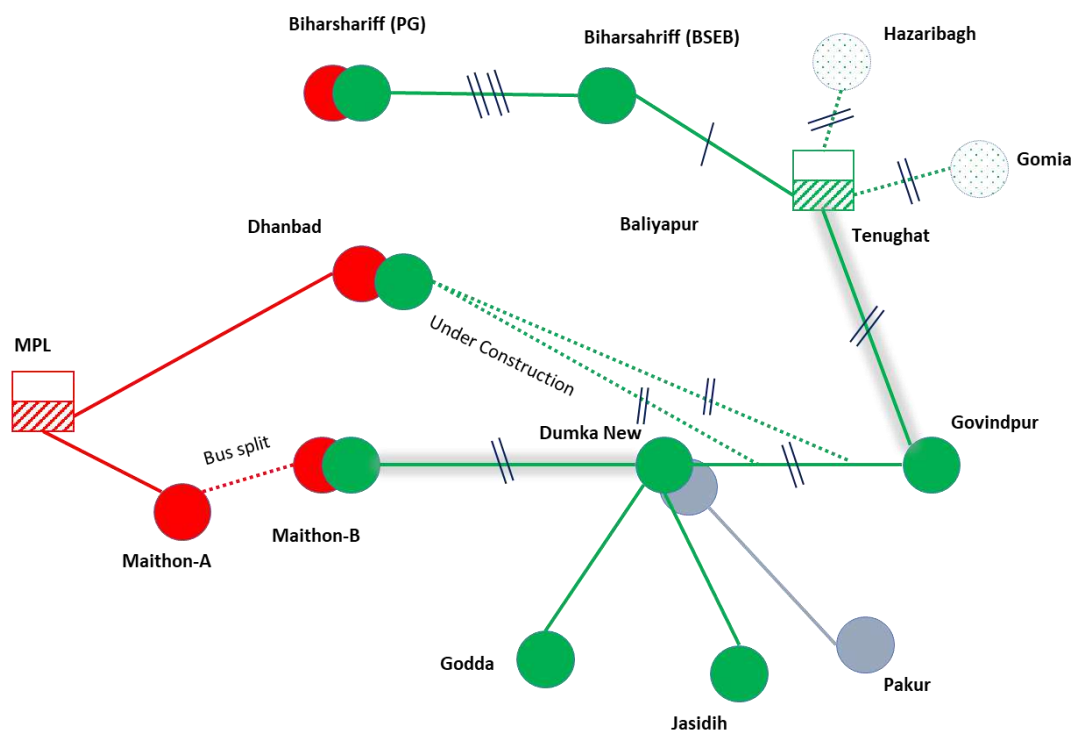


Jharkhand System:

North-Eastern part of Jharkhand network is very sensitive in context of the power evacuation from **Tenughat & Patratu** and demand meet of this area. Several GD/GI events have occurred in this area leading to complete outage of Dumka, Godda, Govindpur, Jasidih & Tenughat. Tripping of any of the side of the network leads to cascade trip and leads to complete blackout. Detailed constraints are below:

Intra State Element(s)	Contingency/Risk in Operation	Actionable points
220 kV Maithon New Dumka D/C (JUSNL)	N violation	Short term measure (Summer '26): ✓ SPS implementation in Maithon - New Dumka- already done. ✓ Continuing 400kV PVUNL- Tenughat link in post PVUNL unit synchronization will support this area. ✓ Restoration of 220 kV Farakka Lalmatia S/C for decongesting 220 kV Maithon Dumka D/C.
220 kV Tenughat-Govindpur D/C (JUSNL)	N-1 violations; Tripping of one circuit leads to cascade effect finally to total blackout.	

		Long term measure (2028 onwards): ✓ LILO of 220 kV Dumka - Govindpur at 400/220 kV Dhanbad. ✓ 220kV Tenughat - Gomia D/C and 220kV Tenughat - Hazaribagh D/C
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Odisha System:

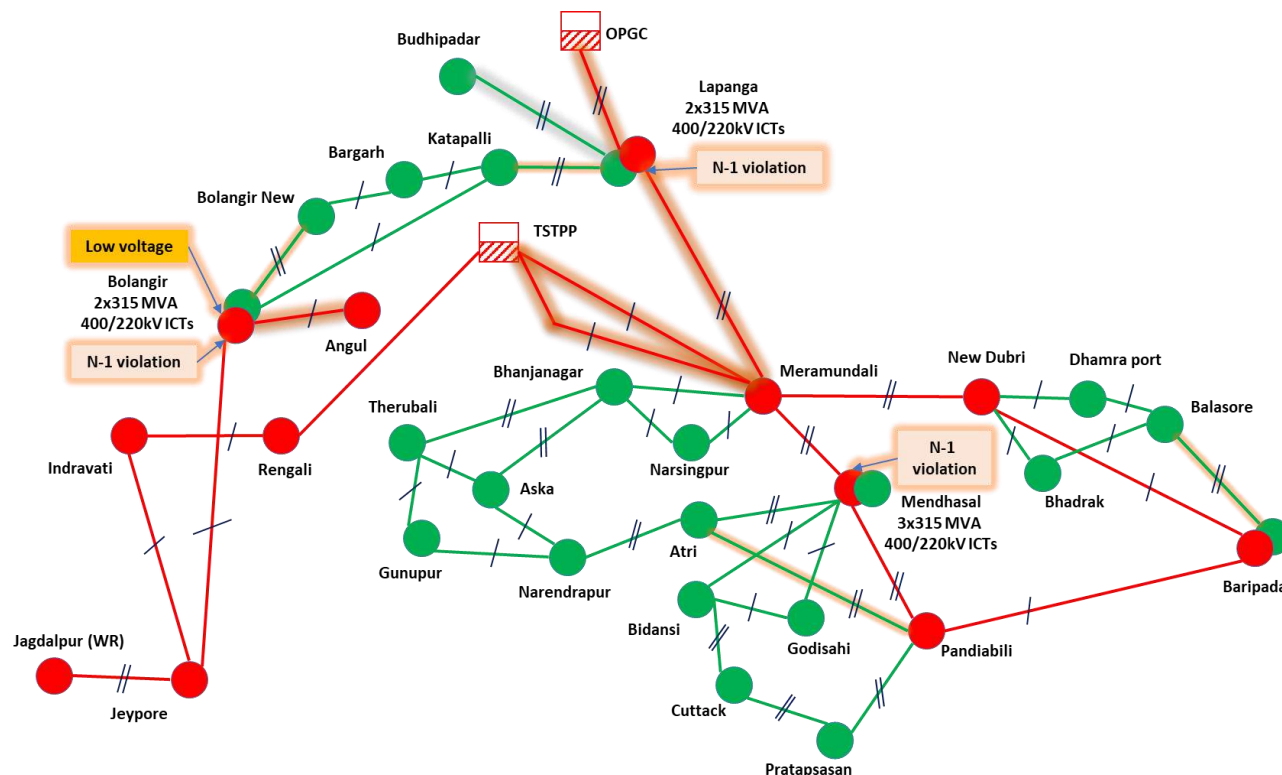
The capital region of Odisha, accounting for nearly 40% of the state's predominantly domestic and commercial demand, experiences very high summer load due to the hot and humid climate and associated space-cooling requirements. Transmission corridors supplying power to this region are already operating at their capacity. During peak summer, voltage in this region has been observed below 390 kV during peak summer even with all corridors in service and has fallen below 370 kV during contingencies. Few near-miss events have already been occurred.

These concerns have been deliberated in several OCC meetings, including the latest 226th OCC, where ERLDC once again recommended implementation of adequate Dynamic compensation devices as long-term measure.

Details of the stressed corridors and contingencies are as follows:

Limiting Constraints	Contingency/Risk in Operation	Remedial Measures
400kV infeed to Bhubaneswar Area	N-1 contingency, Probable voltage	Short term measure (Summer '26):

• 400 kV Lapanga-Meeramundali D/C • 400 kV OPGC- Lapanga D/C • 400kv Talcher – Meramundali D/C • 400kV Meeramundali-Medhashal D/C	collapse scenario in case double circuit outage	✓ Implementation of UVLS scheme in Baripada, New Duburi & Mendhashal area. ✓ Network reconfiguration- 220kV system.
Transformation capacity & 220kV network near Bhubaneswar: • 400/220 kV 315 MVA ICT at Mendhashal -1 & 2 & 3 • 220 kV Pandiabili- Atri D/C		Long term measure (2028 onwards): ✓ Reconductoring of 400kv Talcher – Meramundali D/C approved under ISTS scheme. ✓ OPTCL agreed to install ± 300MVAR STATCOM along with 2x125MVAR MSC and 1x125MVAR MSR at Mendhashal <i>Reconductoring of 400 kV OPGC-Lapanga D/C, 400kV Meeramundali-Medhashal D/C and installation additional ICT or upgradation of ICT capacity may be taken up earliest basis</i>



In western part of Odisha, where majorly industrial loads along with CPP are connected. Budhipadhar, Rourkella, Lapanga and Bolangir are major power sources of this area. 220kV lines connecting Budhipadhar to Lapanga to Bolagir are highly loaded. Detailed contingencies are as follows:

Limiting Constraints	Contingency/Risk in Operation	Remedial Measures
220kV network: • 220 kV Lapanga-Katapalli D/C • 220 kV Bolangir- New Bolangir D/C • 220 kV Budhipadar- Lapanga D/C • 220 kV Budhipadar- Vedanta D/C	▪ N-1 contingency ▪ Low voltage issue.	Short term measure (Summer '26): ✓ Network reconfiguration- 220kV Network. Long term measure (2028 onwards): ✓ Odisha planned for reconductoring this link.
Transformation capacity: • 400/220 kV 315 MVA Lapanga ICT - 1&2 • 400/220 kV 315 MVA ICT at Bolangir -1 & 2	Low voltage issue in this region	<i>Work needs to be expedited to complete early. Augmentation of transformation capacity may be planned.</i>

DVC:

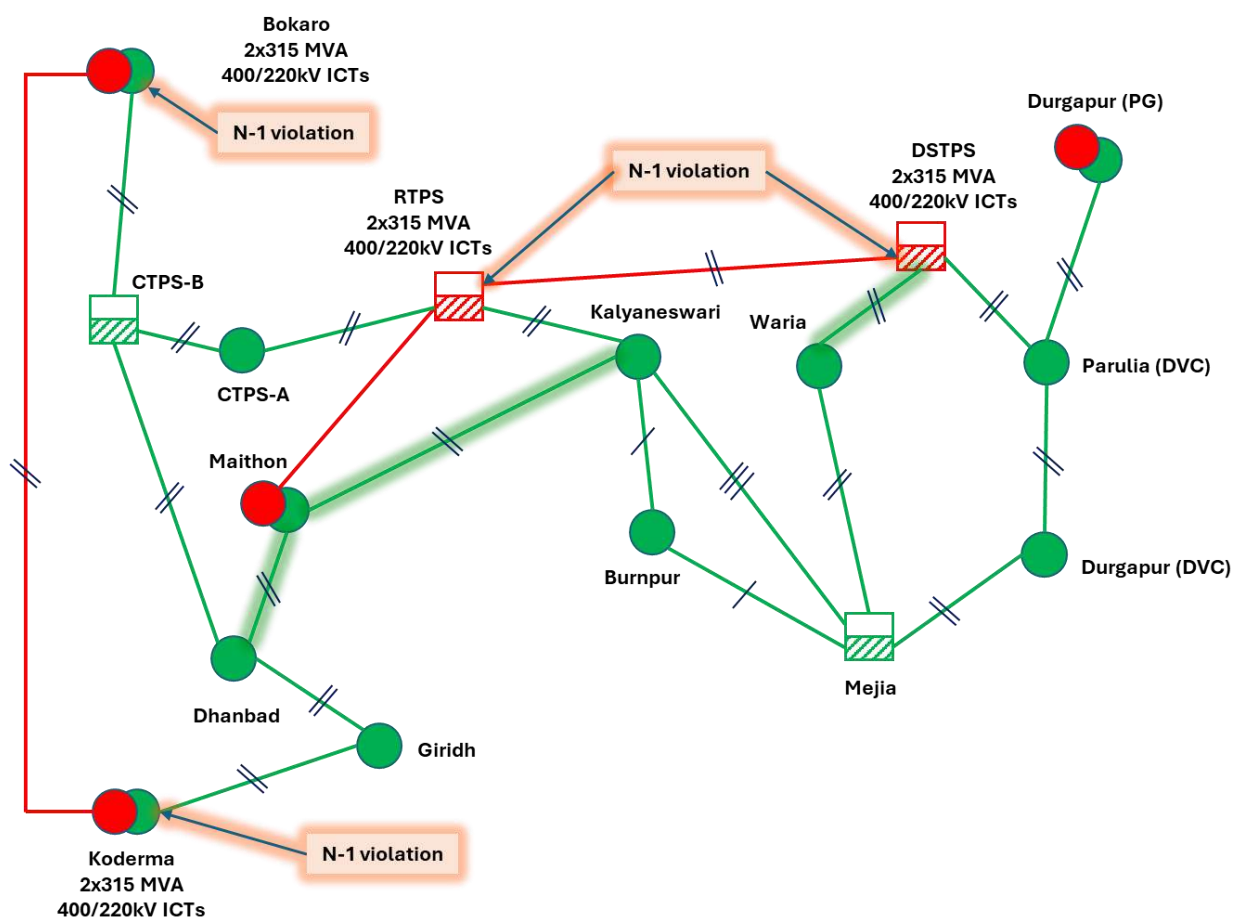
DVC caters mainly to industrial load in the Durgapur, Asansol, and Maithon belt, with demand remaining steady across seasons. Following the decommissioning of 132/220 kV generation, power flow from the 400 kV system into the 220 kV network has increased, leading to heavy loading of 220 kV corridors at Maithon and DSTPS and ICT overloading at Koderma, DSTPS, and Bokaro.

With new 400 kV generation planned and demand growth continuing, the 220 kV network and ICTs are likely to face further congestion, necessitating timely augmentation.

Detailed network constraints are as follows:

Limiting Constraints	Contingency/Risk in Operation	Remedial Measures
400/220kV ICT having generation • 315 MVA ICT at Bokaro -1 & 2 • 315 MVA ICT at Koderma - 1 & 2 • 315 MVA ICT at DSTPS (Andal) -1 & 2	N-1 violation of these ICTs due to high generation in these buses	Short term measure (Summer '26): ✓ Network reconfiguration- 220kV Network. Long term measure (2028 onwards): ✓ DVC planned for an additional 500 MVA ICT at Bokaro. ✓ ICTs will come to Koderma (PH-II) expected by Feb 2028

		More capacity addition is required.
220kV link: • Maithon -Dhanbad D/C • Maithon-Kalyaneshwari D/C • Waria -DSTPS D/C	N-1 violation	Short term measure (Summer '26): ✓ Network reconfiguration- 220kV Network. Long term measure (2028 onwards): ✓ Establishing connectivity between 220 kV and 400 kV of Mejia power plants may reduce the constraint. <i>Long term plan to increase connectivity with 400kV S/S needs to be chalk out</i>



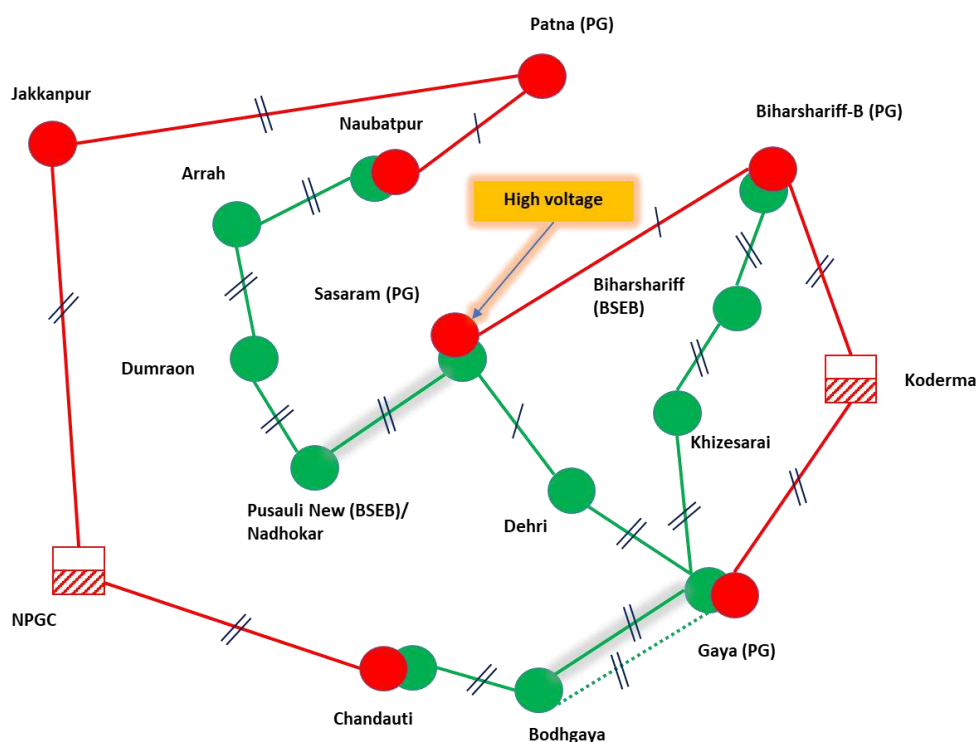
Bihar:

Bihar is undergoing through a massive demand growth over last decade. Its maximum demand met has increased fourfold in this period and is expected to grow further. Both

STU & CTU network in Bihar have grown in this period supplementing the need of the state. The state's demand pattern is largely driven by domestic and agricultural consumption, which leads to sharp variations across different times of the day and seasons. As a result, the grid faces operational challenges, with certain lines getting overloaded during peak periods and over-voltage issues occurring in other parts of the network during lean period.

Detailed network constraints are as follows:

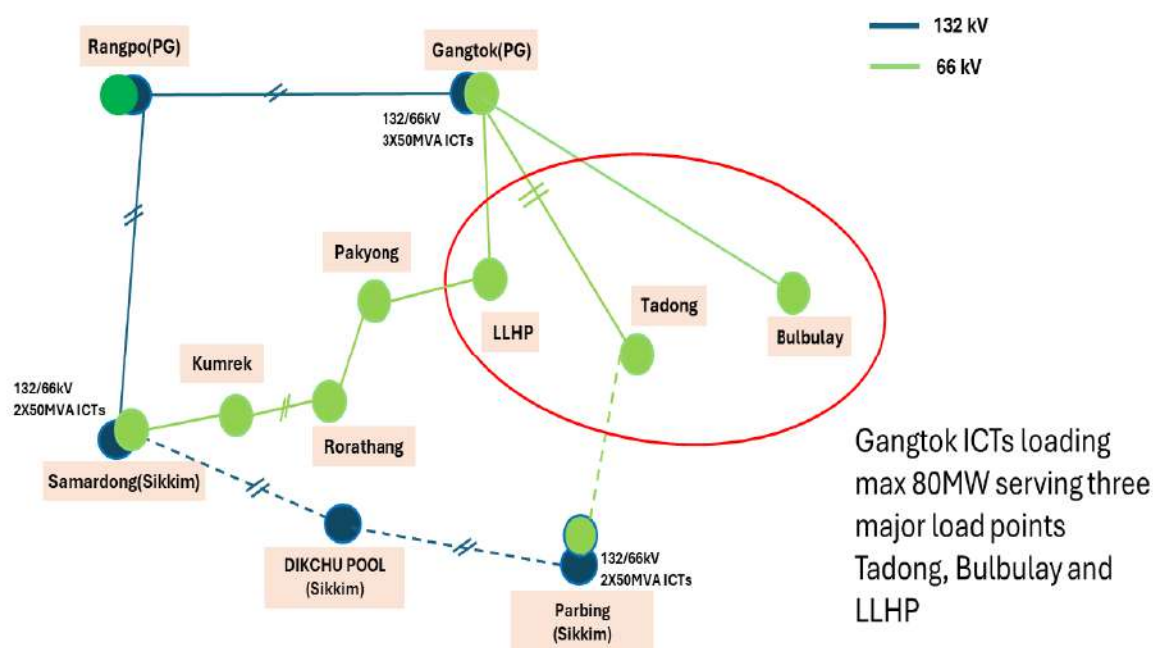
Limiting Constraints	Contingency/Risk in Operation	Remedial Measures
220 kV Sasaram (PG)- Pusauli New D/C 220 kV Gaya- (PG) - Bodhgaya D/C	N-1 violation	Short-term measure: Reconductoring of Sasaram (PG)- Pusauli New D/C has been completed. However, upgradation of bay equipment at State end is pending. Commissioning of 2nd 220 kV Gaya- Bodhgaya D/C post-LILO of 220 kV Gaya-Chandauti D/C at Bodhgaya will make these circuit N-1 secure. <i>Completion of both works needs to be expedited before 2026 Summer.</i>
High voltage issues at 400 kV Sasaram (PG), 400 kV Saharsa (PMTL) (BGCL)	High voltage issues	Short-term measure: As a short-term measure, lightly loaded lines connected to these S/S are opened. Long-term measure: Reactive compensation needs to be planned



Sikkim:

Sikkim demand is mostly centred around the capital city of Gangtok, which reaches its peak during the winter season. Gangtok power is mainly served from 132kV Rangpo (PG) source via 132KV Rangpo-Gangtok D/c. As the demand growth is in an upward trend in Sikkim and all dependency on 132KV Main and Transfer Scheme of 132kV Gangtok (PG) S/S. There is a need of strengthening of the internal network to ensure the reliability of power.

Detailed network constraints are as follows: Limiting Constraints	Contingency/Risk in Operation	Remedial Measures
Single main and transfer scheme at 132kV Gangtok (PG) S/S and Low voltage issues at 66kV system in the downstream of Gangtok	In case of outage (planned/forced) of main bus the power of capital city Gangtok may be put under threat.	Short-term measure: Network re-arrangement Long-term measure: 132/66kV Perbing S/S is under planning stage. Alternate source needs to be planned. Possibilities of Double bus scheme and transfer at Gangtok SS.



Inter/Intra state Transmission Network & Mitigation Measures across Eastern Region states has been carried out, outlining both short-term (**Summer 2026 onwards**) and long-term (**Summer 2028 onwards**) requirements. The same was submitted on 21.08.2025 for deliberation in the 230th OCC meeting. However, during the OCC, it was highlighted that more time is required for states preparedness, suggestions and reviews are essential for effective discussion. Further, OCC has directed to discuss with states and put up the gist of the discussions:

In this context, two meetings were conducted to discuss constraints in detailed:

- **West Bengal -Intra-state system:**

An Online meeting with West Bengal was held on 9th September 2025, where CE, SLDC, ACE (System Operation), SLDC & CE, STU were present. All issues were discussed, and West Bengal recognised the requirement for urgent measures for the upcoming summer and beyond.

Short-term (Summer 2026 onwards):

- Modification of existing SPS scheme of Subhasgram(PG) with under Voltage logic.

Long-term (Summer 2028 onwards):

- Requirement of another 765kV corridor connecting New Jeerat. (Under discussion in the CMETS)
- Requirement of dynamic reactive power compensation was deliberated. WB STU informed that Static Capacitors are being planned instead dynamic reactive compensation. However, ERLDC opined that dynamic reactive compensation is crucial due to the changing load types and increasing AC loads, which pose a significant threat of FIDVR event.

- It was decided to conduct a joint study with ERLDC and WB STU to finalize the plan for dynamic/static reactive compensation.
- **Bihar, Jharkhand, DVC & Sikkim:**

An Online meeting with Bihar, Sikkim, Jharkhand & DVC was held on 11th September 2025, where representatives from SLDCs & STUs/Planning were present in the meeting and all the points and action plan were deliberated.

States	Constraints	Short-term (Summer 2026 onwards):	Long-term (Summer 2028 onwards):
Sikkim	132kV Gangtok (PG) S/S and Low voltage issues at 66kV system in the downstream of Gangtok	- Sikkim load is basically winter pre-dominant. Alternative Source of Gangtok via Parbing (132/66KV SS) is in advance stage of commissioning.	- Reconductoring of 66KV Downstream network Gangtok – Tadong-Sinchey-Bulbulay. - * Requirement of Double main bus scheme at 132KV Gangtok will be explored by Sikkim
Jharkhand	220 kV Maithon New Dumka D/C (JUSNL) and 220 kV Tenughat–Govindpur D/C (JUSNL)	- 220KV Dumka – Gobindpur lilo at Dhanbad (NKTL) will be commissioned by Dec 2025. - 2 nd ICT 400/220KV 315MVA at Tenughat to be commissioned. * Bay equipment of PVUNL – Tenughat will be upgraded as per the line rating to make it capable of carrying power more than 200MW	220KV Tenughat - Gomia & 220KV Tenughat-Hazaribagh Road will be commissioned tentatively on Dec 26.
Bihar	220 kV Sasaram (PG)- Pusauli New D/C 220 kV Gaya-(PG) - Bodhgaya D/C	- Charging of 2nd 220 kV Gaya-Bodhgaya D/C and bay equipment upgradation of 220kV Naodkhar-Psuali-D/C will be expedited.	-

	High voltage issues at 400 kV Sasaram (PG), 400 kV Saharsa (PMTL) (BGCL)	- Buxar Generation will reduce the overloading issue.	
DVC	400/220kV ICTs at Generating Stations: 315 MVA ICT at Bokaro -1 & 2 315 MVA ICT at Koderma -1 & 2 315 MVA ICT at DSTPS (Andal) -1 & 2 220kV link: • Maithon Dhanbad D/C	- 315MVA 400/220KV Koderma ICT replacement (Taken on load basis from MZP) will be completed by Dec 2025. - DVC agreed for SPS scheme for N-1 compliance of ICTs at Bokaro. - Joint Study will be done and scheme will be finalised accordingly. - DVC informed, loading of 220KV Maithon-Dhanbad will be controlled by network rearrangement.	- 400/220KV ICTs upgradation to 500MVA at Koderma. - 3 rd ICT 500MVA at Bokaro. - Connectivity between Mejia B 400KV with 220KV DVC System.

- **Odisha:**

Following to the near miss event on 12.08.2025 and subsequent discussion in the 230th OCC held on 22.09.2025 at Bhubaneswar, it was discussed to form a committee to assess the requirement of short term and long-term transmission adequacy of Odisha system. In this regard, an online meeting has been planned on 18.09.2025 or 19.09.2025.

However, the issue of intra-state network of Odisha was highlighted by ERLDC in their internal Grid Co-ordination Committee meeting held on 3rd September 2025, Bhubaneswar, where all stakeholders of Odisha were present. Following agenda points were discussed:

- SLDC Agreed to implement UVLS scheme in and around Bhubaneswar areas to safeguard the system in next summer.
- STATCOM already planned by Odisha at Mendhashal as long-term measure.
- Reconductoring of 400kV OPGC-Lapanga D/C & 400kV Meramundali-Mendhashal D/C.

ERLDC may update. Members may discuss.

Deliberation in the Meeting

ERLDC presented a detailed overview of the existing vulnerabilities in the Intrastate transmission network of Eastern Region (ER)(**Annexure B.2.1**), particularly focusing on under-voltage, network overloading, and potential power failure scenarios. The State wise deliberation is detailed below: -

WEST BENGAL:

ERLDC highlighted that the state's demand is partly met by WBPDC (intra-state generation), and import through important transmission corridors such as 765 kV New Ranchi–Medinipur–New Jeerat D/C, 400 kV New Ranchi–New PPSP D/C & 400 kV Kahalgaon–Farakka D/C lines.

- During peak demand period, even with all above lines in service voltage occasionally comes down to 380kV level which can result in voltage collapse in the region under **N-2 contingency**(simultaneous outage of both circuits) of the **765kV New Ranchi–Medinipur–New Jeerat D/C**.
- They also apprised the forum about a special on-line meeting which was conducted on 9th September 2025 where these issues were thoroughly discussed and following short term as well as long term measures were suggested to mitigate the concerns arising out of voltage collapse issue.

Short-term (Summer '26) Measure:

• Modification of Existing SPS Scheme at Subhasgram (PG) with Undervoltage Logic

ERLDC informed that a Special Protection Scheme (SPS) is currently operational at Subhasgram(PG) Substation to address overloading issues of the ICTs. The existing scheme involves load curtailment of 350MW from pre-identified feeders to prevent ICT overloading under contingency conditions.

❖ To further enhance system voltage stability, ERLDC proposed that the existing SPS scheme be modified to incorporate undervoltage logic with suitable time delay. Under this modified scheme, stepwise load shedding can be executed when bus voltages fall to critical level to prevent voltage collapse.

❖ They also submitted that as per PV curve studies, implementing an undervoltage load shedding (UVLS) scheme at Subhasgram (PG) with a trigger level of 370 kV and the same pre-identified 350 MW of load, could lead to an estimated voltage improvement of approximately 14 kV.

❖ SLDC, West Bengal submitted that in view of upcoming state elections in upcoming summer of 2026, implementation of UVLS schemes may be avoided.

❖ WBSLDC further suggested that they would discuss the issue of proposed modifications in SPS at Subhasgram with WBSEDCL & CESC after Puja. The outcome of the said meeting shall be shared in the next OCC meeting.

❖ WBSEDCL acknowledged the operational necessity of implementing the Undervoltage Load Shedding (UVLS) scheme from the perspective of grid security and stability. However, they recommended that the scheme be designed with graded load shedding logic, enabling the system to be restored with minimal curtailment, thus balancing grid stability with consumer impact.

❖ PGCIL highlighted concerns regarding differential voltage levels across power system equipment, particularly in relation to their insulation design specifications. They emphasized that voltages must be maintained within a specified range to avoid thermal and electrical stress on equipment, which could accelerate deterioration and reduce the technical useful life of assets.

Long term measure (2028 onwards)

- **Additional 765kV corridor to connect New Jeerat**
- **WB is implementing Static compensation (Capacitor Bank).**

WBSETCL submitted that, based on their system studies, three locations have been identified where static compensation(3*50MVAR) (capacitor banks) can be implemented to provide reactive power support and improve the voltage profile of the system.

❖ However, ERLDC suggested that given the rapid increase in space cooling load in and around Kolkata areas and the anticipated rise in demand in the near future, it would be more appropriate to consider the installation of dynamic compensation to ensure voltage stability under varying system conditions.

❖ WBSLDC raised concern regarding the timely replacement of insulators in the PMJTL transmission lines, emphasizing the need for early action to ensure the healthiness, reliability, and continued availability of these critical transmission corridors.

❖ In response, PGCIL informed that insulator replacement work at identified locations on all 400 kV lines has been completed. Furthermore, the procurement contract for insulators required for 765 kV lines has already been finalized.

❖ PGCIL further assured that insulator replacement at all major identified locations on the 765 kV network is expected to be completed by February 2026.

OCC Decision

OCC acknowledged the operational necessity of modifying the existing Special Protection Scheme (SPS) at Subhasgram (PG) to address potential voltage collapse under N-2 contingency of the 765 kV New Ranchi–Medinipur–New Jeerat D/C.

- OCC advised SLDC, West Bengal, WBSEDCL and CESC to meet after puja for discussing the proposed modified SPS scheme at Subhasgram and share the outcome in next OCC.
- OCC advised to expedite the planning of additional 765 kV transmission corridor to New Jeerat, as a long-term system strengthening measure to address the anticipated surge in electricity demand in the state in the coming years and will significantly contribute to enhancing transmission capacity, ensuring grid reliability, and maintaining voltage stability under high-load conditions.
- OCC suggested that SLDC, West Bengal and WBSETCL may explore the possibility of utilizing the existing 400 kV Haldia-Subhasgram (PG) D/C line, which remains lightly loaded relative to its thermal capacity. It was suggested to explore the option of connecting the said D/C line to nearby sources which would improve its utilization and enhance the reliability of power supply at Subhasgram during exigency conditions. This may ensure power availability to the state capital during any contingencies.
- Additionally, OCC advised WBSLDC to undergo a thorough study and propose scheme for the better utilization of the 400 kV Haldia–Subhasgram (PG) line, outlining the technical feasibility, load flow analysis, and contingency scenarios. The proposed scheme shall be discussed in the OCC forum and if required, shall be flagged at the appropriate level to facilitate early implementation.
- OCC advised PGCIL to expedite the insulator replacement works on the 765 kV PMJTL lines and ensure completion before the onset of summer'26.
- OCC emphasized that all defective insulators of Deccan make, as well as those identified to be at risk of failure in the near future, should be completely replaced. It was specifically recommended that replacement be prioritized to ensure that at least one circuit is fully restored and kept in service, thereby ensuring reliable power supply to meet the summer'26 demand of the state.
- OCC advised PGCIL to include the status of insulator replacement work of PMJTL lines as a follow-up agenda item in upcoming OCC meetings. This will enable continuous monitoring of progress and ensure the timely completion of the works.

ODISHA

Deliberation in the meeting

ERLDC informed that in continuation of the deliberations held during the 230th OCC meeting held on August 22, 2025, Bhubaneswar, a follow-up meeting was held on September 18, 2025, with Sldc, Odisha & OPTCL and deliberation of the meeting are as follows: -

- SLDC Odisha was requested to constitute a committee at the earliest to review the existing protection settings and propose both short-term and long-term measures as recommended by the OCC.
- SLDC Odisha was advised to expedite the implementation of the Under Voltage Load Shedding (UVLS) scheme and in response Sldc, Odisha requested that they will highlight the matter to their higher management and revert back on the same to ERPC & ERLDC within 15 days.
- GM, OPTCL updated that
 - ❖ the reconductoring works of 400kV OPGC–Lapanga D/C, 400kV Meramundali–Mendhashal D/C, 220kV Budhipadar–Lapanga D/C & 220kV Lapanga–Katapali D/C are already approved and currently under tendering stage and expected to be completed by December 2027.
 - ❖ STATCOM at Mendhasal is planned, with commissioning expected in 2028–29.
 - ❖ Talcher Stage III – Meramundali B and Talcher Stage III – Pandiabili Transmission lines are Under Construction and expected to be completed by FY2027-28.

OCC Decision

- OCC directed OPTCL and SLDC Odisha to submit actions taken report i.r.o. near miss event in Odisha event on 12th August 2025.
- OCC advised SLDC Odisha to obtain approval from appropriate authority for implementation of UVLS scheme proposed for safeguarding the load of capital city and nearby area from cascaded failure.
- OCC directed SLDC Odisha to constitute the committee as recommended earlier without any further delay. The committee comprising of members of ERPC, ERLDC, SLDC, OPTCL shall review the protection settings and the short term/long term measures as suggested by ERLDC.

DVC

Deliberation in the meeting

ERLDC informed that an online meeting was held on 11.09.25 for discussing the constraints in DVC network and mitigation measures as follows:

Short term measure

- Replacement of 315MVA 400/220KV Koderma
 - Implementation of SPS scheme for N-1 compliance of ICTs at Bokaro.
- ❖ DVC updated that the 315 MVA ICT at Koderma will be charged by Dec'25 and the SPS scheme for Bokaro will be finalized after joint study with ERLDC.

Long term measure

- ❖ 400/220KV ICTs upgradation to 500MVA at Koderma
- ❖ Installation of 3rd ICT 500MVA at Bokaro
- ❖ Connectivity between Mejia B 400KV with 220KV DVC System

➤ DVC updated that the upgradation of ICT to 500 MVA at Koderma and installation of 3rd ICT at Bokaro will be completed tentatively by May'26 and the connectivity between Mejia B 400KV with 220KV DVC System will be established by Dec'26.

OCC Decision

OCC advised DVC to expedite the implementation of the above suggested measures by ERLDC to improve the system resilience and reliability.

Jharkhand

Deliberation in the meeting

ERLDC informed that an online meeting was held on 11.09.25 for discussing the constraints in Jharkhand network and mitigation measures as follows:

Short term measure

i) 220KV Dumka –Govindpur LILO at Dhanbad

ii) Commissioning of 2nd ICT 400/220KV 315MVA at Tenughat.

➤ JUSNL updated that the LILO of 220KV Dumka –Govindpur at Dhanbad will be completed by Dec'25.

Long term measure

Erection of 220KV Tenughat -Gomia & 220KV Tenughat-Hazaribagh Road transmission line

➤ JUSNL updated that the erection of the above-mentioned line will be completed by Dec'26.

OCC Decision

OCC advised JUSNL to expedite the implementation of the above suggested measures by ERLDC to improve the system resilience and reliability.

2.2 Issues for follow up.

Issue	Reference	Last updated Status	Action Point
Update on Rajarhat GIS (POWERGRID) 400/220kV S/S: 2x500MVA	Vide 226th OCC dated 22.04.2024: The need to prioritize the installation of a 3rd 400/220KV,500MVA ICT at Rajarhat (PG) with the same urgency as Subhasgram (PG) to prevent a recurrence of similar critical situations in the future If the proposed 3rd ICT is not operational by the summer of 2026, severe	As per 230th OCC: PowerGrid informed that the transportation of the ICT has already commenced from Kolaghat. Regarding the GIS package, it was informed that only a single bid has been received from M/s Siemens as of 17th July 2025. Consequently, the bid submission deadline has been extended by one month, after which a final decision will be taken. The updated. status on the GIS	Powergrid updated that they would go for fresh bidding since only one bid has been received. OCC Decision: OCC took a serious note of the matter and observed that all the efforts to ensure the availability of 3rd ICT at Rajarhat

	congestion is likely to affect the ICTs at Rajarhat (PG).	package tender will be shared in the next OCC meeting. OCC advised PGCIL to expedite the bidding process for the GIS package and finalize the same at the earliest to avoid any further delays in project implementation.	<i>would go in vain due to delay in implementation of GIS Bay for the said transformer.</i> <i>OCC strictly advised PGCIL to complete the bidding process at the earliest to avoid any further delays in commissioning of ICT.</i>
Near Miss event in Odisha System on 12.08.2025	On 12.08.2025 at 22:36 hrs, the Odisha system experienced a near-miss event while meeting its all-time maximum demand of 7302 MW (recorded at 22:33 hrs). the events led to the tripping of 400 kV Lapanga-Sterlite D/C and 400 kV OPGC Lapanga D/C transmission lines. This led to critical loading of 400 kV Talcher-Meramundali D/C lines putting the power supply to capital city of Bhubaneswar and adjoining areas in jeopardy.	As per 230th OCC: OPTCL was advised to submit the Action Taken Report to ERPC/ERLDC within a month. OCC advised SLDC Odisha to constitute the committee at the earliest. The committee shall review the protection settings and the short term/long term measures as suggested by ERLDC. ➤ OCC also advised SLDC Odisha to expedite the identification of HT and LT feeders in coordination with DISCOMs and share the details of the identified feeders for the implementation of the UVLS scheme at the earliest.	<i>OPTCL/SLdc, Odisha updated that for implementation UVLS at OPGC approval from Higher management is awaited.</i> <i>They also updated that the committee i.r.o OPGC-Lapanga Grid disturbance shall be formed shortly in co-ordination with OPTCL.</i> <i>OCC took a serious note of the matter and advised SLDC,Odisha to submit the action taken report,Details of the committee formation and status of UVLS</i>

			<i>implementation in Odisha network to tackle the Under voltage issues at the earliest.</i>
Update on IB Valley TPS Islanding Scheme	- IB valley TPS Islanding scheme has also been put on hold for long time. The status regarding the same has been sought on urgent basis by Ministry of Power (Govt of India). 229th OCC Decision: ➤ OCC advised ERLDC to conduct the dynamic studies of the OPGC network at the earliest in association with SLDC Odisha and OPGC. ➤ OCC advised OPTCL to prepare the DPR after the completion of dynamic studies. ERLDC has done dynamic study of the proposed islanding scheme, and one online meeting was arranged on 19.08.25 to discuss about the study result, where SLDC Odisha, OPGC	As per 230th OCC: SLDC Odisha informed that the DPR for the IB Valley TPS Islanding Scheme is planned to be submitted by September 2025, and the same will be communicated to ERPC.	<i>SLDC Odisha informed that the updated DPR is yet to be prepared by OPTCL. OCC advised OPTCL to expedite the preparation of the DPR and intimate the same to ERPC at the earliest</i>

	(IB Thermal) & OPTCL were present.		
Commissioning of an additional 250 MVA ICT at Tenughat	On 06.08.2025 a special meeting was convened to decide on the future course of the 400 kV Tenughat–PVUNL interim line.	As per the Deliberation of the Meeting, JUSNL and SLDC Jharkhand have decided to keep the 400 kV Patratu–Tenughat line continuously in service to improve the reliability of the Jharkhand power system. SLDC highlighted that the line helps relieve loading on the 220 kV Maithon–Dumka D/C and supports the Dumka–Govindpur loop under varying Tenughat unit availability, including during outages. However, during high demand periods with both Tenughat units offline, loading on the line may exceed 200 MW. This is a concern due to the single moose conductor termination (limiting capacity to 200 MW) and the aging ICT at Tenughat (safe limit: 180 MW due to a 40-year-old CT).	<i>JUSNL informed the forum that the electrical clearance for the ICT is currently pending due to the non-submission of the transformer test report to the Energy Department, Government of Jharkhand.</i> OCC Decision: OCC advised JUSNL to expedite submission of the required transformer test report to the concerned authority to avoid further delays in obtaining electrical clearance and ensure timely commissioning and integration of assets into the grid.
Implementation of AGC in Intra-state generating units.	218th OCC dated 13.08.24 With the increasing penetration of renewable energy, managing frequency is expected to become more	As per 230th OCC: Bihar updated that Closed loop communication link has been established however testing is yet to be performed.	Deliberation: NTPC updated that NOC has been received from SLDC, Bihar to carry out open loop testing.

	challenging in the future. Therefore, it is crucial to enhance frequency control and stability through increased participation from intra-state AGC.		<i>NTPC further updated that They will approach SLDC, Bihar for NOC for carrying out, Close loop testing.</i>
Scheduling of PVUNL units BY SLDC, Jharkhand	In 229 th OCC Meeting, It was decided that PVUNL should be treated as a regional entity for the purpose of scheduling and accounting till it is under control area jurisdiction of ERLDC as per CERC directives till December 2025. Thereafter All the Scheduling & other related issue shall be dealt as per the State Regulations.		<i>JUSNL updated that SAMAST scheme is under Implementation in SLDC and is expected to be completed by January 2026. Thereafter scheduling of PVUNL may be done.</i>

All concerned may update. Members may discuss.

2.3 Extension of AMC of SCADA System and OSI Support in Eastern Region, maintained by M/s Chemtrols Industries Pvt. Ltd: BSPTCL

This is regarding the Annual Maintenance Contract for services of round the clock comprehensive maintenance of software and hardware of overall EMS/SCADA system of BSPTCL. Present SCADA/EMS system in BSPTCL (Eastern Region) is commissioned in March 2016 by M/s Chemtrols and M/s OSI under ULDC Upgradation Scheme through POWERGRID and was under AMC contract till 02nd July 2023. The AMC was further extended till 02nd July 2025 on the same rates and terms and condition in all ER Constituents on mutually agreed basis with great effort of ERPC. As Annual Maintenance Contract (AMC) for the SCADA system at SLDC, Bihar, maintained by M/s Chemtrols Industries Pvt. Ltd., has expired. Similar situations may also arise in other Eastern Region constituents' states and DVC. The matter is of critical importance as the SCADA system forms the backbone of real-time grid monitoring and control. Given the situation, and the requirement of continued support from OSI for ensuring reliable and secure operations, it

is proposed to discuss the matter in 230th OCC meeting with all Eastern Region constituents, ERLDC, M/s OSI and M/s Chemtrols to address the issues of extension of AMC for SCADA so that the modalities, cost, terms and conditions, support in case of end of life of hardware and software may be fixed and uniform rates and terms and conditions may be discovered.

In 230th OCC Meeting:

OCC opined that a separate meeting may be convened by PGCIL involving all ER states, ERLDC, M/s Chemtrols, and M/s OSI for detailed deliberations

PGCIL may explain. Members may discuss.

Deliberation in the meeting

PowerGrid informed that two meetings have already been held with Chemtrol for the finalization of modalities, cost, terms and conditions, and support arrangements in case of end-of-life (EOL) of hardware and software.

➤ *ERLDC updated that they have also conducted a meeting with Chemtrol, and the reference rate is expected to be finalized within 10 days, after which it will be communicated to PowerGrid.*

OCC Decision

OCC advised PowerGrid to arrange a separate meeting involving all ER stakeholders for finalization of modalities, terms & conditions, and cost, taking into consideration the reference rate provided by ERLDC.

2.4 Sorting of 400 kV Farakka-Purnea and 400 KV Purnea - Gokarna lines to establish the old configuration of Farakka-Gokarna 400 kV line either for better Reliability: WB SLDC

With present flooded condition in the Sahebganj around area, 400 kV Farakka-Purnea and 400 KV Purnea - Gokarna lines are kept off by ERLDC. If this status to continue for a considerable period of time, then it was proposed to make shorting of these two lines in a suitable place before the flooded area to establish the old configuration of Farakka-Gokarna 400 kV line either for better redundancy or for additional option to use in case of any exigency condition.

WB SLDC may update. Members may discuss.

Deliberation in the meeting

WBSLDC informed that due to ongoing flooding in Sahibganj area, the 400 kV Farakka–Purnea and 400 kV Purnea–Gokarna lines have been kept in open condition.

➤ *As an interim measure, shorting of these two lines has been proposed to enhance redundancy in the West Bengal network during exigencies.*

➤ *Power Grid ER-I submitted that the location for shorting of the above lines will be finalized shortly and the same shall be communicated to WBSLDC.*

➤ In response, ERLDC agreed with the proposal. However, it was observed that it may increase the risk of N-1 violation on the 400 kV KHSTPP–FSTPP corridor during summer peak.

➤ PGCIL submitted as the length of the line is getting modified due to shorting of above said lines, distance protection settings may be modified accordingly.

OCC Decision

OCC acknowledged the operational need for the shorting of the 400 kV Farakka-Gokarna line for increasing the redundancy of the WB network.

➤ OCC advised PG ER-I to identify the location for shorting of the above-mentioned lines at the earliest and the same maybe shared with ERLDC, WBSLDC and ERPC.

➤ OCC advised PGCIL to change the protection setting of the reconfigured line in co-ordination with ERLDC.

2.5 Bringing Purnea-Dalkhola lines to service either in through bus mode or in split bus mode: SLDC, WB

Present Bus arrangement at Dalkhola (PG) sub-station is very often not confirming (n-1) contingency criteria in respect of Kishanganj-Dalkhola 220 kV lines. Hence Purnea-Dalkhola lines need to be brought back to service either in through bus mode or in split bus mode depending upon the system study results. This is extremely important to implement before the festival period to avoid any undesired outcome during Durga puja festival and onwards.

WB SLDC may update. Members may discuss.

Deliberation in the meeting

WBSLDC highlighted the need to close 220 kV Purnea–Dalkhola lines before the festive season, due to high loading in Dalkhola and frequent (N-1) violations on the 220 kV Kishanganj–Dalkhola lines.

➤ WBSLDC further informed that Samsi is at present radially connected to Gazole & it is expected that load at Samsi during festive season will rise so in order to enhance the reliability of power supply to Samsi, Bus splitting at Dalkhola (PG) may be implemented along with closing of 220KV Purnea-Dalkhola (PG) D/C.

➤ ERLDC raised the following risks:

1. 50% chance of (N-1) violation if line is closed, with each circuit reaching up to 240 MW under peak conditions.

2. Possible cascading tripping in case one New Purnea–Purnea CKT trips, leading to:

❖ Tripping of second New Purnea–Purnea CKT

❖ Tripping of Kishanganj–Dalkhola D/C

❖ Resulting in loss of load at Samsi, Purnea, and Dalkhola

➤ ERLDC mentioned that while HTLS reconductoring has already completed, the bay equipment upgradation is still pending.

- PowerGrid clarified that both line and terminal equipment upgrades have been completed.
- WBSLDC remarked that HTLS reconductoring alone is insufficient to solve the issue and erection of new lines would have been a more sustainable solution.
- ERLDC and WBSLDC proposed the following for implementation of the scheme:
 1. Samsi load (which is radially fed from Gazole through splitting) may be shifted towards 220kV Purnea (PG) source through 220kV New Purnea (PG)-Purnea(PG)-Dalkhola(PG)-Gazole link.
 2. Dalkhola (WB) load may be kept on 220kV Kishanganj (PG) source through 220kV Kishanganj (PG)-Dalkhola (PG)-Dalkhola (WB) link.

OCC Decision

- OCC acknowledged WBSLDC's apprehensions, noting that the objective of the upgradation project stands defeated due to lack of proper planning and coordination among stakeholders.
- OCC advised PowerGrid and ERLDC to jointly confirm completion of terminal equipment upgradation and update the same in ERLDC's operational database.
- OCC consented to the proposal by ERLDC and WBSLDC for reconfiguring load flows as outlined above.

2.6 Providing multiple user access (Edit Option) for operating WBES to ensure smooth and uninterrupted operations: JIPL

JIPL received communication from ERLDC dated **01.09.2025** regarding **Implementation of login in WBES through verified OTP and the requirement to update user mobile numbers and email IDs.**

After internal discussion, JIPL responded via email on **02.09.2025**, highlighting a key operational challenge: **multiple users currently operate under the same credentials**, and receiving OTP on only one registered mobile number and email ID is creating difficulties in accessing the system efficiently.

We kindly request your guidance on how to proceed in such cases. Specifically, we would like to know if there is any provision to **accommodate multiple mobile numbers and email addresses for OTP delivery** under a single user account.

Additionally, we request you to kindly consider providing **multiple user access (Edit Option)** for operating WBES to ensure smooth and uninterrupted operations.

JIPL may update. Members may discuss.

Deliberation in the meeting

JITPL requested that multiple login access to the Web-Based Energy Scheduling (WBES) portal be provided to facilitate their operational convenience.

- ERLDC informed that any additional login with 'Edit' rights shall not be issued so as to uphold system security, avoid duplication of control, and ensure clear accountability of all actions performed in the system.

➤ ERLDC also appraised that additional logins with View Only rights may be provided, through which all details available in the 'Reports and Analytics' menu of WBES can be viewed and downloaded.

OCC Decision

OCC advised ERLDC to take up the issue of multiple WBES login with NLDC for exploring the possibility of proving multiple edit rights.

2.7 SgTPP is facing software license related and other issues during the meter data download from Commercial meters of some 400 KV feeders maintained by PGCIL: WBPDCCL

Line Feeder	Line Jurisdiction	Main Meter	Check Meter	Remarks
Farakka1	PGCIL	Genus	L&T	The check meter data download may be stopped any time due to expiry of the meter data software validity.
Berhampore-1	PGCIL	Genus	L&T	The check meter data download may be stopped any time due to expiry of the meter data software validity.
Berhampore-2	PGCIL	Genus	L&T	1. The main meter data download not possible from Optical port of the meter 2. The check meter data download may be stopped any time due to expiry of the meter data software validity.
Jeerat-2	PGCIL/WBSETCL	Genus	L&T	As per ERLDC, it is intra state feeder. Existing meter of PGCIL may be replaced with SAMSAT meter of WBSETCL

The same downloaded meter data is forwarded to the SLDC & ERLDC on periodical basis. Due to the above-mentioned issues, the existing practise may be discontinued at any time.

WBPDCCL may update & PGCIL may explain. Members may discuss.

Deliberation in the meeting

WBPDCCL informed that because of software issues, the data downloading from L&T meter (check meters) may be stopped at any time.

•In response, ERLDC mentioned that as L&T meters of Farakka 1, Berhampore 1 and Berhampore 2 are installed as check meters in the tie lines of West Bengal and as per the provision of metering regulations, these check meters data are not required at

ERLDC for accounting purpose. Main/Standby meters installed at these lines are sufficient for accounting.

- Regarding Genus meters, ERLDC informed that downloading of data is not required as the data is already being reported to them through AMR.
- Regarding the L&T meter replacement at Jeerat-2 feeder, WBPDCCL requested that PowerGrid may be directed to replace the L&T meter. However, ERLDC informed that since the said line is an intrastate line of West Bengal, (both substations belong to West Bengal), Accordingly, if required, Necessary action may be taken by West Bengal regarding replacement of the L&T meter

OCC Decision

OCC opined that the L&T meters installed at outgoing feeders of SgTPP may be replaced by WBPDCCL at their own cost.

2.8 Shutdown proposal of generating units for the month of October 2025: ERPC

Maintenance Schedule of Thermal Generating Units of ER during 2025-26									
System	Station	Unit	Capacity (MW)	CEA Approved		No. of Days	Revised proposed		Reason
				From	To		From	To	
DVC	KODARMA TPP	1	500	5-Oct-25	1-Nov-25	28			AO H-Blr, LPT & Gen
CESC	SOUTHERN REPL. TPS	2	67.5	27-Oct-25	5-Nov-25	10			AO H

Members may discuss/update.

Deliberation in the meeting

- DVC submitted that:
 - Koderma Unit#1 shutdown is being postponed and its shutdown will now be availed from 30.10.25 to 27.11.25.
 - Mejia Unit#4 shutdown will be availed from 23.10.25 to 26.11.25.
- CESC informed that the proposed shutdown of Southern TPS Unit#2 will not be availed.
- WBPDCCL informed that shutdown of Sagardighi Unit#3 is being preponed, and it will now be under shutdown from 26.10.25 for a period of 28 days.

OCC Decision

The OCC forum approved the shutdown proposals with the modifications as noted above.

2.9 Shutdown Program of Generating Units: NTPC

a) Shutdown of TSTPS Unit-1(500MW) (500 MW)

As per the approved schedule in the 228th OCC, the planned shutdown of **TSTPS Unit-1(500MW)** was from **16th October 2025 to 30th November 2025** (45 days).

However, due to delay in receipt of certain critical turbine spares being imported from Germany, it will not be feasible to commence the overhaul as per the original plan, these spares are essential to complete the overhaul and to ensure reliable operation of the unit after resumption.

In view of the above, it is kindly requested to consider the revised proposed outage schedule for TSTPS Unit-1 as follows:

Revised Dates: 15th December 2025 to 28th January 2026 (45 days)

b) Shutdown of FSTPS Unit#4 (500 MW)

FSTPP, NTPC will avail the Shutdown of Unit-4 (500 MW) from 01.11.2025 to 05.12.2025 as per scheduled Overhauling Plan (LGBR 2025-26).

c) Shutdown of NKSTPP Unit#3(660MW)

Planned shutdown of from 05.10.2025 for six days Reason: Preparation for major CAT-1 PG test like boiler efficiency, Turbine cycle heat rate, ACC condenser vacuum test etc. Some of the defects which will be taken up during the planned shutdown are as follows:

1. Attending various leakages in HPBP system
2. Attending various leakage in spray system in boiler
3. Attending defects in SOFA damper
4. Installation of power transducer in ECP for load measurement.

NTPC may update. Members may Discuss.

OCC Decision

The OCC forum approved the shutdown proposal as per the requested time period.

2.10 Resolving issues of repeated line faults in 220kV Keonjhar (PG) – Turumunga D/C line and enabling Auto-recloser function at Turumunga (OPTCL) end: PG ODISHA

The first time charging of 220kV Keonjhar (PG) – Turumunga D/C line on 04.06.2025. There has been several instances of line faults in these lines, however, each time the line has tripped from Turumunga (OPTCL) end. On further enquiry from OPTCL end, it came to the notice that, the Auto-Recloser function is not in service for both 220kV Keonjhar

(PG) – Turumunga#1 & 2 at OPTCL end resulting in tripping of the line from OPTCL end only, whereas auto-recloser function successfully operated from Keonjhar (POWERGRID) end.

Due to repeated faults in 220kV Keonjhar (PG) – Turumunga D/C line, 315MVA, 400/220kV ICTs at Keonjhar Substation along with associated bay equipment's have been subjected to both electrical and mechanical stresses. Recently, on 05.08.2025, 315 MVA ICT#2 at Keonjhar Substation tripped due to the operation of Buchholz relay and fault gases have been noticed during the DGA analysis. As only Buchholz relay was operated without any operation of electrical protection, it indicates internal fault and replacement this ICT is under progress.

As a precautionary measure we have now kept the auto-recloser function of 220kV Keonjhar (PG) – Turumunga D/C line in “Non-Auto” mode at Keonjhar (PG) end, till rectification/normalization of Auto-Recloser function of both the lines at Turumunga (OPTCL) end. A letter has been sent to The General Manager, EHT (O&M) circle, Jajpur Road (**Annexure B.2.10**) with a request to resolve the issue.

Hence, OPTCL may please be instructed to resolve the issue at the earliest for smooth and reliable operation of grid.

Further, functionality of operationalization of Auto-Recloser scheme for the lines, where two ends belong to different utilities may also be ensured by RLDC before according FTC Clearance.

PGCIL may update. Members may discuss.

OCC Decision

OCC opined that the matter maybe deliberated in PCC meeting.

2.11 Inordinate delay in extension of CT and CVT inputs of 400kV Angul-JITPL#1 & 2 at JITPL end for the reporting of Analog PMU Data to ERLDC: PG ODISHA

PMU has been installed by POWERGRID at JITPL, Angul plant under URTDSM package for the reporting of real time data to ERLDC. PMU was installed at JITPL and SAT was signed on 12.10.2018 with the remark “CT & PT connection will be done by JITPL during opportunity shutdown” (**Annexure B.2.11.1**). The optical link from JITPL to POWERGRID, Angul Substation has been commissioned, and the PMU is reporting to ERLDC since 07.03.2025 with digital signal only. Analog signals are not reporting to ERLDC due to the non-extension of CT & CVT inputs of 400kV Angul-JITPL#1 & 2 at JITPL end. In this regard, copy of e-mails and letters issued to M/s JITPL, Angul are attached(**Annexure B.2.11.2**) for ready reference. The validation of PMU data by ERLDC SCADA team has not been possible due to non-extension of inputs for Analog data, which in turn affecting the DOCO of PMU installed at M/s JITPL end. Also, contract closing of the executing agency (M/s GE) is getting affected due to pending validation of PMU data at ERLDC end due to delay in extension of CT and CVT inputs.

As commissioning and complete data reporting of JITPL PMU to ERLDC is a critical long pending issue, hence M/s JITPL may be suitably instructed to extend CT and CVT inputs to the installed PMU for both 400kV Angul-JITPL#1 & 2 at the earliest.

PGCIL may update. Members may discuss.

Deliberation in the meeting

- *Powergrid informed that the analog signals are not reporting to ERLDC due to the non-extension of CT & CVT inputs of 400kV Angul-JITPL#1 & 2 at JITPL end.*
- *JITPL submitted that it will be taken care of during the upcoming overhauling scheduled for 1st week of December.*

OCC Decision

OCC urged JITPL to extend CT and CVT inputs to the installed PMU for both 400kV Angul-JITPL#1 & 2 at the earliest.

2.12 Ensuring Reliable Operation of +/- 450 MVar STATCOM for Voltage Control at 400/220kV Jeypore Substation: PG ODISHA

In July 2018, (+/-) 405 MVar STATCOM was commissioned and charged at the POWERGRID 400/220kV Jeypore substation. The STATCOM consists of two voltage source converters (VSCs), each rated at (+/-) 100 MVar, along with two mechanically switched reactors (MSRs) rated at (-) 125 MVar each and two mechanically switched capacitors (MSCs) rated at 125 MVar each. The MSRs and MSCs are automatically switched “ON” and “OFF” based on the reactive power requirements.

All these components are connected to the medium-voltage (MV) side of a coupling transformer rated at 500 MVA, 400/28 kV. Since the MSR and MSC branches operate at lower voltages, they generally handle higher currents compared to the same MVar capacity of the 400 kV reactors and capacitors.

The primary purpose of the STATCOM is to provide support during transient stability rather than steady-state stability; however, it can also be used for steady-state stability for shorter periods. The Short Circuit Level (SCL) at the Jeypore substation is very low (6 GVA), causing frequent voltage fluctuations on the bus even with minor disturbances in the grid. These voltage fluctuations lead to a high number of operations of the MSRs and MSCs, contributing to increased stress on the circuit breakers.

On October 15, 2021, the Y-phase pole of the MSR 1 circuit breaker experienced a failure, resulting in significant economic losses and availability issues for the STATCOM. Following this incident, measures were implemented to limit the number of operations by adjusting the voltage band of the STATCOM.

In June 2025, a double circuit line (Jeypore – Jagadalspur) was added to the Jeypore bus, which increased the SCL to around 9 GVA; however, due to lightly loaded conditions, the bus voltage consistently remained high, ranging from 415 to 420 kV. This high voltage caused the STATCOM to continuously inject inductive reactive power between (-) 280 and (-) 430 MVar into the grid. As a result, both MSRs were in service for more than three months, which led to multiple hot spots at the clamps of the MSRs and the melting of conductors in several locations due to the prolonged high currents.

POWERGRID availed the shutdown of the STATCOM on 03rd July'2025 to address these hot spots and replace the melted conductors. On 27th August 2025, the R-phase of MSR I failed due to an internal flashover in the air core reactor coil resulting in an approximate financial loss of 1.5 crore and outage of MSR-I for 2 days. Subsequently, the voltage band was adjusted to a higher range to minimize the continuous operation of the MSRs.

Later, due to a change in load patterns, the number of MSR operations has increased again. To reduce these operations, the STATCOM's voltage band needs to be raised further. However, penalties imposed on OPTCL for injecting VAr at voltages greater than 1.03 PU are preventing this adjustment (as per information from GRID INDIA), leading to the continued operation of the MSRs.

As a result of these challenges, the utilization of the STATCOM is straying from its initial intended function. It is now primarily being used for steady-state stability, causing either continuous operation of the MSRs or an increased number of operations, leading to forced outages and economic losses due to equipment failure, as well as problems with hot spots and the melting of conductors.

To mitigate these issues, it is crucial to control the voltage at Jeypore to stabilize the grid, ensuring reliable operation of the STATCOM. This will allow for greater utilization during transient stability without risking forced outages due to equipment failure or conductor damages from prolonged service or excessive operations of the MSRs.

PGCIL may update. Members may discuss.

Deliberation in the meeting

- *PowerGrid updated that the ± 450 MVar STATCOM installed at the 400/220 kV Jeypore Substation for voltage control is currently being underutilized, as it is only being operated for steady-state stability.*
- *They also highlighted that the limited mode of operation of STATCOM has resulted in*
 - ❖ *continuous or frequent operation of Mechanically Switched Reactors (MSRs),*
 - ❖ *Forced outages and economic losses due to equipment failure,*
 - ❖ *Hot spots and conductor melting due to operational stress.*
- *To address the issue, PowerGrid requested that the forum consider raising the voltage band at Jeypore, which would reduce the frequency of MSR operations allow greater utilization of the STATCOM for transient stability purposes*
- *ERLDC submitted that raising the voltage band may result in penalties being imposed on OPTCL, as it could cause deviations from voltage limits at other nodes.*
- *ERLDC further suggested that the matter be taken up for detailed discussion in the upcoming CMETS meeting.*

OCC Decision

OCC noted the operational challenges highlighted by PowerGrid and agreed that the proposal to revise voltage band for better STATCOM utilization may be referred to the CMETS forum for detailed technical evaluation and stakeholder consultation.

2.13 De-commissioning of 3X16.67 MVar (50MVAR), Bus Reactor-1 at Jamshedpur Sub-station w.e.f. 25.08.2025: PG ER1

3X16.67 MVar (50MVAR), Bus Reactor-1 at Jamshedpur Sub-station was commissioned under Farakka Transmission System (FATL) on dated 29.07.1993. Further one no. 125MVAR Bus reactor was also connected in parallel to this Reactor w.e.f. 01.12.2021 under ERSS-IX project.

Residual Life Analysis (RLA) of the said single phase reactor units were carried out from CPRI, Bangalore in 2019. As per RLA report, the condition of Y-Phase and B- Phase reactors are critical and recommended for replacement.

As 02 units of single phase reactor were suggested to be taken out of service in RLA, the matter was taken up in 8th Consultation Meeting for Evolving Transmission Schemes in Eastern Region (CMETS-ER) where in it was agreed (as per para 6.5(c) of minutes) that “As long as the 50MVA bus reactor-1 (3x16.67MVA single phase reactors) at Jamshedpur S/s (which is presently installed in parallel with a 125MVA bus reactor) is serviceable, the same may be operated and thereafter it may be decommissioned”.

The Reactor Bank is being maintained by POWERGRID as per its maintenance strategy; however, its condition is getting deteriorated day by day. Recently during its routine oil testing, critically low BDV and high moisture content observed despite filtration done on many occasions.

Further, emergency shut down of 3x16.67 MVA Bus Reactor-1 at Jamshedpur was taken on 25.08.2025 at 16:08Hr due to critically low BDV and high moisture content. The drying out of units is planned but vacuum could not be achieved due to multiple leakages from various gasket joints as well as welded joints.

In view of the above, it is prudent that condition of said reactor bank is critical due to aging and is not serviceable. Further sudden failure of any unit may result in consequential damage to nearby equipment. Therefore, 3X16.67 MVA (50MVA), Bus Reactor-1 at Jamshedpur Sub-station is proposed for de-commissioning w.e.f. 25.08.2025.

PGCIL may update. Members may note.

OCC Decision

OCC noted.

2.14 Critical condition of tower at location no. 1100 (DD + 0) of 400kV D/C New Purnea – Farakka & New Purnea -Gokarna Transmission Line due to unprecedented flood in river Ganga: PG ER1

400kV D/C New Purnea-Farakka and New Purnea-Gokarna Line was commissioned in Nov'2019. The Line passes through river Ganga in Bhagalpur district. As per prevailing condition of the river, Pile foundations were made for the locations 1103 to 1111 (9 nos.) during construction of the Line. Later, due to encroachment of river Ganga, 2 no. Pile foundations at location no. 1101 & 1102 were also built and tower shifted on the Pile foundation.

Before onset of this monsoon on dtd. 30.06.2025, the tower at location no. 1100 was 430 Meter away from riverbank of Ganga and average encroachment of the river as per recent past was around 50-60 meter per year. During current monsoon season, heavy unprecedented flood is observed in the Ganga River and entire location was filled with approx. 2mtr. water, which further accelerated the soil cutting in the river at faster rate. (Annexure B.2.14)

During the flood, regular patrolling of the location through boats was done. Entire location till location no. 1083 was waterlogged till 22/08/2025. As the water level decreased, and riverbank became visible, location No 1100 was only 140mtr from riverbank. After this, in consultation with Flood Control Department, Bhagalpur, bamboo bundling along the riverbank was done in the location to limit further soil erosion.

Despite continuous efforts to limit the soil erosion through Bamboo bundling and Drum floating technique, the location was only 90 mtrs away from bank of river ganga on 06.09.2025. Due to further soil erosion, location no. 1100 was only 42 mtrs away on 13.09.2025 and it was then decided to switch off the Lines and proceed for de-stringing of the tower to avoid cascading damage to adjacent towers.

As on 18/09/2025, main river course is only 19 meters from tower no. 1100. Also, due to heavy rain, Location No. 1100 is flooded with water. As of now, de-stringing of four conductors could only be done in such adverse climatic conditions and tough task for making back stay. Nevertheless, the de-stringing shall be completed for rest of the conductor at the earliest possible time.

The location is under continuous monitoring by teams deployed at site. In the meantime, proposal for shifting the towers at location no. 1100 and 1099 on Pile foundation has been also processed and is under tendering.

Since the tower is in endangered condition, possibility of washing out of the tower due to above explained force majeure condition cannot be ruled out and, in such case, it will take considerable time in making the Line ready for charging.

PGCIL may update. Members may discuss.

Deliberation in the meeting

PowerGrid highlighted:

- ❖ *the critical condition of the transmission tower at location no. 1100 (DD+0) on the 400 kV D/C New Purnea – Farakka and New Purnea – Gokarna lines due to unprecedented flooding in the Ganga River.*
- ❖ *De-stringing of conductors on the location 1099 side has already been completed as a precautionary measure.*
- ❖ *A proposal for shifting towers at locations 1099 and 1100 to Pile foundations has been processed and is currently under tendering.*

OCC Decision

➤ *OCC noted the update and appreciated the proactive steps taken by PowerGrid. The forum advised PowerGrid to closely monitor the site condition and to expedite the tendering and implementation of the pile foundation shifting to ensure long-term stability and reliability of the transmission line.*

2.15 Availability Certification of deemed ISTS Lines of WBSETCL & OPTCL-ERPC

In reference to the directions issued by the Hon'ble Central Electricity Regulatory Commission (CERC) under:

- Para 63 of its Order dated 29.05.2025 in Petition No. 309/TT/2023 (Annexure B.2.15.1),
- Para 43 of its Order dated 31.05.2025 in Petition No. 324/TT/2023,
- and as per CTUIL letter dated 26.08.2025 (**Annexure B.2.15.2**),
- along with the relevant provisions of the CERC Tariff Regulations, 2024,

SLDC, West Bengal is required to submit the verified tripping details for the following transmission lines from the year 2014 onwards to ERPC at the earliest,. This is essential to enable the issuance of Availability Certification for the following deemed ISTS/natural ISTS lines of WBSETCL, which are carrying Inter-State Power:

1. 400 kV S/C Kolaghat–Baripada Transmission Line – 98.60 km
(Kharagpur–Baripada ISTS Portion including Loop-out ISTS part)
2. 220 kV S/C Santaldih–Chandil Transmission Line – 73.9 km

Further, in reference to the Hon'ble CERC Order dated 26.02.2025 in Petition No. 288/TT/2023, and CTU letter no. CTU/BCD/AVC/01 dated 26.08.2025 (**Annexure B.2.15.2**), SLDC, Odisha is also requested to submit the verified tripping details for 9 (nine) non-ISTS lines (**as listed in Annexure B.2.15.3**), which are carrying ISTS power for the tariff block 2024–29. These details are required to issue the Availability Certification for the relevant deemed ISTS/natural ISTS lines of OPTCL. All concerned SLDCs are requested to treat this matter as urgent and submit the required data to ERPC at the earliest to ensure timely issuance of the certifications.

WBSETCL & OPTCL may update. Members may discuss.

Deliberation in the meeting

➤ *ERPC Secretariat informed that WBSETCL and OPTCL are required to submit the verified tripping details of the aforementioned transmission lines, duly verified by their respective SLDCs, at the earliest and submission is necessary for the issuance of transmission availability certificates in compliance with relevant CERC orders.*

OCC Decision

OCC advised WBSETCL and OPTCL to expedite the submission of SLDC-verified tripping reports to ERPC Secretariat to avoid any delay in processing the transmission availability certification.

2.16 Methodology for computation of Average Monthly Frequency Response Performance, Beta 'β': ERLDC

NLDC vide letter dated 07.01.2025 and an addendum dated 10.01.2025 had forwarded stakeholders comments to CERC for amendment in CERC approved Beta 'β' methodology.

CERC vide letter to Grid-India dated 19.08.2025 has clarified that no changes are required in the already approved methodology for computation of Beta 'β'.

Proposed	Existing	CERC Response	Proposal by
In case, there was no reportable event during the billing month, Beta 'β' for that billing month shall be the median of last ten (10) reportable events.	Beta is zero if no reportable event in the billing month.	No Change	NTPC (also raised in ER OCC) & NHPC
In case of generators whose tariff is determined by CERC and are falling under the jurisdiction of SLDCs, concerned SLDC will certify the Beta 'β'	Beta is furnished by concerned LDCs and certified and issued by the RPC	No Change	Proposed by SRPC. ERPC agreed in-principle in 222nd OCC meeting.

Members may note.

OCC Decision

OCC noted.

2.17 Ensuring Data Availability and Accuracy in RTG Monitoring Portal: ERLDC

NLDC is assessing the quantum of secondary and tertiary reserves on year-ahead basis, to be maintained at the inter-state and intra-state levels. As per Grid Code, the states shall ensure the availability of reserves and if state falls short of maintaining reserve capacity as allocated to it, the NLDC through RLDC shall procure such Secondary reserve capacity on behalf of the State under advance intimation to the concerned state and allocate the cost of procurement of such capacity to that state based on the methodology as per the detailed procedure which is to be issued by the NLDC. Hence monitoring of reserve is of utmost importance from the present grid operation point of view. For this purpose, a Real-Time Generation (RTG) monitoring portal has already been developed to monitor the real-time data of Declared Capacity (DC), Schedule and actual generation (in MW) of all intra-state thermal plants. Thus, all states/thermal plants are requested to Ensure availability and accuracy of data (DC, Schedule, Margin) in the RTG portal for effective reserve monitoring. Members may discuss.

OCC Decision

OCC asked states to timely submit updated data in the prescribed format.

2.18 Submission of data for National Short-term Resource Adequacy Assessment FY 2026-27: ERLDC

As per Clause 3.2 of Section-3 of CEA Guidelines for Resource Adequacy, "NLDC shall annually publish a one-year look-ahead Short-term National Resource Adequacy Plan". Further as per the provisions of Regulation 5.3(d) of the Indian Electricity Grid Code (IEGC), the concerned agencies are required to furnish the requisite technical data to NLDC through respective RLDCs every year, for carrying out a national level generation resource adequacy assessment. Accordingly, NLDC has already published National Resource Adequacy Plan for 2024-25 and 2025-26. For preparation of National Resource Adequacy plan for 2026-27 the concerned utilities have been requested to submit the following data vide mail dated 09th September 2025:

1. Electricity Demand Data (as per format RA-1)
2. Reserve Requirement (as per format RA-2)
3. Tie Line Details (as per format RA-3)
4. Transfer Capability (as per RA-4)
5. Generation Data (as per RA-5)

However, ERLDC is yet to receive data from concerned agencies except Sikkim and Jharkhand. All concerned utilities are requested to submit the data at the earliest for timely completion of the National Resource adequacy assessment.

Members may discuss.

Deliberation in the meeting

ERLDC informed that requisite data is yet to receive data from concerned agencies except Sikkim and Jharkhand.

OCC Decision

OCC urged all concerned utilities to submit the data at the earliest.

2.19 ESP R&M activities at Kolaghat TPS Unit#6,5 & 4 M to comply with the environmental emission as on 01.01.2025 norms- WBPDCCL

Kolaghat TPS Unit#6,5 & 4 will be performing ESP R&M to comply with the environmental emission as on 01.01.2025 norms. The final shut down dates and the proposed generation in the intermediate period is given in the attached schedule.

The same S/d dates will be reflected in the LGBR for FY 2026-27 & FY 2027-28.

However, for FY 2025-26, the shutdown dates may please be noted for future reference.

UNIT NO.	ACTIVITIES	START DATE	END DATE	GENERATION (MW)	SENT OUT (MW)
UNIT#6	1ST S/D	05.12.2025	11.12.2025	0	0
	SINGLE PASS OPERATION	12.12.2025	15.04.2025	140	125
	2ND S/D	16.04.2026	22.04.2026	0	0

	SINGLE PASS OPERATION	23.04.2026	15.08.2026	140	125
	FINAL S/D	16.08.2026	15.09.2026	0	0
UNIT#4	1ST S/D	05.01.2026	11.01.2026	0	0
	SINGLE PASS OPERATION	12.01.2026	19.05.2026	130	115
	2ND S/D	20.05.2026	26.05.2026	0	0
	SINGLE PASS OPERATION	27.05.2026	03.10.2026	130	115
	FINAL S/D	01.12.2026	31.12.2026	0	0
UNIT#5	1ST S/D	22.08.2026	29.08.2026	0	0
	SINGLE PASS OPERATION	30.08.2026	28.12.2026	140	125
	2ND S/D	01.01.2027	08.01.2027	0	0
	SINGLE PASS OPERATION	09.01.2027	06.06.2027	140	125
	FINAL S/D	07.06.2027	07.07.2027	0	0

WBPDCCL may update. Members may discuss.

Deliberation in the meeting

WBSEDCL informed the forum that shutdown of Kolaghat TPS Unit#6,5 & 4 will be needed as per the attached schedule for performing ESP R&M to comply with the environmental emission norms.

OCC Decision

OCC advised WBSEDCL to avail the requisite shutdowns after consulting with WBSLDC.

2.20 Successful VAPT assessment completion of the AMR system – for kind information: PGCIL

On 222nd OCC meeting, the proposal of conducting a VAPT assessment on the AMR application was agreed. PGCIL initiated the Bid process and other subsequent activities for conducting the VAPT. On 11th Sep, the VPAT assessment was started, and it was carried out by a CERT-In certified vendor. Hardware and application both have been scanned under VAPT process.

VAPT was completed successfully, and the report is under final stage of preparation. There is no critical risk observed in the system. 04 number of High Vulnerability and 12 number of medium Vulnerability risks are identified.

The High Vulnerabilities are related to HTTP protocol being used and remote desktop port is opened. Since the AMR application is only accessible via LAN (not accessible over internet), using HTTP protocol will not create any risk. However, we are envisaging the process to implement token-based HTTPS protocol to eliminate this risk permanently.

The second point i.e. usage of RDP (Remote Desktop Protocol) which is currently being used only within the Servers which are connected over LAN and present at ERLDC. There is no RDP used over internet. In the present AMR architecture, there are multiple network interfaces being created where Servers are connected. Now, to connect from one Server to another from a different network zone to perform operations, RDP is getting used.

The medium category points are recommendation types on network related areas which are being analyzed by the team concerned.

As per CEA & CERT-In guidelines, conducting VAPT is a must requirement. We successfully carried out the same for the AMR system with zero critical issues and some recommendations. VPAT will be conducted every 06 months now onwards. (VAPT preliminary report is attached in **Annexure B.2.20**).

This is for kind information to the members.

PGCIL may update. Members may note.

Deliberation in the meeting

PGCIL informed that VAPT for the AMR application was successfully conducted by a CERT-In certified vendor with no critical risks identified and also updated that four high and twelve medium vulnerabilities were found, primarily related to HTTP usage and RDP within LAN, both being addressed.

➤ *He further appraised that VAPT will be conducted every six months as per CEA and CERT-In guidelines.*

OCC Decision

OCC noted the successful completion of the VAPT assessment for the AMR application by a CERT-In certified vendor, with no critical vulnerabilities observed.

➤ *OCC advised PGCIL to implement mitigation measures for the identified high and medium vulnerabilities, particularly transitioning to HTTPS protocol and reviewing RDP access within LAN.*

➤ *OCC opined that that VAPT shall be conducted every six months in compliance with CEA and CERT-In guidelines.*

2.21 Development of prototype of AMR SLDC data sharing application- WB utility for pilot project – for kind information: PGCIL

As approved in 54th ERPC meeting, sharing of SEM data over AMR system is getting developed. As decided, the first phase of development has been done for WB utility. Team PGCIL, WBSLDC & TCS have conducted multiple joint meetings, discussions and demonstrations on the application development.

The prototype has been developed with some dummy set of data and successfully demonstrated the same at WB/SLDC on 16.09.2025. It is expected that, post festive season after mutually agreed setup, the application for WB utility will be declared Go-Live. A presentation is attached here to get an overview of the system.

PGCIL may update. Members may discuss.

Deliberation in the meeting

PGCIL informed that as per the deliberation decided in the 54th ERPC meeting, development of SEM data sharing over the AMR system was initiated, with Phase-1 focused on implementation for the West Bengal (WB) utility.

➤ PGCIL appraised that multiple joint meetings have been held with WBSLDC, and TCS held for finalizing the application framework and a prototype using dummy SEM data was successfully demonstrated at WB SLDC on 16.09.2025.

➤ TCS gave a presentation (**Annexure B.2.21**) showcasing the functionality and architecture of the 1min AMR meter for real time seamless data communication.

OCC Decision

➤ OCC noted the progress on SEM data sharing over the AMR system as per the 54th ERPC meeting decision and appreciated the successful demonstration of the prototype for WB utility.

➤ OCC advised PGCIL, WBSLDC, and TCS to finalize the setup and proceed with Go-Live post the festive season through mutual coordination.

➤ OCC further advised PGCIL to share the presentation with all other Eastern Region utilities for familiarization with the 1-minute AMR prototype for real-time data communication, enhancing system operation.

2.22 Implementation of ADDCAP activities at Malda SS for complete protection upgradation along with replacement of old Isolators & Bus Conductors -PGCIL ER2

Under ADDCAP complete upgradation is approved by CERC. Under subject upgradation package, old & obsolete CRP by Universal Electric (UE) & English Electric (EE) under Chukha Transmission System (CTS) are envisaged for replacement. Initially CTS scheme was commissioned in 1984 and already elapsed 41 Years of operation. Also, centralized Bus Bar system will be replaced by distributed Bus Bar system in the package.

Along with old CRP, old Pantograph and DBR isolators are also envisaged for replacement for enhancing system reliability.

Under present configuration, 06 No's SPR (Switchyard panel room) and new control room building is envisaged and related civil works are almost on the verge of completion. For upgradation purposes, feeder wise S/D are planned and attached as separate presentation for further review.

Considering the festive ensuing season, initial S/D's are planned from last part of October-2025 and will continue till January-2026.

As Malda SS having DMT scheme for both 400 KV & 220 KV system, for bus bar commissioning complete bus outage are proposed only on two occasions in December & January.

The detail shutdown planning covering various Elements are attached in **Annexure-B.2.22**.

PGCIL may update. Members may discuss.

OCC Decision

OCC acknowledged the operational requirement by PowerGrid of the above shutdowns under ADCAP and advised PowerGrid to plan the shutdowns in coordination with WBSLDC and ERLDC taking real time grid conditions into consideration.

2.23 Temporary bypassing of Ranchi (POWERGRID) – Raghunathpur (DVC) Ckt-2&3 and Ranchi (POWERGRID) –Ranchi-New (POWERGRID) ckt-3 & 4 lines outside Ranchi S/s to facilitate shifting of Bus reactor and Bay equipments at Ranchi SS: PGCIL

➤ Vide Minutes of 17th CMETS-ER held on 29-03-2023:

- a. CTU informed that presently the fault level at 400kV bus at Ranchi (POWERGRID) 400/220kV S/s is observed to be about 44kA (designed limit is 40kA) which is expected to increase to about 51kA by 2027-28 timeframe.
 - b. Accordingly, it has been decided that “Ranchi (POWERGRID) –Raghunathpur TPS (DVC) 400kV D/c (Quad) line (of DVC) is to be bypassed from Ranchi (POWERGRID) with any two ckts of Ranchi (POWERGRID) –Ranchi New (POWERGRID) 400kV 2xD/c (Quad) lines (of POWERGRID) so as to form Ranchi New (POWERGRID) – Raghunathpur (DVC) 400kV (Quad)D/c line”.
 - c. It was noted that above bypassing arrangement reduces the fault level of Ranchi (POWERGRID) substation to about 39kA from 51kA in 2027-28time-frame. The arrangement will also give operational flexibility such that the lines can be terminated at Ranchi (POWERGRID) S/s under exigencies as per instructions of grid operators.
 - d. The ISTS licensee shall coordinate with DVC for shifting of Ranchi (POWERGRID) – Raghunathpur (DVC) 400kV D/c (Quad) line to bays 431 and 434. During the implementation period of the above scheme, ERS can be used for temporary bypassing of Ranchi (POWERGRID) – Raghunathpur (DVC) and Ranchi (POWERGRID) –Ranchi-New (POWERGRID) ckt-3 & 4 lines outside Ranchi S/s so as to facilitate shifting of bus reactor and bay equipment, if required.
- Civil work under future bay area under Substation Package SS-121 for (a) Extension of 400 KV Ranchi (PG) substation including transmission line works and (b) Extension of 400 KV Ranchi (New) substation under Eastern region Expansion Schedule XXXVIII (ERES-XXXVIII) Project has been completed and long shutdown of 400kV Ranchi (POWERGRID) – Raghunathpur (DVC) 400kV D/c (Quad) line required for shifting of bay equipment.
- Accordingly, for optimization of outage for 400kV Ranchi-Raghunathpur D/C Line, ERS expert committee members of POWERGRID visited the site and recommended that:

a. 400 kV Raghunathpur (DVC) Circuit-2 (between Tower Location-432 and 433) will be connected to New Ranchi (POWERGRID) Circuit-3 (between TowerLocation-10 and 11) through 01 ERS tower with twin moose conductor.

b. 400 kV Raghunathpur (DVC) Circuit-3 (between Tower Location-432 and 433) will be connected to New Ranchi (POWERGRID) Circuit-4 (between TowerLocation-08 and 09) through 01 ERS tower with twin moose conductor.

ERS arrangement shown as under:

➤ Technical Parameters of the present Line are as under:

Line Length:

a. 400kV Ranchi-Raghunathpur-2&3= 155 KM

b. 400kV Ranchi-New Ranchi-3&4= 78.6 KM

PLCC Details at Ranchi End:

1. 400kV Ranchi-Raghunathpur-2: Ch-1 BPL-V 95 (420/380kHz),
Ch-2 BPL-V 95 (432/396kHz)

2. 400kV Ranchi-Raghunathpur-3: Ch-1 ABB- ETL41-NSD50 (156/300kHz)
Ch-2 ABB- ETL41-NSD50 (204/312kHz)

3. 400kV Ranchi-New Ranchi-3: Ch-1 ABB- ETL41-NSD50 (312/308kHz)
Ch-2 ABB- ETL41-NSD50 (444/440kHz)

4. 400kV Ranchi-New Ranchi-4: Ch-1 ABB- ETL41-NSD50 (160/140kHz)
Ch-2 ABB- ETL41-NSD50 (432/428kHz)

Wave Trap Configuration:

1. 400kV Ranchi-Raghunathpur-2: R and B-phase

2. 400kV Ranchi-Raghunathpur-3: Y and B-phase

3. 400kV Ranchi-New Ranchi-3&4: R and Y-phase

➤ Before implementation of the ERS scheme, following shall be ensured:

a. Revised relay setting adoption for New Ranchi and Raghunathpur(DVC) end according to actual Line length (Line length will be 232km approx.).

b. Diversion of 02 no's ABB Make ETL41-NSD50 PLCC panels (Available at POWERGRID Ranchi SS spare stock) with protection coupler to DVC Raghunathpur end for the 400kV Raghunathpur (DVC) Circuit-2 -New Ranchi (POWERGRID) Circuit-3 Line since the existing PLCC at Raghunathpur end is BPL make, and New Ranchi end is ABB make.

c. Finalization of PLCC frequencies.

d. Shifting of Wave Trap at New Ranchi end according to Raghunathpur end in both circuits.

ERS installation work is planned to be start from **06-10-2025** and final charging through ERS is expected by **13.10.2025**.

Members may note.

OCC Decision

OCC noted.

3. PART-C: ITEMS FOR UPDATE/FOLLOW-UP/INFORMATION

3.1. ER Grid performance during August 2025

The average and maximum consumption of Eastern Region and Max/Min Demand (MW), Energy Export for the month August-2025 were as follows:

AVERAGE CONSUMPTION (MU)	MAXIMUM CONSUMPTION(MU)/ DATE	MAXIMUM DEMAND (MW)	MINIMUM DEMAND (MW)	SCHEDULE EXPORT	ACTUAL EXPORT
		DATE / TIME	DATE / TIME	(MU)	(MU)
617 MU	673 MU, 12.08.2025	31331 MW, 16.08.2025 at 23:01 Hrs.	21717 MW, 24.08.2025 at 07:32 Hrs.	2247	2356

ERLDC/ERPC may highlight the performance of the ER grid.

OCC Decision

OCC noted ER grid performance during August 2025.

3.2. **Non-Submission of FRC data in stipulated time-frame: ERLDC**

Adhering to IEGC clauses 30.8 and 30.10.(a) to 30.10.(q), generating stations within the Eastern region are required to submit essential data to ERLDC within two days of receiving a notification regarding a reportable frequency event. Additionally, according to clause 30.10.(n), all control areas within the eastern region must assess their frequency response characteristics and share the evaluation, along with high-resolution data, with the ERLDC. Therefore, timely submission of primary response data is crucial for compliance with the IEGC.

In 230th OCC Decision: -

All generators were advised to regularly share high resolution data against each reportable frequency event with ERLDC on time to facilitate accurate assessment of FRP for respective control areas.

All generating utilities were also urged to update the google sheet (link mentioned above) with email address where notifications of reportable events will be shared.

Hence all are again requested to follow the stipulated timeline and submit the data to ERLDC and also fill the google sheet below to include the email address where notifications of reportable events should be sent.

The latest data receipt status is given below: (as on 11.09.2025):

STATIONS		11-05-2025 16:51 HRS	12-06-2025 13:34 HRS	16-06-2025 11:51 HRS	22-07-2025 19:46 HRS	29-07-2025 14:55 HRS
FSTPP #STG 1 & 2	ISGS					
FSTPP # STG 3	ISGS					
KhSTPP #STG 1	ISGS					
KhSTPP #STG 2	ISGS					
TSTPP #STG 1	ISGS					
Barh stage-1	ISGS					
Barh stage-2	ISGS					
BRBCL	ISGS					
Darlipalli	ISGS					
North Karanpura	ISGS					
NPGC	ISGS					
TEESTA V	ISGS					
Dikchu						
IBEUL (JSW UTKAL)/INDBHARAT	IPP					
GMR	CPP					
MPL	CPP					
ADHUNIK	CPP					
JITPL	CPP					
TEESTA III	CPP					
Bihar	STATE					
Jharkhand	STATE					
DVC	STATE					
OPTCL	STATE					
WB	STATE					
Updated as on	11.09.2025					
	Received					
	Not Received					
	Plant Out					
	Data freeze at plant					

Hence all are again requested to follow the stipulated timeline and submit the data to ERLDC and also fill in the google sheet below to include the email address where notifications of reportable events should be sent.

https://docs.google.com/spreadsheets/d/1slvAOmQIEQVIMn0LnB78eKMa2sz2QYICZ-sPEpeV_jk/edit?usp=sharing

ERLDC may explain and all SLDCs may update. Members may discuss.

OCC Decision

- All generators were advised to regularly share high resolution data against each reportable frequency event with ERLDC on time to facilitate accurate assessment of FRP for respective control areas.
- All generating utilities were also urged to update the google sheet (link mentioned above) with email address where notifications of reportable events will be shared.

3.3. Regarding Non-Submission of Forecasting Data from States: ERLDC

Clause 2 of Regulation 31 of IEGC 2023 has mandated all the SLDCs to timely submit the demand estimate data to the respective RLDC and RPC.

229th OCC Decision

- Latest Forecast and Resource Adequacy Data receipt status at ERLDC is shown below:

Latest Forecast and Resource Adequacy Data receipt status at EREDC is shown below:																																
Bihar																																
Jharkhand																																
DVC																																
Odisha																																
West Bengal																																
Sikkim																																
Date	31-08-25	30-08-25	29-08-25	28-08-25	27-08-25	26-08-25	25-08-25	24-08-25	23-08-25	22-08-25	21-08-25	20-08-25	19-08-25	18-08-25	17-08-25	16-08-25	15-08-25	14-08-25	13-08-25	12-08-25	11-08-25	10-08-25	09-08-25	08-08-25	07-08-25	06-08-25	05-08-25	04-08-25	03-08-25	02-08-25	01-08-25	
Status of Furnishing of Day Ahead Forecast data by ER States as on 11.09.25																					Received	Not Received	Delayed									

[illegible]

Status of Furnishing of Day Ahead Forecast data by ER States on 23/08/2025						Status of Furnishing of Day Ahead Forecast data by ER States on 23/08/2025				
	Status of Furnishing of Week Ahead Forecast data by ER States					Status of Furnishing of Week Ahead Resource Adequacy data by ER States				
Bihar										
Jharkhand										
DVC										
Odisha										
West Bengal										
Sikkim										
Date	25-08-25 to 31-08-25	18-08-25 to 24-08-25	11-08-25 to 17-08-25	04-08-25 to 10-08-25	2025-07-28 to 2025-08-03	25-08-25 to 31-08-25	18-08-25 to 24-08-25	11-08-25 to 17-08-25	04-08-25 to 10-08-25	2025-07-28 to 2025-08-03
				RECEIVED		NOT RECEIVED			DELAYED RECEIVED	
	Status of Furnishing of Month Ahead Forecast data by ER States					Status of Furnishing of Month Ahead Resource Adequacy data by ER States				
Bihar										
Jharkhand										
DVC										
Odisha										
West Bengal										
Sikkim										
Date	01-08-2025 to 31-08-2025 (August 2025)					01-08-2025 to 31-08-2025 (August 2025)				
				RECEIVED		NOT RECEIVED			DELAYED RECEIVED	

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OCC Decision:

- *OCC advised all SLDCs for strictly adhering to the schedule of demand estimation as mandated in IEGC 2023, timely sharing with ERLDC in specified format as well as uploading of forecasting error on their respective websites.*
- *SLDCs who are submitting day ahead forecast were advised to also share the forecasting data for their respective control areas on weekly as well as monthly basis with ERLDC.*
- *All SLDCs were urged to regularly furnish resource adequacy data besides demand forecast.*

4. PART-D: OPERATIONAL PLANNING

4.1. Anticipated power supply position for October -2025

The abstract of peak demand (MW) vis-à-vis availability and energy requirement vis-à-vis availability (MU) for the month of October-2025 is prepared by ERPC Secretariat (**Annexure D.1**) on the basis of LGBR for 2025-26 and feedback of constituents, keeping in view that the units are available for generation and expected load growth etc.

Members may update.

Deliberation in the meeting

All states were requested to provide their anticipated power supply position for October 2025 within a week. Updated anticipated power supply position will be shared accordingly.

4.2. Major Thermal Generating Units/Transmission Element outages/shutdown in ER Grid (as on 14-09-2025)

a) **Thermal Generating Stations outage report:**

SL No	STATION	STATE	AGENCY	UNIT NO	CAPA CITY (MW)	REASON(S)	OUTAGE DATE
1	KHSTPP	BIHAR	NTPC	2	210	Annual overhauling	28-Aug-2025
2	NABINA GAR(BR BCL)	BIHAR	NTPC	4	250	Annual Overhauling	25-Aug-2025
3	SOUTHERN	WEST BENGAL	CESC	2	67.5	For carrying out planned maintenance jobs related to unit reliability	30-Aug-2025
4	OPGC3	ODISHA	OPGC	3	660	Annual Overhauling	24-Aug-2025
5	BAKRES HWAR	WEST BENGAL	WBPDCL	1	210	Boiler Overhauling activities	24-Aug-2025
6	KBUNL	BIHAR	NTPC, BSPHCL	2	195	TG shaft high vibration	12-Sep-2025
7	BARH	BIHAR	NTPC	1	660	Abnormal sound from boiler.	12-Sep-2025
8	JSWEUL	ODISHA	JSWEUL	2	350	To attend issue	08-Sep-2025

						observed in the turbine lube oil system.	
9	OPGC	ODISHA	OPGC	4	660	Tripped on boiler tube leakage	12-Sep-2025
10	CHANDR APURA TPS	DVC	DVC	7	250	Boiler Tube Leakage	11-Sep-2025
11	RTPS	DVC	DVC	1	600	Initially tripped due to boiler tube leakage. Later found crack and oil leak in a turbine main oil tank.	02-Sep-2025

All Generating stations are requested to update expected restoration time and reason outage to ERLDC/ERPC on weekly basis in case of any change at their end.

b) Major Generating stations Out on Reserve Shutdown due to low system demand:

SL No	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
NIL							

c) Hydro Unit Outage Report:

S. NO	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	TEESTA STG III Hep	SIKKIM	TUL	1	200	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in Teesta River and damage of Teesta III Dam & downstream Powerhouses	04-Oct-2023
2	TEESTA STG III Hep	SIKKIM	TUL	2	200		
3	TEESTA STG III Hep	SIKKIM	TUL	3	200		
4	TEESTA STG III Hep	SIKKIM	TUL	4	200		
5	TEESTA STG III Hep	SIKKIM	TUL	5	200		

6	TEESTA STG III Hep	SIKKIM	TUL	6	200		
7	TEESTA HPS	SIKKIM	NHPC	1	170	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in Teesta River and damage of Teesta III Dam & downstream Powerhouses	04-Oct-2023
8	TEESTA HPS	SIKKIM	NHPC	2	170		
9	TEESTA HPS	SIKKIM	NHPC	3	170		
10	RANGIT HPS	SIKKIM	NHPC	1	20	Due to high inflow and high PPM	14-Sep-2025
11	RANGIT HPS	SIKKIM	NHPC	2	20	Due to high inflow and high PPM	14-Sep-2025
12	RANGIT HPS	SIKKIM	NHPC	3	20	Due to high inflow and high PPM	14-Sep-2025
13	BALIMELA HPS	ODISHA	OHPC	5	60	Repair and maintenance work	16-Jan-2025
14	BALIMELA HPS	ODISHA	OHPC	6	60	Initially unit was out due to Severe water leakage from turbine, later unit was taken under Repair and maintenance work from 00:00 hrs of 16.01.25	06-Jan-2025
15	CHIPLIMA HPS / HIRAKUD II	ODISHA	OHPC	1	24	Capital Overhauling	15-Dec-2023

d) Long outage report of transmission Element (MORE THAN 01 WEEK) (As on 14.08.2025):

Transmission Element / ICT	Outage From	Reasons for Outage
220/132 KV 100 MVA ICT II AT LALMATIA	22-01-2019	220/132KV, 100MVA Transformer (NTPC side) is charged on 07.02.2024 from HV side on no load. Now, it is in idle charged condition
220KV-FSTPP-LALMATIA-I	21-04-2021	Two nos. of tower collapsed on 29.05.2024 near to Lalmatia GSS in the Loc. No. 246 & 247. Presently 220 kV Farakka-Lalmatia line is charged (from loc no 241 to loc 84) at 132 kV voltage level for anti-theft purpose by tapping at loc. No. 100-101.
220KV-WARIA-BIDHANNAGAR-1	08-06-2022	To control overloading of 220 kV Waria-DSTPS (Andal) D/C line

220KV-WARIA-BIDHANNAGAR-2	08-06-2022	To control overloading of 220 kV Waria-DSTPS (Andal) D/C line
132KV-BARHI-RAJGIR-1	25-03-2023	Dismantling of tower no. 227, 228, and 229 crossing the premises of Mahabodhi Cultural centre along with Destraining of conductor of both circuits and Earthwire between tension tower no. 218-237 in same line.
132KV-NALANDA-BARHI(DVC)-1	25-03-2023	Dismantling of tower no. 227, 228, and 229 crossing the premises of Mahabodhi Cultural centre along with Destraining of conductor of both circuits and Earthwire between tension tower no. 218-237 in same line.
400KV-RANGPO-TEESTA-V-1	04-10-2023	Tower near gantry of Teesta V powerhouse collapsed due to sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in TEESTA river and damage of Teesta III Dam & downstream Powerhouses
400KV-RANGPO-TEESTA-V-2	04-10-2023	Tower near gantry of Teesta V powerhouse collapsed due to sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in TEESTA river and damage of Teesta III Dam & downstream Powerhouses
132KV-RANGPO-SAMARDONG-1	22-05-2024	Rangpo:Y_N fault with fault distance 0.157 KM ,14.562kA Samardong: NA
132KV-CHANDIL-MANIQUEI-1	05-06-2024	Power assistance withdrawn
400KV/220KV 315 MVA ICT 1 AT NORTH KARANPURA	12-09-2024	Tripped on Differential protection
132KV-MADHEPURA (BH)-SAHARSA(PMTL)-1	23-09-2024	To control loading on 132kV Madhepura-Saharsa line
400KV/220KV 315 MVA ICT 1 AT TSTPP	01-11-2024	Tripped on PRD protection
132KV-PATRATU-PATRATU-1	16-11-2024	Diversion/Heightening of line due to inadequate clearance from under construction railway Line by PVUNL
132KV-PATRATU-PATRATU-2	16-11-2024	Diversion/Heightening of line due to inadequate clearance from under construction railway Line by PVUNL
400KV/220KV 315 MVA ICT 2 AT MEJIA-B	20-01-2025	Tripped during charging of ICT#1 bay with cable from 220 kv GIS side
400KV/220KV 315 MVA ICT 1 AT LATEHAR	22-04-2025	R phase LA of 400/220/33 KV ICT - I got bursted

400KV/220KV 315 MVA ICT 2 AT LATEHAR	16-04-2025	Transformer REF protection operated
400KV/220KV 315 MVA ICT 2 AT KODERMA	02-06-2025	Transformer Differential Protection operated
132KV-RAXAUL(NEW)-PARWANIPUR-2	03-07-2025	To carry out Gantry erection works at near by Parsauni 132/66/33 kV Substation of Nepal
132KV-RAXAUL(NEW)-PARWANIPUR-1	03-07-2025	To carry out Gantry erection works at near by Parsauni 132/66/33 kV Substation of Nepal
220KV-RAJARHAT-NEW TOWN(AA-II)-1	10-07-2025	Emergency shutdown for BCU replacement work at Rajarhat. Charging attempted but tripped on SOTF. B_ph cable faulty
220KV-PATNA-KHAGAUL-1	02-08-2025	Tower No. 63 has bent significantly on one side
400KV/220KV 315 MVA ICT 2 AT KEONJHOR(PG)	05-08-2025	Buchholz relay operated
400KV MAIN BUS - 2 AT DIKCHU	05-08-2025	Bus bar protection operated
400KV-DIKCHU-RANGPO-2	05-08-2025	Damaged insulator replacement work. While charging the line bus bar protection operated at Dikchu
400KV-ALIPURDUAR (PG)-PUNASANGCHUN-1	09-06-2025	CONNECTION OF ALL PHASE JUMPERS AT LOC. 177
400KV-PVUNL-PATRATU-2	19-08-2025	Patratu end:General trip,DT received & overvoltage protection.Repair work going on isolator chamber,expected by 15.9.2025
400KV-ARAMBAGH-NEW PPSP-2	04-01-2025	Damage in GIS chamber at new ppsp ss for Arambag ckt.Expected in January 2026
220/132 KV 100 MVA ICT II AT LALMATIA	22-01-2019	220/132KV, 100MVA Transformer (NTPC side) is charged on 07.02.2024 from HV side on no load. Now, it is in idle charged condition
220KV-FSTPP-LALMATIA-I	21-04-2021	Two nos. of tower collapsed on 29.05.2024 near to Lalmatia GSS in the Loc. No. 246 & 247. Presently 220 kV Farakka-Lalmatia line is charged(from loc no 241 to loc 84) at 132 kV voltage level for anti-theft purpose by tapping at loc. No. 100-101.
220KV-WARIA-BIDHANNAGAR-1	08-06-2022	To control overloading of 220 kV Waria-DSTPS (Andal) D/C line
220KV-WARIA-BIDHANNAGAR-2	08-06-2022	To control overloading of 220 kV Waria-DSTPS (Andal) D/C line

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132KV-NALANDA-BARHI(DVC)-1	25-03-2023	Dismantling of tower no. 227, 228, and 229 crossing the premises of Mahabodhi Cultural centre along with Destraining of conductor of both circuits and Earthwire between tension tower no. 218-237 in same line.
400KV-RANGPO-TEESTA-V-1	04-10-2023	Tower near gantry of Teesta V powerhouse collapsed due to sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in TEESTA river and damage of Teesta III Dam & downstream Powerhouses
400KV-RANGPO-TEESTA-V-2	04-10-2023	Tower near gantry of Teesta V powerhouse collapsed due to sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in TEESTA river and damage of Teesta III Dam & downstream Powerhouses
132KV-RANGPO-SAMARDONG-1	22-05-2024	Rangpo:Y_N fault with fault distance 0.157 KM ,14.562kA Samardong: NA
132KV-CHANDIL-MANIQUEI-1	05-06-2024	Power assistance withdrawn
400KV/220KV 315 MVA ICT 1 AT NORTH KARANPURA	12-09-2024	Tripped on Differential protection
132KV-MADHEPURA (BH)-SAHARSA(PMTL)-1	23-09-2024	To control loading on 132kV Madhepura-Saharsa line
400KV/220KV 315 MVA ICT 1 AT TSTPP	01-11-2024	Tripped on PRD protection
132KV-PATRATU-PATRATU-1	16-11-2024	Diversion/Heightening of line due to inadequate clearance from under construction railway Line by PVUNL
132KV-PATRATU-PATRATU-2	16-11-2024	Diversion/Heightening of line due to inadequate clearance from under construction railway Line by PVUNL
400KV/220KV 315 MVA ICT 2 AT MEJIA-B	20-01-2025	Tripped during charging of ICT#1 bay with cable from 220 kv GIS side
400KV/220KV 315 MVA ICT 1 AT LATEHAR	22-04-2025	R phase LA of 400/220/33 KV ICT - I got bursted
400KV/220KV 315 MVA ICT 2 AT LATEHAR	16-04-2025	Transformer REF protection operated

400KV/220KV 315 MVA ICT 2 AT KODERMA	02-06-2025	Transformer Differential Protection operated
132KV-RAXAUL(NEW)-PARWANIPUR-2	03-07-2025	To carry out Gantry erection works at near by Parsauni 132/66/33 kV Substation of Nepal
132KV-RAXAUL(NEW)-PARWANIPUR-1	03-07-2025	To carry out Gantry erection works at near by Parsauni 132/66/33 kV Substation of Nepal
220KV-RAJARHAT-NEW TOWN(AA-II)-1	10-07-2025	Emergency shutdown for BCU replacement work at Rajarhat. Charging attempted but tripped on SOTF. B_ph cable faulty
220KV-PATNA-KHAGAUL-1	02-08-2025	Tower No. 63 has bent significantly on one side
400KV/220KV 315 MVA ICT 2 AT KEONJHOR(PG)	05-08-2025	Buchholz relay operated
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400KV-DIKCHU-RANGPO-2	05-08-2025	Damaged insulator replacement work. While charging the line bus bar protection operated at Dikchu
400KV-ALIPURDUAR (PG)-PUNASANGCHUN-1	09-06-2025	CONNECTION OF ALL PHASE JUMPERS AT LOC. 177
400KV-PVUNL-PATRATU-2	19-08-2025	Patratu end:General trip,DT received & overvoltage protection.Repair work going on isolator chamber,expected by 15.9.2025
400KV-ARAMBAGH-NEW PPSP-2	04-01-2025	Damage in GIS chamber at new ppsp ss for Arambag ckt.Expected in January 2026

Transmission licensees/ Utilities are requested to update expected restoration date & work progress regarding restoration regularly to ERPC/ERLDC on monthly basis by 5th of each month so that status of restoration can be reviewed in OCC. Utilities are also requested to update outage of any elements within their substation premises like isolator/breaker to ERPC/ERLDC regularly. (Reported as per Clause 5.2(e) of IEGC).

Deliberation in the meeting

Members noted.

4.3.Commissioning of new units and transmission elements in Eastern Grid in the month of July-2025.

The details of new units/transmission elements commissioned in the month of May-2025 based on the inputs received from beneficiaries:

NEW ELEMENTS COMMISSIONED DURING July, 2025							
□□□□□□□ □□□□□□□ / GENERATING UNITS							
S I. N o.	□□□□□ Location / Pooling Station	□□□□□/ □□□□□ □□ □□□ OWNER/UN IT NAME	□□ □□□ □□ □□ □□/ □□ □□ □□□ Unit No/S ourc e	□□□□□ □ □□□□ □□ (□□□□ □□□) Capacity added (MW)	□□□/□□ □□□□□ □□□□□□ (□□□□□□ □) Total/Insta lled Capacity (MW)	□□ □□□□ DATE	□□□□□□□ Remarks □□□
1	Chausa, Buxar, Bihar	Buxar TPP - UNIT 1	1/Coa l	660	660	21-08- 2025 20:55	
अप.□□.□□/□□.□□/□□.□□ / ICTs/ GTs / STs							
S I. N o.	□□□□□□/□□□□□ Agency/O wner	□□- □□□□□□□ SUB- STATION	□□□□ □□ □□□□ □□ ICT NO	□□□□ □□□ (□□ □□) Votlag e Level (kV)	□□□□□□ (□□□□□) CAPACIT Y (MVA)	□□ □□□□ DATE	□□□□□□□ Remarks
1	OPTCL	NEW DUBURI	400KV/2 20KV 500 MVA ICT 3 AT NEW DUBUR I	400/22 0	500	29-08- 2025 21:53	First time loaded
□□□□□□ □□□□ / TRANSMISSION LINES							
S I.	□□□□□□/□□□□□ Agency/O wner	□□□□ □□ □□□ LINE NAME	□□□□□ (□□□□) Length (KM)	□□□□□□ □ □□□□□□ Conducto r Type	□□ □□□□ DATE	□□□□□□□ Remarks	

N o .						
1	JUSNL	400KV-PVUNL- PATRATU-2	6.6	ACSR Moose	03-08- 2025 16:31	
2	JUSNL	400KV-PVUNL- PATRATU-1	6.6	ACSR Moose	03-08- 2025 16:40	
□□□□ / □□□□□□ □□□□ □□ □□□□□□□□□□□□ / LILO/RE-ARRANGEMENT OF TRANSMISSION LINES						
□□ □ S I. N o .	□□□□□□/□□□□□ Agency/O wner	□□□□ □□ □□□□ / □□□□ □□ Line Name/LILO at	□□□□□□ (□□□□) Length (KM)	□□□□□□ □ □□□□□□ Conductor Type	□□ □□□□ DATE	□□□□□□□ Remarks
1	PGCIL ER_II	132 KV Siliguri-Melli TL	92.346		08-08- 2025 16:47	Back to original Configuration
2	PGCIL ER_II	132 KV Rangit- Kurseung TL	61.205		08-08- 2025 16:50	Back to original Configuration
3	BSPTCL	132KV- CHANDAUTI (PMTL)- Daudnagar(New)-1	LILO length- 24.2 kms/total length- 106 kms	HTLS equivalent to panther	24-08- 2025 19:57	Idle charged from Chandauti(P MTL) end).
□□/□□□□ □□□□□□□□□□ / BUS/LINE REACTOR						
□□ □ S I. N o .	□□□□□□/□□□□□ Agency/O wner	□□□□□□□□ □□ □□□ Element Name	□□- □□□□ □□□ SUB- STATION	□□□□□□ □ (□□□□) Voltage Level (kV)	□□ □□□□ DATE	टिप्पणी Remarks
1	PGCIL ER- I	63MVAR SWITCHABLE L/R OF 400KV- DURGAPUR- KHSTPP-1 AT KHSTPP	400	63	29-08- 2025 23:59	

2	PGCIL ER-I	63MVAR SWITCHABLE L/R OF 400KV- DURGAPUR- KHSTPP-2 AT KHSTPP	400	63	18-08- 2025 17:57	
3	SJVN	125MVAR 400KV B/R-1 AT BUXAR TPP	400	125	11-08- 2025 16:12	
N						
□□ / BUS						
□□ □ . S I. N o .	□□□□□□/ □□□□□ Agency/O wner	□□□□□□□ □□ □□□ Element Name	□□- □□□□ □□□ SUB- STATION	□□□□□□ □ (□□□□) Voltage Level (kV)	□□ □□□□ DATE	□□□□□□□ Remarks
NIL						
□□.□□.□□.□□/□.□□ □□□□□□ □□□□/□□□□□□□ □□□□□□ □□□□□□□ / HVDC /AC Filter bank / FACTS DEVICE associated System						
□□ □ . S I. N o .	□□□□□□/ □□□□□ Agency/O wner	□□□□□□□ □□ □□□ Element Name	□□- □□□□ □□□ SUB- STATION	□□□□□□ □ (□□□□) Voltage Level (kV)	□□ □□□□ DATE	□□□□□□□ Remarks
NIL						
□□ / BAYS						
□□ □ . S I. N o .	□□□□□□/ □□□□□ Agency/O wner	□□□□□□□ □□ □□□ Element Name	□□- □□□□ □□□ SUB- STATION	□□□□□□ □ (□□□□) Voltage Level (kV)	□□ □□□□ DATE	□□□□□□□ Remarks
1	SJVN	400KV MAIN BAY OF GT-1 AT BUXAR TPP	BUXAR TPP	400	21-08- 2025 20:55	

2	WBSETCL	400KV MAIN BAY OF TURGA-2 AT NEW PPSP	NEW PPSP	400	19-08-2025 20:16	
3	WBSETCL	400KV MAIN BAY OF TURGA-1 AT NEW PPSP	NEW PPSP	400	19-08-2025 20:11	
4	WBSETCL	400KV TIE BAY OF (TURGA-1 AND TURGA-2) AT NEW PPSP	NEW PPSP	400	19-08-2025 17:48	
5	SJVN	400KV TIE BAY OF (500 MVA ICT 1 AND 125MVAR B/R-1) AT BUXAR TPP	BUXAR TPP	400	07-08-2025 13:55	
6	SJVN	400KV MAIN BAY OF 125MVAR B/R-1 AT BUXAR TPP	BUXAR TPP	400	07-08-2025 13:54	
7	PVUNL	400KV MAIN BAY OF PATRATU(JUSNL)-2 AT PVUNL	PVUNL	400	04-08-2025 23:03	
8	OPTCL	400KV MAIN BAY OF 500 MVA ICT 3 AT NEW DUBURI	NEW DUBURI	400	29-08-2025 21:53	
9	OPTCL	220KV MAIN BAY OF 500 MVA ICT 3 AT NEW DUBURI	NEW DUBURI	220	28-08-2025 10:16	

Members may note.

Deliberation in the meeting

Members noted.

4.4. UFR operation during the month of August 2025

Frequency profile for the month as follows:

MONTH	MAX	MIN	% LESS IEGC BAND	% WITHIN IEGC BAND	% MORE IEGC BAND
	(DATE/TIME)	(DATE/TIME)			
August, 2025	50.35 Hz on 28-08-2025 at 13:01 Hrs	49.46 Hz on 21-08-2025 at 19:14 Hrs	6.63	76.22	17.15

Hence, no report of operation of UFR has been received from any of the constituents.

Members may note.

Deliberation in the meeting

Members noted.

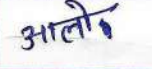
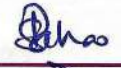
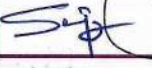
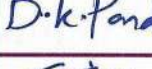
Annex-A

Participants in 231st OCC Meeting of ERPC

Venue: ERPC Secretariat, Kolkata

Time: 10:30 hrs.

Date: 22.09.2025 (Monday)

Sl No	Name	Designation/ Organization	Contact Number	Email	Signature
1	N.S Mondol	Member Secretary ERPC		mserpc-power@gov.in	
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4	Manjunatham	EE, ERPC	9972023842	manjunatham.cca @gov.in	
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9	Shrimalya Ghosal	A.M., ERLDC	9038711099	sghosal@grid-india.in	
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11	Partha Ghosh	DGM/AM	9434748263	partha.sharma@powergrid.in	
12	Padmalochan Sahoo	GM, GRIDCO	9438907671	ele. plsahoo@gridco.co.in	
13	Bibhuprasad Mohapatra	GM, GRIDCO	9438907671	ele. bprasad@gridco.co.in	
14	Ajaya Kumar Sahu	DGM, POWERGRID, Odisha Project	9437962180	ajaya.sahu@powergrid.in	
15	Gagan Kumar	ETE, SUDC BSPTCL	7992486100	gagankumishra@gmail.com	
16	S.K. Sharma	SGM, DVCLDC	9434539423	SanjaySharma@dlk.sv.in	
17	P.V. RAUT.	MPL.	9223501513	rautpv@tatapower.com	
18	Amit Kumar Gupta	JIPL	9681335196	JIPLSchedule@Jindal group.com	
19	Diptikanta Panda	GMR	8114918762	Diptikanta.Panda@ gmrgroup.in	
20	Shiladitya Chatterjee	APNRL	9748357917	Shiladitya.Chatterjee@ adshunik@nrc.co.in	

"Coming together is a beginning, staying together is progress, and working together is success." –Henry Ford

Participants in 231st OCC Meeting of ERPC

Venue: ERPC Secretariat, Kolkata

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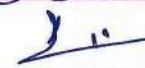
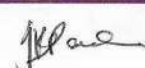
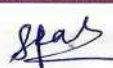
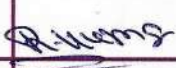
Sl No	Name	Designation/ Organization	Contact Number	Email	Signature
21	MANASH PROTIM NATH	DGM/ERLDC	7005694122	mpnath@gridindia.in	
22	SOURAV MANDAL	CM/ERLDC	9402102354	souravmandal@grid-india.in	
23	DEBASUDE	DGM/CESC	9230521123	debanshi.de@rpsc.in	
24	Sumet Narang	SENGR/NTPC	8005493953	sumetnarang@ntpc.co.in	
25	Ludipray BARMAN	AGM/NTPC	9424140773	shaybarmann@ntpc.co.in	
26	MANOJ PODDER	AGM-WBPDCL	8336901077	mpodder@wbpdcl.co.in	
27	RASAT KUMAR KOLEY	Sr. Mg ² (OS) WBPDCL	9474860642	rk.koley@wbpdcl.co.in	
28	SAJAL K. BAG	ACE, SLDC, WB	9434910265	sajalkbag74@gmail.com	
29	RITA CHAKRABORTY	CE, SLDC, WB	9434910141	ce.wb@slde@gmail.com	
30	SHOUYIK BANERJEE	ACE/SLDC/WB	9434910379	svkbanerjee@yahoo.com	
31	R-DAS	CE, CPD	9434910740	cpd.wbsetel@gmail.com	
32	J. Sengupta	ACE, CPD	9434910547	cpd.wbsetel@gmail.com	
33	Preetam Banerji	ACE, WBSEDL	7003871189	preetban72@gmail.com preetam.banerji@wbsecl.in	
34	Ch. Eswara Rao	GM, Powergrid Odisha	9437462478	eswar@powergrid.in	
35	Sudeep Kumar	DGM, POWERGRID Patna	9431820338	sudeepkumar@powergrid.in	
36	Manas Das	DGM, ERLDC GRID-INDIA	7007070925	manasdas@grid-india.in	
37	Bilash Achari	DGM-ERLDC GRID-INDIA	7003472016	bilash.achari@grid-india.in	
38	Rakesh K. Pradhan	CM/ERLDC GRID-INDIA	9831337570	rkpradhan@grid-india.in	
39	Suranjana Das	NTPC-TSTPS	9434041089	suranjandas@ntpc.co.in	
40	Abhishek Murai	NTPC - Barh	9470894493	abhishekmurai@ntpc.co.in	

Participants in 231st OCC Meeting of ERPC

Venue: ERPC Secretariat, Kolkata

Time: 10:30 hrs.

Date: 22.09.2025 (Monday)

Sl No	Name	Designation/ Organization	Contact Number	Email	Signature
41	Suresh Babu-K	AGM NTPC Dastipali	9493546143	ksureshbabu@ntpc.co.in	
42	D. BISWAS	GM (SO) ERLDC	9434740041	dbiswas@grid-india.in	
43	R. SUTRADHAN	ED ERLDC	9436302714	rajibsubradhan@grid-india.in	
44	S. C. De	CGM, ERLDC	9436335369	scode@grid-india.in	
45	B. KUMAR	EEE	6299998789	BK202vnu@gmail.com BK202vnu@gmail.com	
46	I. Mondal	AE, CPD, WBSCTCL	9434910924	cpd.wbsctcl@gmail.com	
47	Shabari Pramanick	CM, ERLDC	9007058764	shabari.pramanick@grid-india.in	
48	Sourabh Agarwal	DM, ERLDC	9939042657	sourabh.agarwal@grid-india.in	
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50	S. K. Panda	SISC, DVC	6370134294	skpanda@dvc.gov.in	
51	Manoj Tannir	Adani Power	9099005872	manoj.tannir@adani.com	
52	SHIVAJI DEWANJEE	CHEMTROLS	9748704685	shivaji.dewanjee@chemtrols.com	
53	ANIRUDHIYA SAHA	CHEMTROLS	6290942597	anirudhiya.saha@chemtrols.com	
54	Rajukumar	APNRL Asst. Manager	9971948767	rajukumarstherma@apnrl.co	
55					
56					
57					
58					
59					
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Annexure B.2.1

ERLDC 231st OCC Meeting

22nd September 2025, ERPC, Kolkata

Intra-State Network Constraints

Part 2



West Bengal Demand Scenario

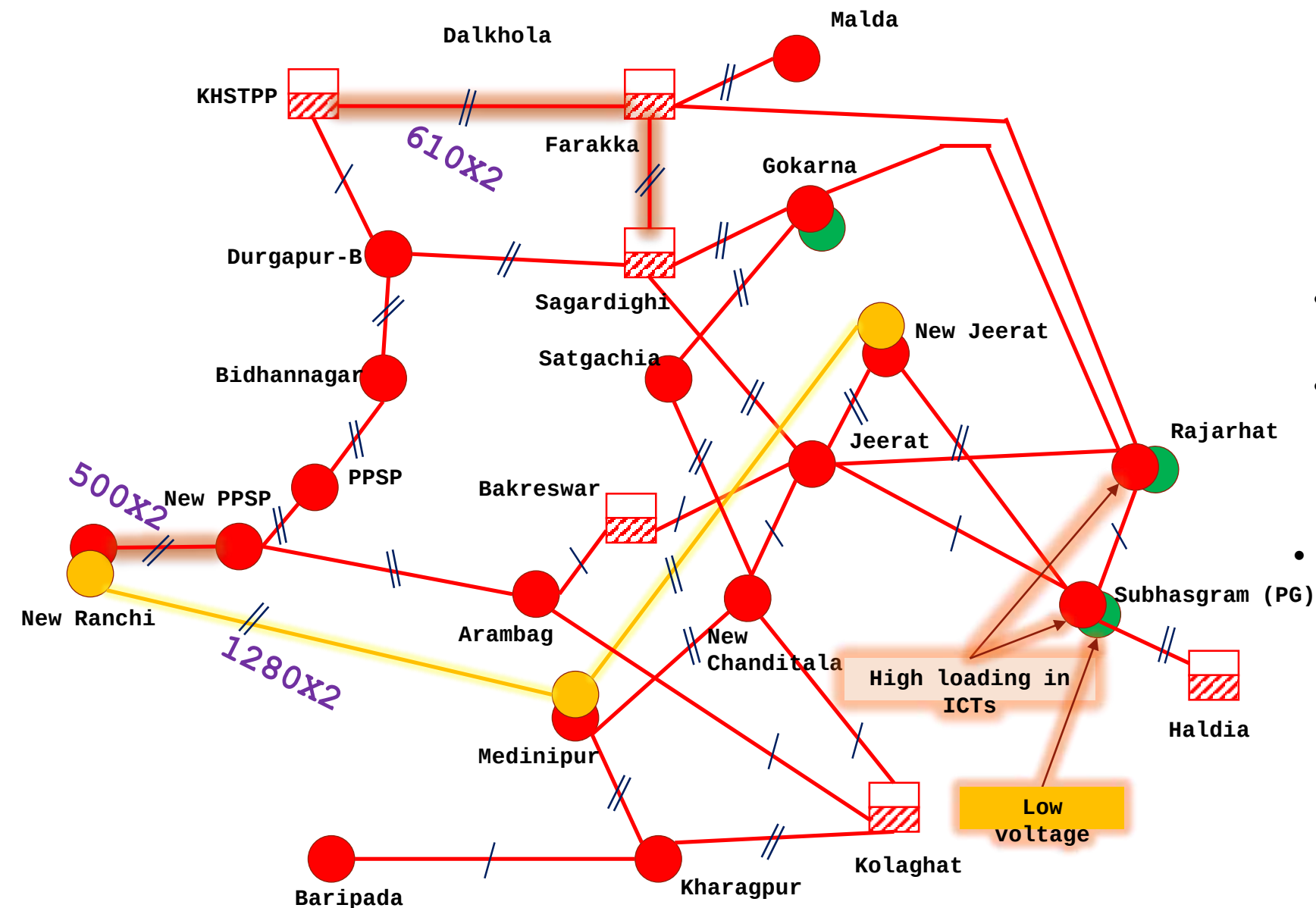
WB So far Highest: 13GW

Internal Gen: 7GW

Import from Grid: 6GW

- Kolkata demand at instance of WB Max demand: **3460MW**
- Kolkata Region demand So far highest: **3650 MW**

- 765kV New_Ranchi-Medinipur D/C at WB Max demand time: **1280X2 MW**





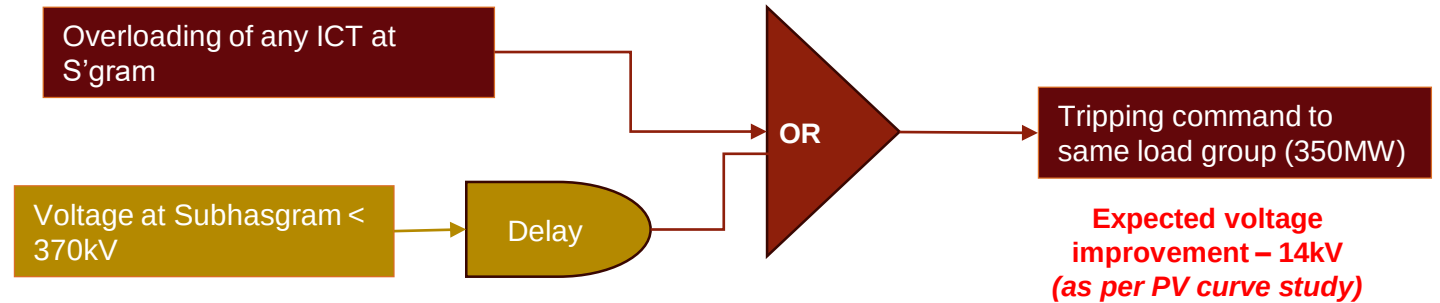
765kV New Ranchi–Medinipur-New Jeerat Corridor

Meeting on
9th Sept 2025



Short-term (Summer '26)

- Modification of existing SPS scheme of Subhasgram(PG) with under Voltage logic.



Long-Term (2028 onwards)

- Additional 765kV corridor to connect New Jeerat
- WB is implementing Static compensation (Capacitor Bank).



But Dynamic compensation is needed

Joint study will be done



Alternate source of 132kV Gangtok (PG) S/S

Meeting on
11th Sept 2025



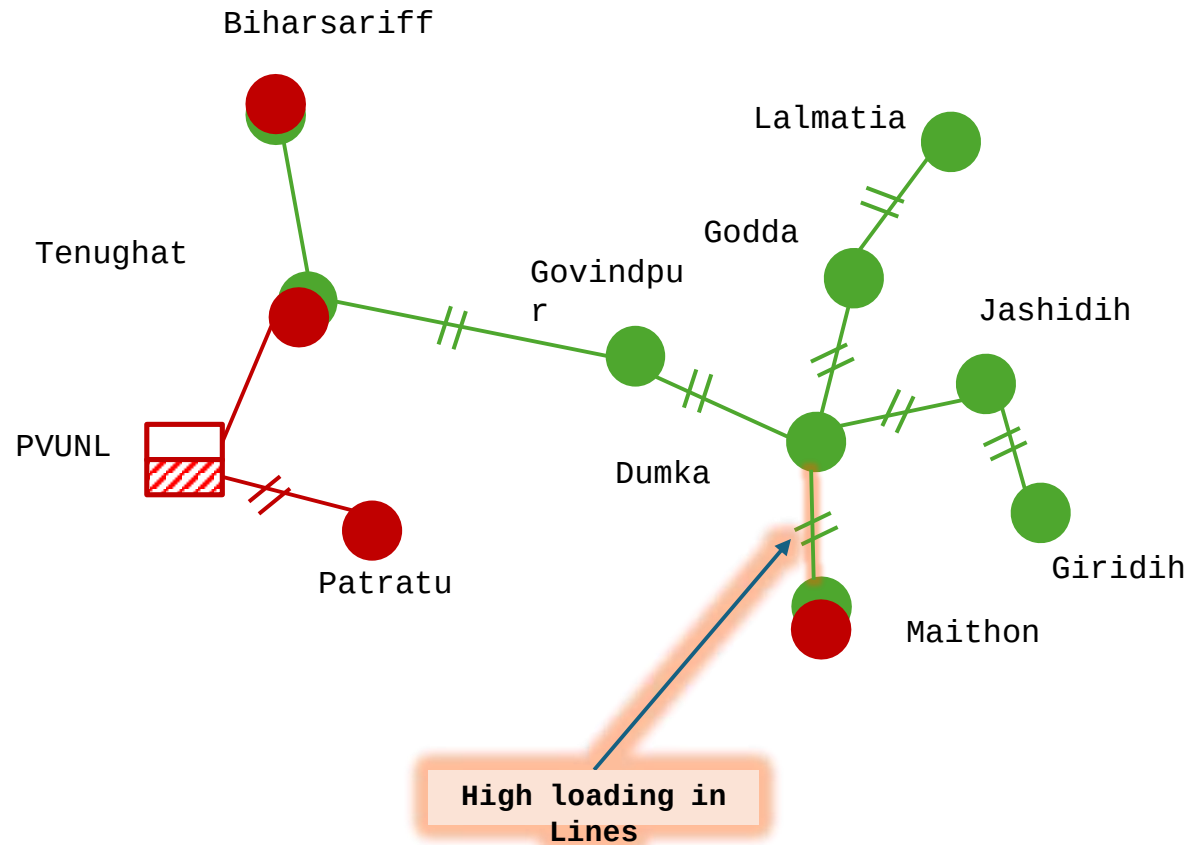
Short-term

- Parbing station is in advanced stage of commissioning → 132/66kV ICT of **Parbing** is under transportation. It is stuck due to repeated landslide.

Long-Term (2028 onwards)

- Reconductoring of 66KV Downstream network Gangtok –Tadong-Sinchey-Bulbulay
- Requirement of Double main bus scheme at 132KV Gangtok will be explored by Sikkim

Jharkhand Constraints

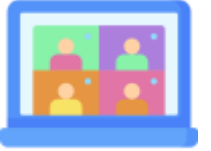


Tripping of any one circuit leads to cascade tripping to this area leads to blackout of Jasidih, Giridih, Dumka, Govindpur, including generation of Tenughat



220 kV Maithon-New Dumka D/C (JUSNL) and 220 kV Tenughat–Govindpur D/C (JUSNL)

Meeting on
11th Sept 2025



Short-term (Summer '26)

- > 220KV Dumka –Govindpur LILO at Dhanbad (NKTL)



Dec '25

- > 2nd ICT 400/220KV 315MVA at Tenughat to be commissioned.



Bay equipment of PVUNL need to be upgraded from single moose to twin moose similar to line

Long-Term (2028 onwards)

- > 220KV Tenughat -Gomia & 220KV Tenughat-Hazaribagh Road



Dec '26

400/220kV ICTs in DVC Generating plants

Meeting on
11th Sept 2025



Short-term (Summer '26)

- > 315MVA 400/220KV Koderma ICT replacement
- > SPS scheme for N-1 compliance of ICTs at Bokaro.



Dec '25



Joint Study will be done and
scheme will be finalised

Long-Term (2028 onwards)

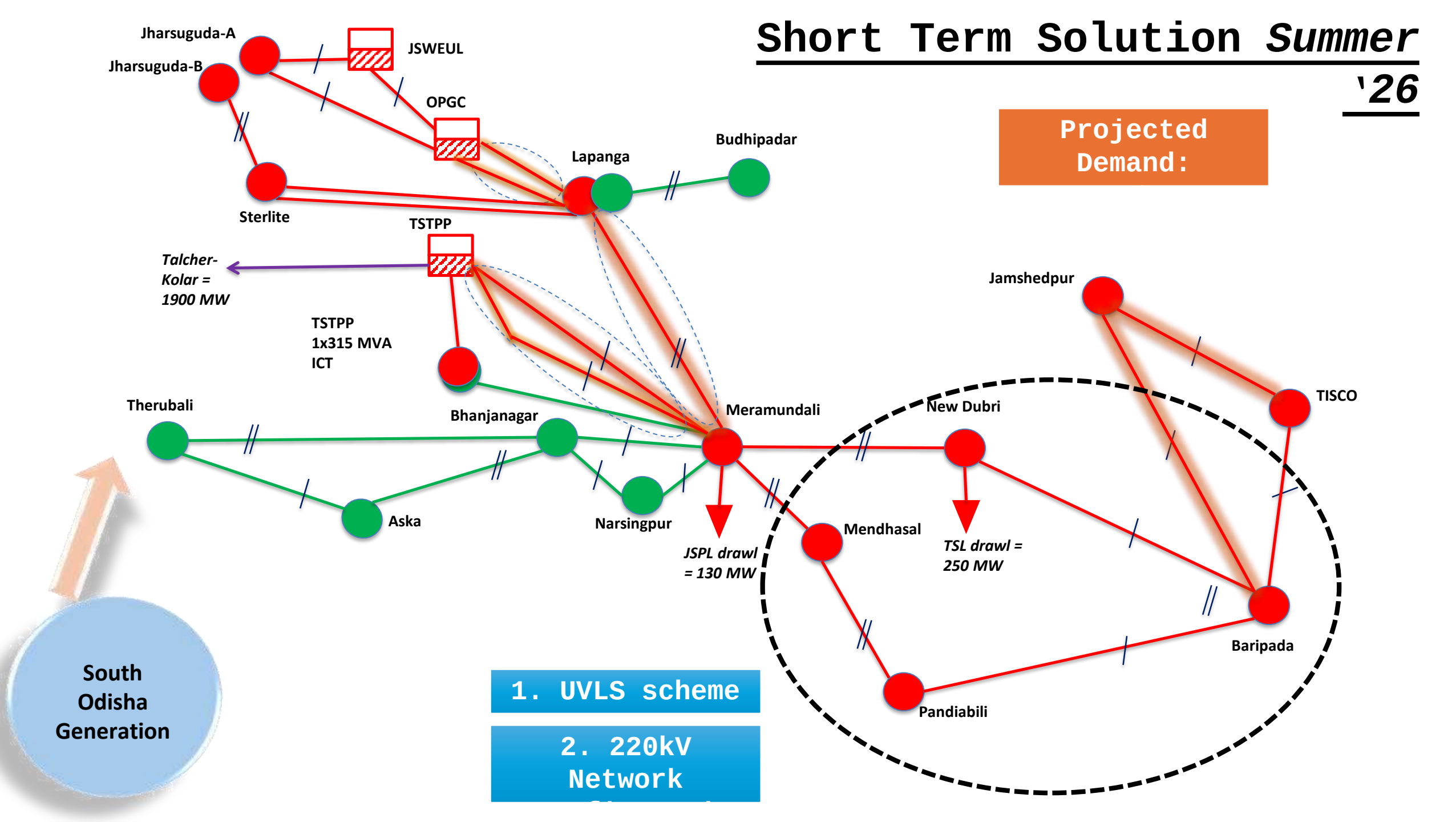
- > 400/220KV ICTs upgradation to 500MVA at Koderma
- > 3rd ICT 500MVA at Bokaro.
- > Connectivity between Mejia B 400KV with 220KV DVC System.



May '27

Dec- 2028

Projected
Demand:





Bhubaneswar Area reliability

Meeting on
18th Sept 2025



Short-term (Summer '26)

- Implementation of SPS scheme (Load restriction) with under Voltage logic in and around Bhubaneswar area

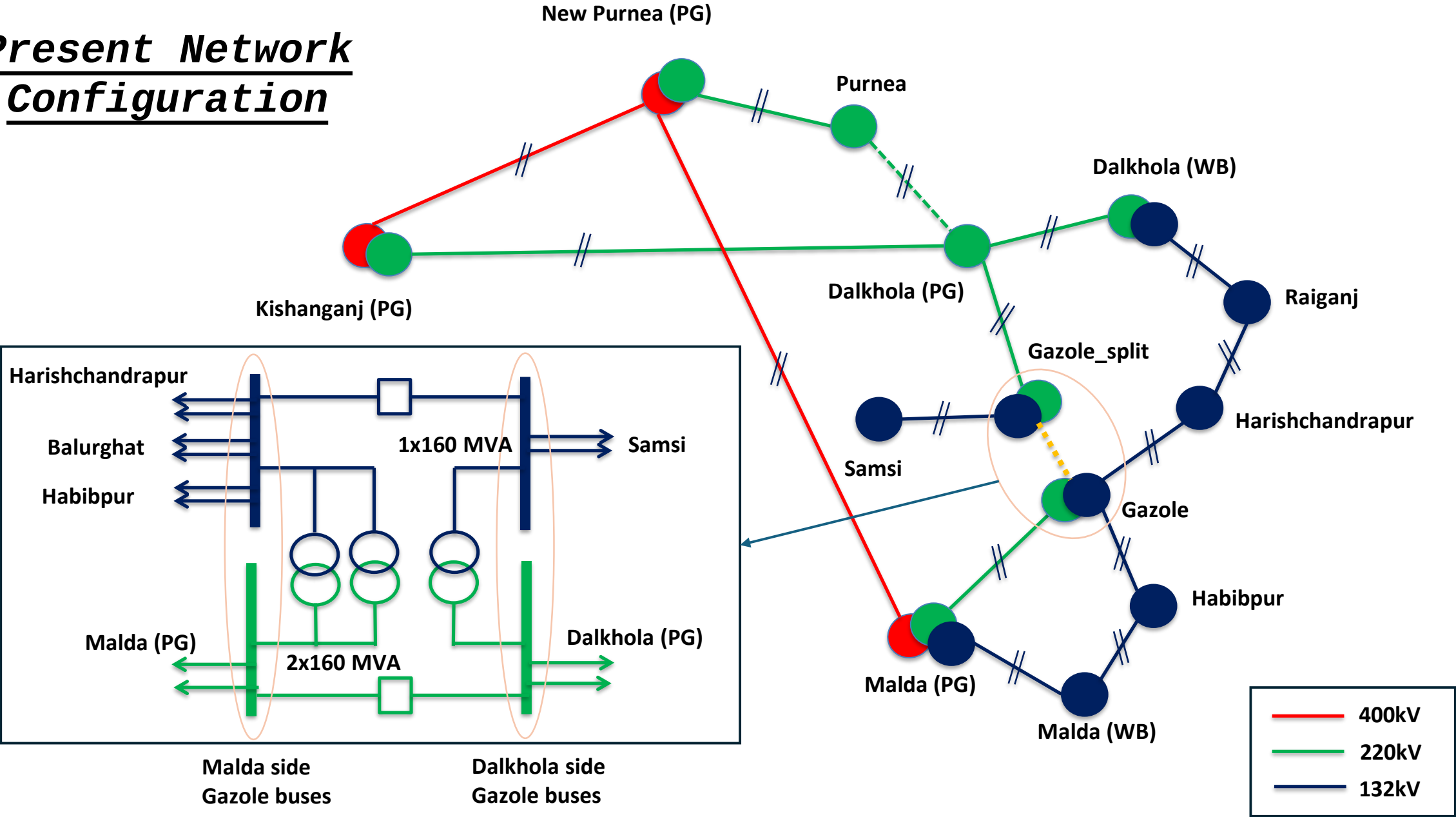
Long-Term (2028 onwards)

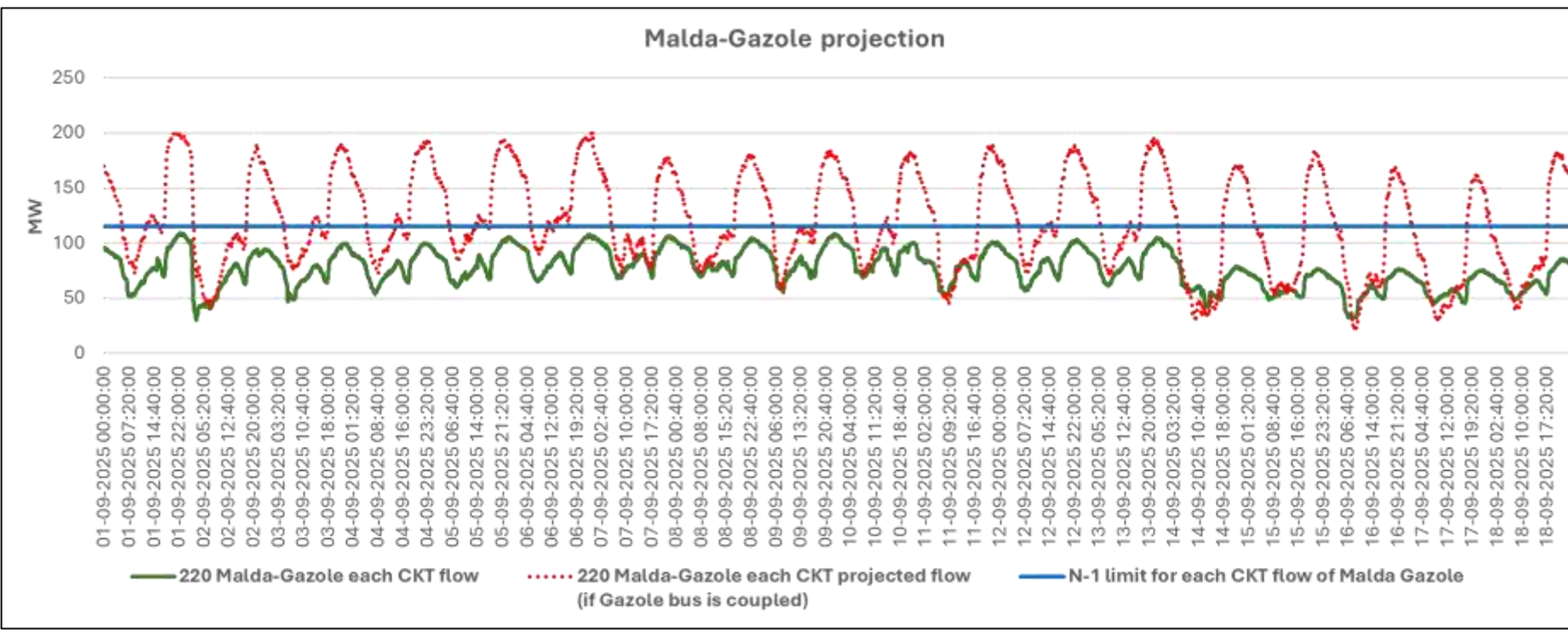
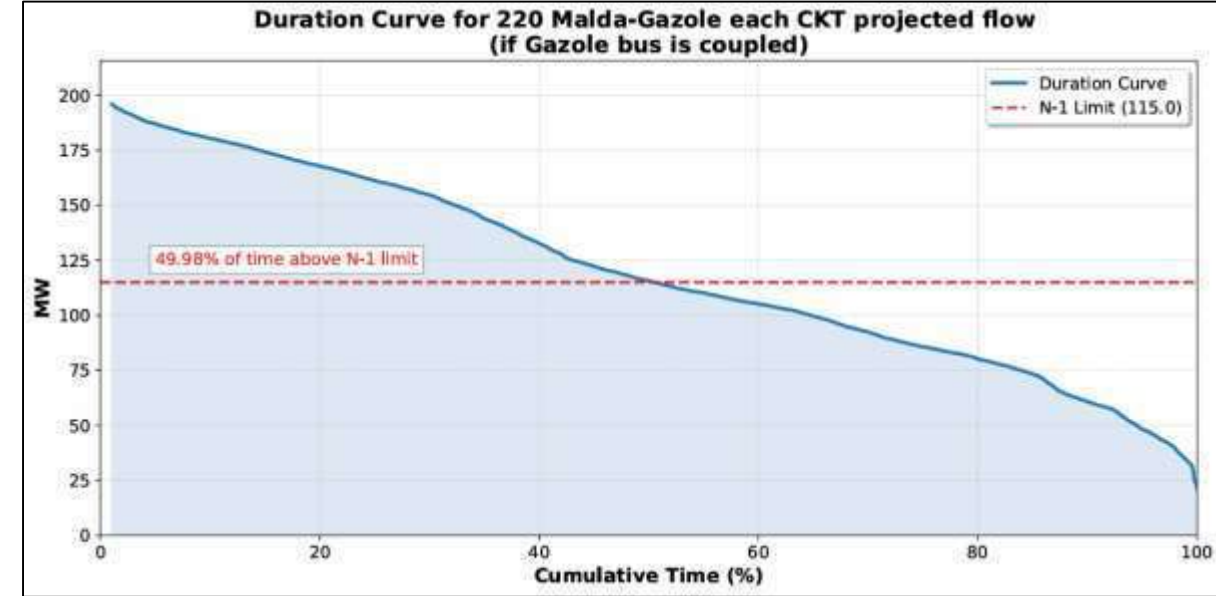
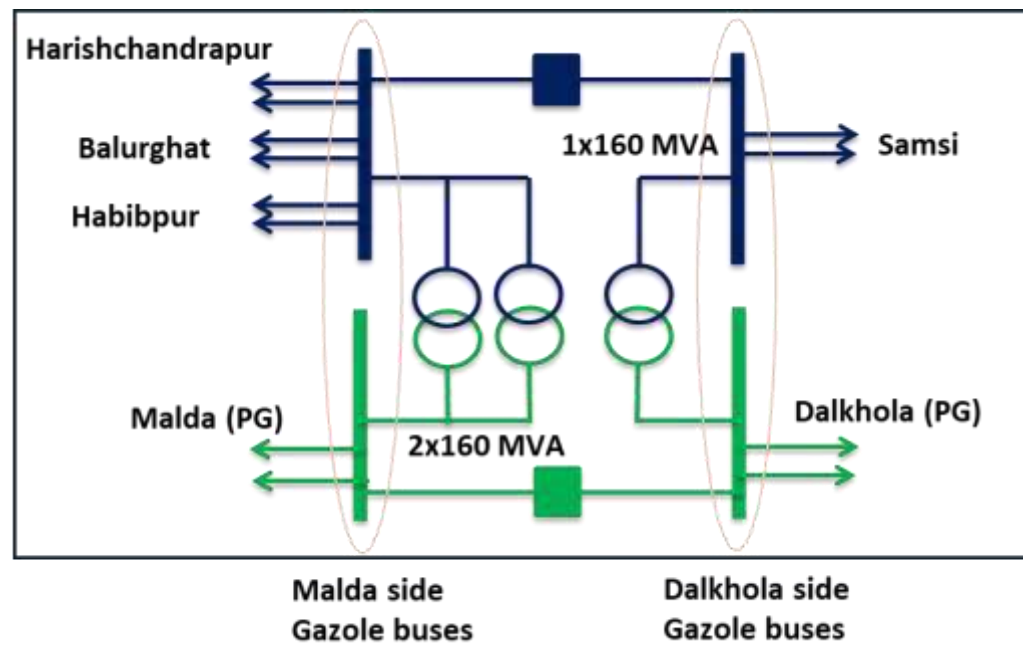
- Implementation of STATCOM at Mendhashal
- Reconductoring of 400kV OPGC-Lapanga, Meramundali -Medhashal
- Commissioning of Khuntuni S/S, 400kV Talcher_III-Meramundali-D/C
- Commissioning of 765kV Duburi OPTCL S/S



Constraints Near Dalkhola Area

Present Network Configuration



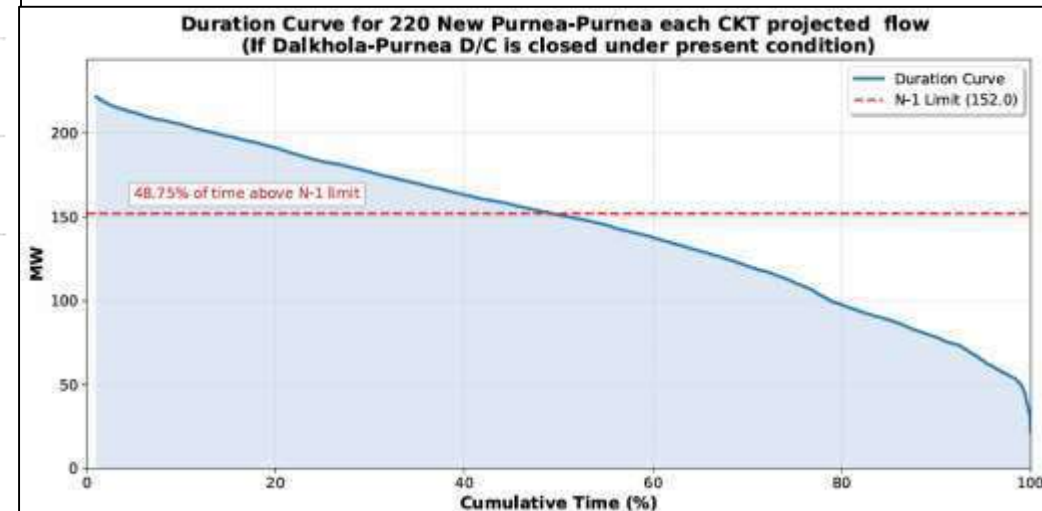
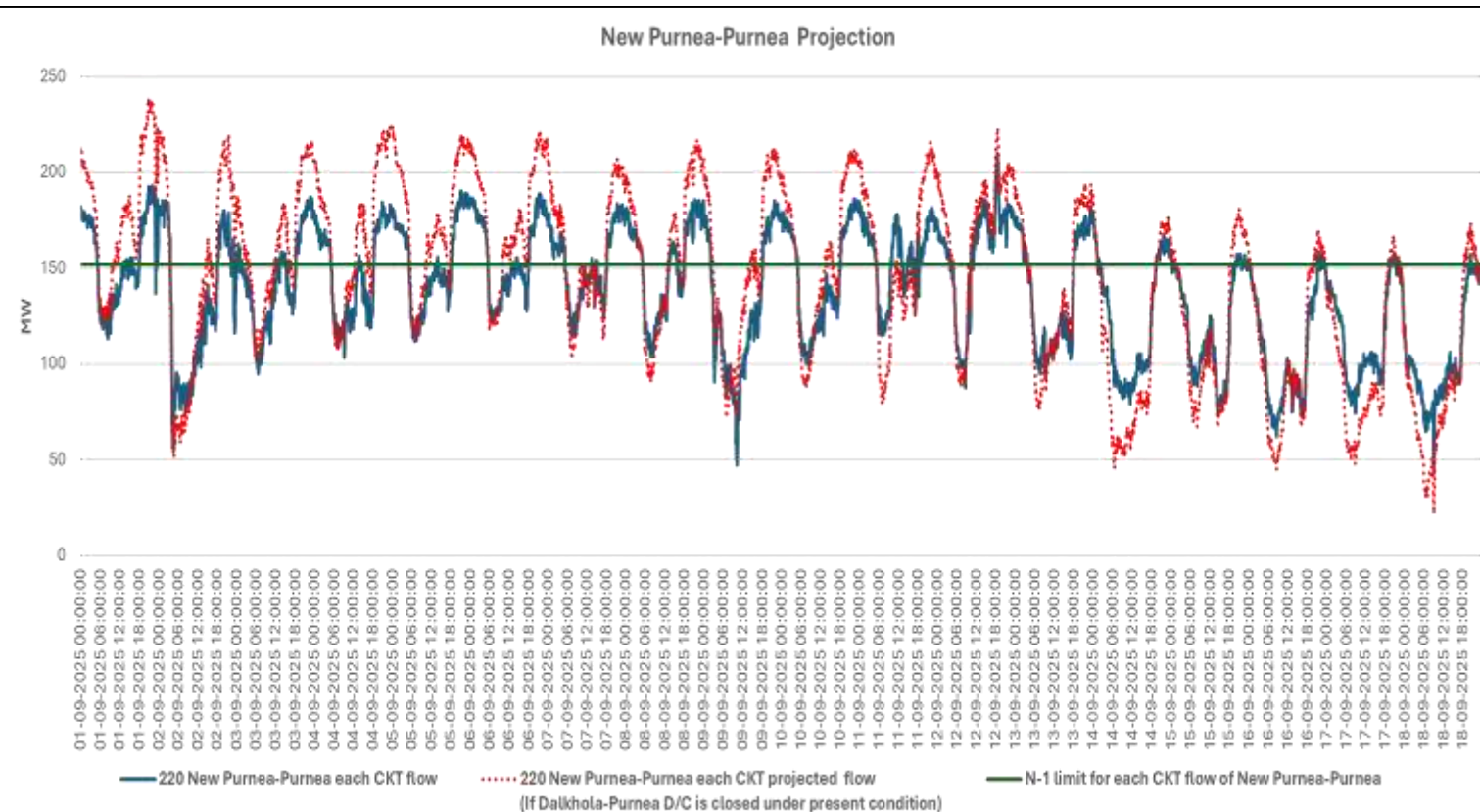
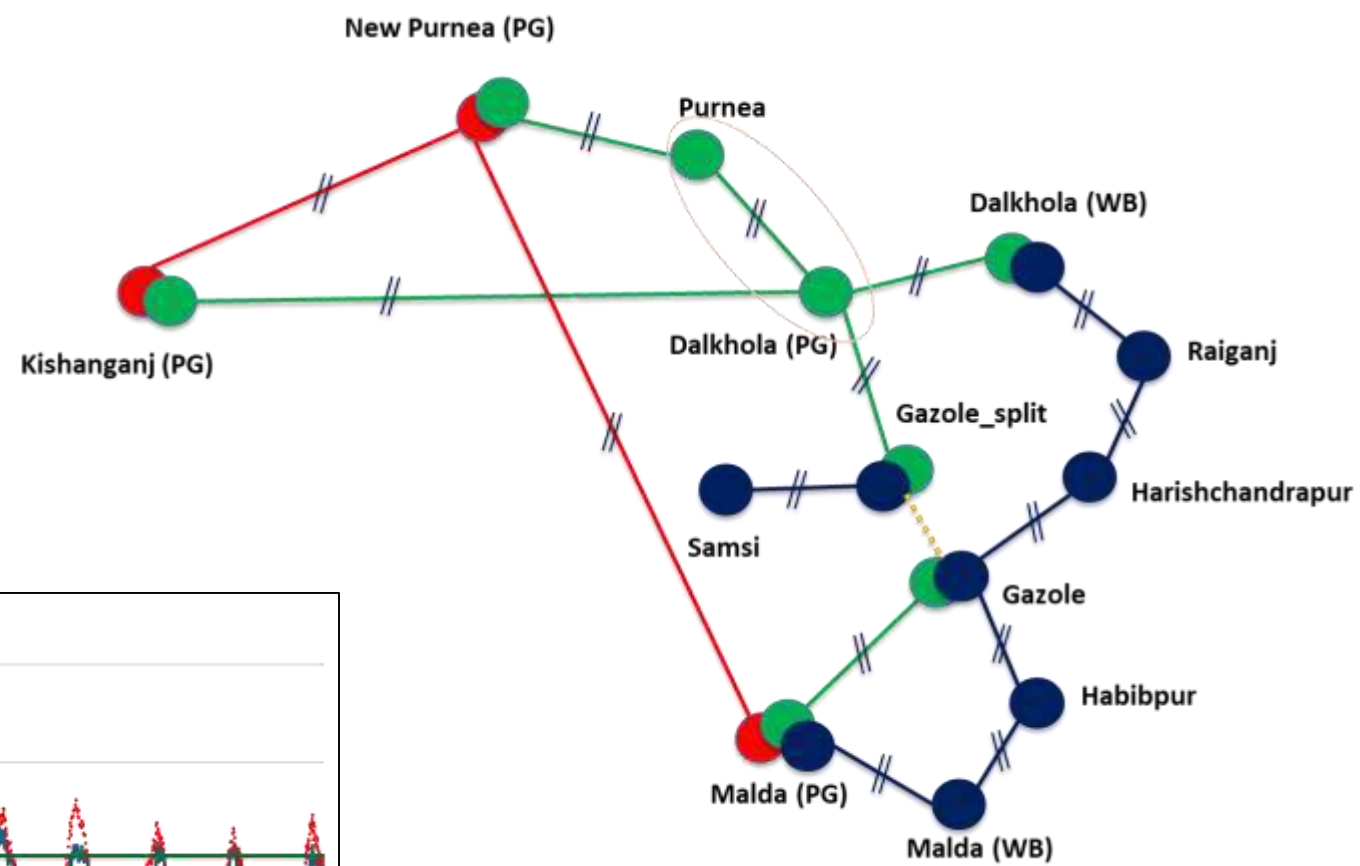


What If:
Gazole bus is
coupled

What If: Dalkhola-Purnea is closed

violating N-1 almost 50% of times.

- In Peak demand condition each circuit flow reaching up to 240MW.

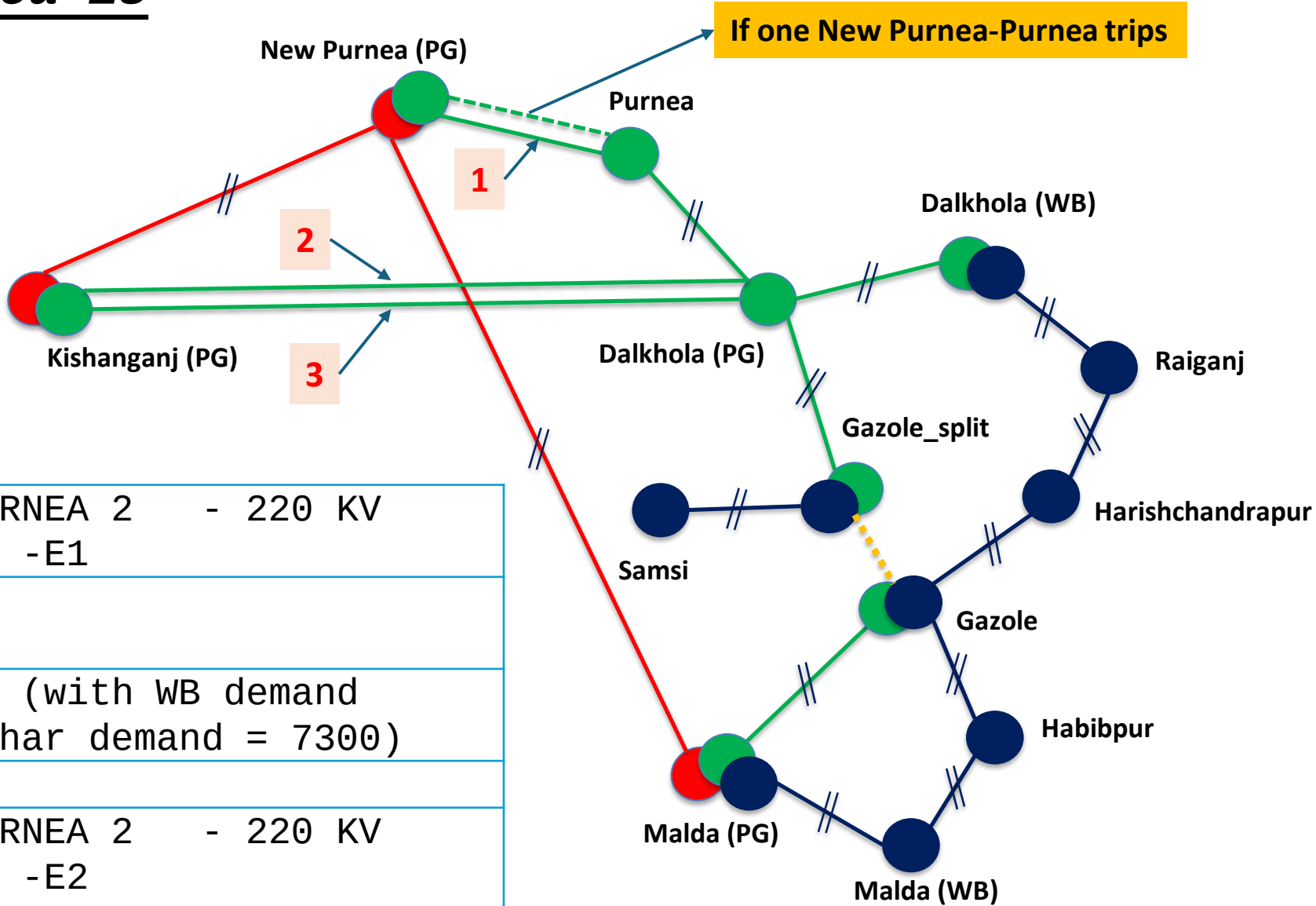


What If: Dalkhola-Purnea is closed

If one 220kV New Purnea-Purnea CKT trips:

- Then other 220kV New Purnea-Purnea CKT will also trip.
- After that 220kV Kishanganj-Dalkhola D/C will also trip.
- Hence, **total load of Samsi, alongwith loads of Purnea and Dalkhola area will be lost.**

Description of Contingency	220 KV N PURNEA 2 - 220 KV PURNEA 2 -E1
Depth of Cascade	L4
Total Load Loss	502.2863288 (with WB demand =10300 & Bihar demand = 7300)
Total Gen Loss	0
Cascade Path	220 KV N PURNEA 2 - 220 KV PURNEA 2 -E2 132 KV BANMANKHI - 132 KV SAHARSHA -1 220 KV KISHAN 2 - 220 KV DALKOLA -1

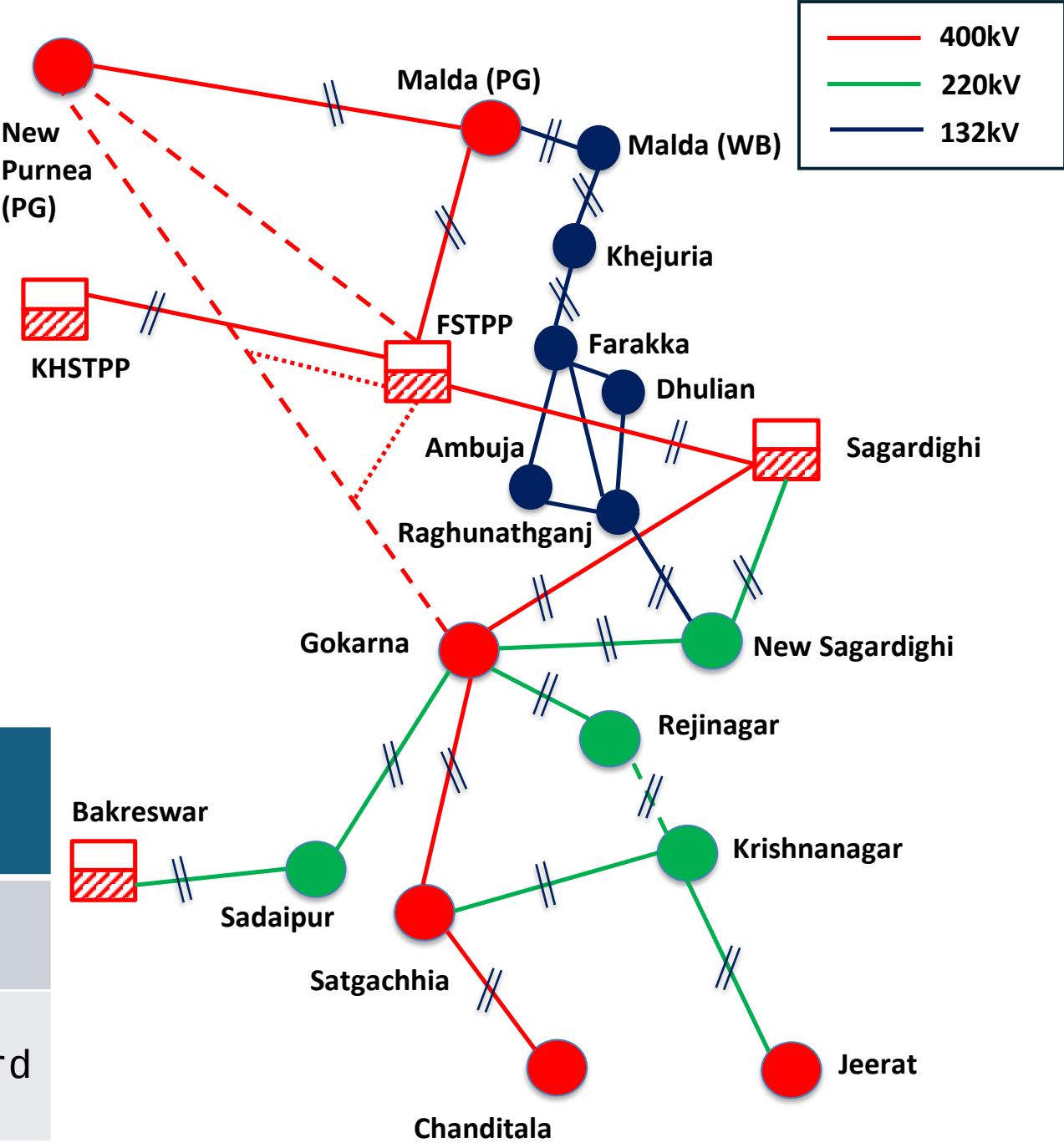


Restoration of Old Configuration: 400kV FSTPP- Gokarna

Present
Network
Configuration

- 400kV New Purnea-Gokarna and 400kV New Purnea-FSTPP are under long outage due to soil erosion near the tower structure.
- Some portion of 400kV New Purnea-Gokarna can be charged as 400kV FSTPP-Gokarna as per old configuration.
- The parameters of the line are

Parameters	400kV New Purnea-Gokarna	400kV FSTPP-Gokarna
Line length	250 km	119.7 km
Conductor type	Triple sonwbird	Triple sonwbird



Observati ons

- Hence, closing of 400kV FSTPP-Gokarna:
 - Increases chance of N-1 violation of 400kV KHSTPP-FSTPP in summer.
 - Also, as KHSTPP-FSTPP reconductoring is planned, closing this CKT will further aggravate the loading of remaining KHSTPP FSTPP CKT.
 - Re-Configuration may be done and may be closed as and when required during any contingency from Gokarno or during any Planned Shutdown.

Methodology for computation of Average Monthly Frequency Response Performance, Beta ' β '

Proposed	Existing	CERC Response	Proposal by
In case, there was no reportable event during the billing month, Beta ' β ' for that billing month shall be the median of last ten (10) reportable events.	Beta is zero if no reportable event in the billing month.	No Change	NTPC (also raised in ER OCC) & NHPC
In case of generators whose tariff is determined by CERC and are falling under the jurisdiction of SLDCs, concerned SLDC will certify the Beta ' β '	Beta is furnished by concerned LDCs and certified and issued by the RPC	No Change	Proposed by SRPC. ERPC agreed in-principle in 222nd OCC meeting.

Ensuring Data Availability and Accuracy in RTG Monitoring Portal

Real Time Resource Monitoring																		
☐ Enable Auto Refresh Select Region ALL Select Fuel Type THERMAL Download Data																		
Plant Name	Fuel Type	Re. No.	State Name	Utility Type	Installed Capacity	Effective Capacity	Actual Generation	Schedule	Schedule Revised	Declared Capacity	Re-derived	Percentage of Re-derived	Capacity on Hold	Actions	Unavailability Power/Hours	Primary Unavailability Power/Hours	Queue Hold	Planned Outage
☐ BRCC	THERMAL	ERLDC	Bihar	ISGS	1000	1000	669	375	0	602	0	100	750		307	0	0	250
☐ KSTPP ST-I	THERMAL	ERLDC	Bihar	ISGS	840	840	730	315	0	570	0	100	630		251	0	0	210
☐ Barn St.-I	THERMAL	ERLDC	Bihar	ISGS	1320	1320	585	694	0	1244	0	47.83	1320		590	610	0	0
☐ Barn St.-II	THERMAL	ERLDC	Bihar	ISGS	1320	1320	1067	684	0	1244	0	100	1320		580	0	0	0
☐ RPL	THERMAL	ERLDC	Jharkhand	Regional_EPP	1050	1050	1034	541	0	904	0	100	1050		481	0	0	0
☐ NSTPP(WPSC)	THERMAL	ERLDC	Bihar	ISGS	1900	1900	1901	1026	0	1066	0	100	1900		500	0	0	0
☐ FSTPP ST-III	THERMAL	ERLDC	West Bengal	ISGS	500	500	469	250	0	471	0	99.50	500		252	0	0	0
☐ FSTPP ST-I & II	THERMAL	ERLDC	West Bengal	ISGS	1600	1600	1134	566	0	940	0	100	1600		401	250	0	0
☐ IB TPS	THERMAL	ERLDC	Odisha	State	420	420	310	300	0	300	0	94.47	420		0	27	0	0
☐ KSTPP ST-II	THERMAL	ERLDC	Bihar	ISGS	1500	1500	1005	770	0	1410	0	96.39	1500		430	17	0	0
☐ SANTALDIN TPS	THERMAL	ERLDC	West Bengal	State	500	500	445	300	0	470	0	100	500		0	0	0	0
☐ KSTPP	THERMAL	ERLDC	Jharkhand	ISGS	1320	1320	1057	1021	0	1056	0	67.73	1320		833	0	200	0
☐ BANDEL TPS	THERMAL	ERLDC	West Bengal	State	275	275	203	153	0	130	0	100	275		37	0	0	0
☐ BAKRESHWAR	THERMAL	ERLDC	West Bengal	State	1050	1050	950	955	0	955	0	100	940		0	0	0	210
☐ OMR-3	THERMAL	ERLDC	Odisha	State_EPP	350	350	330	247	0	342	0	100	350		0	0	0	0
☐ BUDGE-BUDGE	THERMAL	ERLDC	West Bengal	State	750	750	520	500	0	604	0	76.00	750		194	162	0	0
☐ HALDIA ENERGY L	THERMAL	ERLDC	West Bengal	State	600	600	463	566	0	566	0	85.34	600		0	60	0	0

National Short-term Resource Adequacy



1. Existing and Planned conventional generators Parameters:

[illegible]

*Units under outage due to reserve shutdown & coal outages at the time of carrying out of study shall not be considered under planned or forced outages.

Historic Generation Profile of Hydro Plants for last 5 years

Plant Name	Date & Time	Generation (MW)
Plant A	01-04-2018 00:00	
Plant B	01-04-2018 01:00	
Plant C	01-04-2018 02:00	
Plant D	01-04-2018 03:00	
Plant E	01-04-2018 04:00	
Plant F	01-04-2018 05:00	
Plant G	01-04-2018 06:00	
Plant H	01-04-2018 07:00	
Plant I	01-04-2018 08:00	
Plant J	01-04-2018 09:00	
Plant K	01-04-2018 10:00	
Plant L	01-04-2018 11:00	
Plant M	01-04-2018 12:00	
Plant N	01-04-2018 13:00	
Plant O	01-04-2018 14:00	
Plant P	01-04-2018 15:00	
Plant Q	01-04-2018 16:00	
Plant R	01-04-2018 17:00	
Plant S	01-04-2018 18:00	
Plant T	01-04-2018 19:00	
Plant U	01-04-2018 20:00	
Plant V	01-04-2018 21:00	
Plant W	01-04-2018 22:00	
Plant X	01-04-2018 23:00	

1. Existing and Planned Renewable Plants:

[illegible]

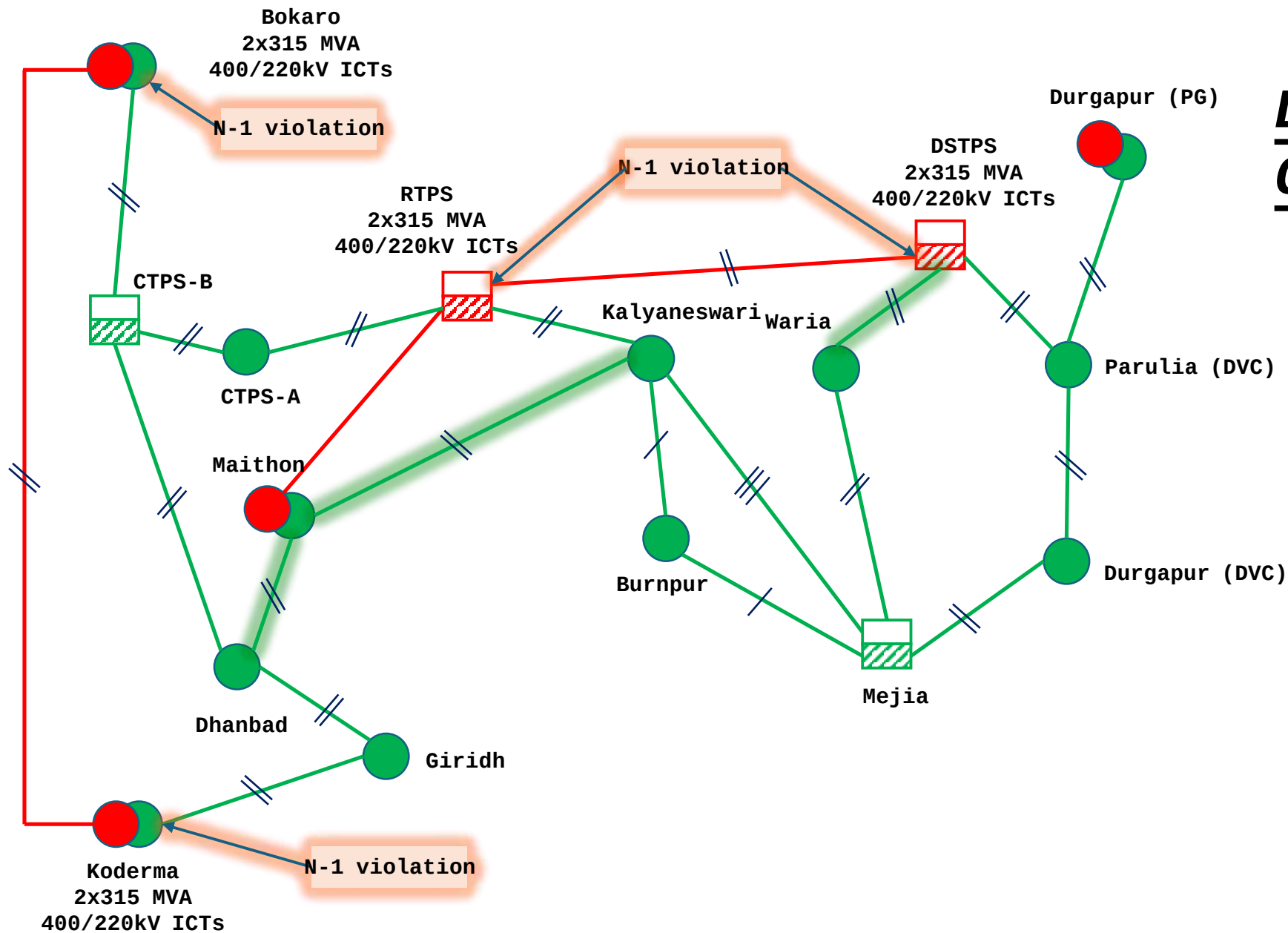
2. Existing and Planned Hydro (DAM, Run of river, and Pump Storage) Generator Parameters

[illegible]

Production			Expansion		
Variable Charge (Rs/MWh)	Dispatchable/ Non-Dispatchable	Energy Limits (Yearly, Monthly, Weekly, Daily as applicable)	Fixed Charge (Rs/MW/year) (if applicable)	Commissioning Date	Decommissioning Date (for both existing unit and planned unit)

Renewable historic generation profiles for last five years

Name	Datetime	Available Capacity (MW)	PV/ Wind Generation profile (MW)	Curtailed quantum (MW)
	01-01-2018 00:00			
	01-01-2018 01:00			
	01-01-2018 02:00			
	01-01-2018 03:00			
				
				
	31-03-2023 23:00			
	31-03-2023 24:00			



DVC Constraints

Ref: PG/ODP/RTAMC/ 1005

Dated: 25.08.2025

To
The General Manager,
EHT (O&M) Circle, Jaipur Road,
OPTCL, AT: TTS Colony, Dhabalagiri,
Odisha - 755019
Mob: 9438907974

Sub: Repeated Faults in 220kV Keonjhar(PG)-Turumunga Transmission Line-I

Dear Sir,

In the period between 18.06.2025 to date, there have been five instances of line fault in 220kV Keonjhar(PG)-Turumunga line-I. Trip details are as mentioned below:

220kV Keonjhar(PG)-Turumunga line-I tripping history from 18.06.2025 Onwards				
Sr. No.	Date	Relay details at PG end	A/R at PG end	A/R at OPTCL end
1	18.06.2025	Z1 Optd., R-Ph to Ground Fault, IF- 2.88 kA, Fault Distance - 12.1KM	Successful	Unsuccessful
2	23.06.2025	Z1 Optd., R-Ph to Ground Fault, IF- 5.12kA, Fault Distance - 6.9KM	Successful	Unsuccessful
3	15.08.2025	Z1 Optd., R-Ph to Ground Fault, IF- 2.92kA, Fault Distance - 11.8KM	Successful	Unsuccessful
4	23.08.2025	Z1 Optd., R-Ph to Ground Fault, IF- 2.91kA, Fault Distance - 11.6KM	Successful	Unsuccessful
5	24.08.2025	Z1 Optd., R-ph to Ground Fault, IF- 2.98kA, Fault Distance - 11.3KM	Successful	Unsuccessful

This is causing stress on the 315 MVA, 400/220kV ICTs of our Keonjhar Substation and other bay equipments. It is also observed that the Auto-reclosure function at OPTCL end is not functional for this line.

You are requested to take immediate urgent action towards making the line fault-free and also intimate us about the A/R status your end.

Matter urgent

Thanking You

Yours faithfully


(Ajaya Kumar Sahu)

25.08.2025
DGM (RTAMC), Bhubaneswar

Copy for kind information: Sr GM(AM), BBSR

पावर ग्रिड कारपोरेशन ऑफ इंडिया लिमिटेड
POWER GRID CORPORATION OF INDIA LIMITED

G-9, Outer Institutional Area, Katwaria Bazar, New Delhi-110 019



Ref: No. C-LDC-URTDSM-Alstom-167

Date: Dec 9, 2015

To

The Project Director
M/s Alstom T&D India Ltd., NMS-EMI, Ground Floor
A-7, Sector 65, Noida - 201301, Uttar Pradesh

Attn: Mr. Nilesh Raje

Subject: - WAMS System for Package-I and Package-II under Unified Real Time Dynamic State Measurement (URTDSM) Project - Approval of PMU Site Acceptance test (SAT) Procedure.

Ref: Supply Contract Agreement No. CC-CS/363-NR1/SCADA-2161/3/G1/R/NOA-114157
and
Supply Contract Agreement No CC-CS/363-WR1/SCADA-2162/3/G1/R/NOA-114159

Dear Sir,

This is with reference to the documents submitted vide your transmittal No. 3818-ATDIL-PG-URTDSM-TECH-0353 dated 09.12.2015. Approval for the following document is attached at Annexure-I.

1. WAMS- PMU Site Acceptance test (SAT) Procedure) - Rev D (Doc No. POWERGRID-URTDSM-A/GEN-3-600)

Approval/Comments conveyed herein neither relieves the Contractor of his contractual obligations and his responsibilities towards weights, qualities, design details, assembly fits, performance particulars and conformity of the supplies with the Indian Statutory Laws as may be applicable nor does it limit the purchaser's right under the contract.

Yours faithfully,

(Sunil Kumar)
Asst. General Manager (L&C)

Corporate office: "Saudamini" Plot No. 2, Sector-29, Gurgaon- 22001 Haryana, India
Phone: +91 124 2571700-19, Extn: 3109 Fax: +91 124 2571981, E-Mail: sunil.kumar@powergridindia.com

पावर ग्रिड कारपोरेशन ऑफ इंडिया लिमिटेड
(एनएसई एनएसएल)
POWER GRID CORPORATION OF INDIA LIMITED

(A Government of India Enterprise)
B-55, Qutab Institutional Area, Kirti Nagar, New Delhi-110 016



Ref: No. C-LDC-URTDSM-Alstom-167 Dated Dec 9, 2015

Annexure-I

Sl. No	Document Title Reference Number	Rev	Category	Remarks
1.	2. WAMS- PMU Site Acceptance test (SAT) Procedure) – Rev D (Doc No. POWERGRID-URTDSM-A/GEN-3-600)	D	I	

Category I, II, III, IV means:

- I: Approved/Released for implementation.
- II: Approved/Released for implementation subject to incorporation of comments. Revised drawing/document required
- III: To be resubmitted for approval after incorporating the comments.
- IV: For information and record.

Note: - The approvals of the drawing/document vide this letter does not cover the approval of sub-vendor for make of the items. The approval in this respect is to be obtained from QA&I deptt.

Corporate office: "Saudamini" Plot No. 2, Sector-29, Gurgaon-122001 Haryana, India
Phone: +91 124 2571700-19, Extn : 3109 Fax: +91 124 2571981, E-Mail: sunilkumar@powergridindia.com

Compliance for Comments Dated :08.10.2015 & 08.12.2015

S.No	M/s PGCIL Comments	Alstom Reply
1	Annexure-I : Comments on compliance for comments dated 25.08.2015	a,b This test will be conducted during system availability test & is included in CC Availability test procedure.
2	Annexure-I : Point No - 1	Reference documents included in section 1.1.3: 1. Approved BOQ of the substation 2. Approved typical scheme referred in BOQ 3. Installation guide 4. PMU database (DB) (System generated copy)
3	Annexure-I : Point No - 2	Complied
4	Annexure-I : Point No - 3.2	Complied
5	Annexure-I : Point No - 3.3	This software shall be utilized for local configuration of Intelligent Electronic Devices (PMU)
6	Annexure-I : Point No - 4.1	Approved BoQ and system generated PMU database is included in Document reference System generated PMU database will be furnished as a S/S specific document
7	Annexure-I : Point No - 4.2.1	Complied as 4.2.1 Check for BOQ, Technical details, & Wiring as per approved typical scheme referred in BOQ
8	Annexure-I : Point No - 4.2.3	Dual DC supply check has been included in 6.1
9	Annexure-I : Point No - 5.1	4. This alarm will be reported to Alarm Management System & the same is mentioned in PDC system design document. This will be demonstrated in integrated CC FAT in factory 6 Complied & Added in procedure.
10	Annexure-I : Point No - 6.1	Dual DC supply check has been included
11	Annexure-I : Point No - 6.2	Complied
12	Annexure-I : Point No - 6.3	Complied
13	Annexure-I : Point No - 9.1.1	Point No. 9 (Tests at PDS) has been removed from SAT procedure
		15. Complied
14	Comments on the table in Annexure-I	Complied
15	General compliance	Visibility of data at PDS/PDC is subject to availability of link between PMU and PDS/PDC at control center. The test stands completed with the data visibility at PMU HMI in the absence of link between PMU and PDS/PDC at control center



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Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD.	Rev	0
Project	WAMS - PKG-I & II UNDER URTDSM	Date	9.12.2011
MDL Ref.	POWERGRID-URTDSM-AGEN-3-600	Page	2 of 20

ALSTOMTRANSMISSION & DISTRIBUTION
Network Management Solutions**Unified Real Time Dynamic State Measurement (URTDMS)**CC-CS/363-NRI/SCADA-2161/3/G1/R/CA-I/4857 dt 31 03 2014
CC-CS/363-NRI/SCADA-2161/3/G1/R/CA-II/4858 dt 31 03 2014**Transmittal No. 3818-ATDIL-PG-URTDMS-TECH- 353**From: **RAJASEKHAR REDDY**Date: **09/12/2015**Original Sent By: **By Hand**Action Requested: **For Approval**

S.No	Designation	Document N°	Revision N°
1	PMU SITE ACCEPTANCE TEST(SAT) PROCEDURE	POWERGRID-URTDMS-A/GEN-3-600	0

<u>DISTRIBUTION LIST</u>	Original Set	Copy With enclosures	Copy Without enclosures
POWERGRID			
Mr. Sunil Kumar (Asst. GM, LD&C)	1		
LD&C Wing, Ground Floor, Plot no. 2, Saudamini, Sector 29, Gurgaon-122001			


(Rajasekhhar Reddy)
Project Manager
For ALSTOM T&D

SITE ACCEPTANCE TEST PROCEDURE FOR PHASOR MEASUREMENT UNIT, PANELS WITH GPS & LAN SWITCH UNDER URTDSM PROJECT

CUSTOMER : POWERGRID CORPORATION OF INDIA LTD.

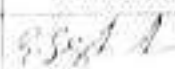
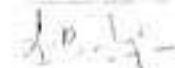
TITLE : SITE ACCEPTANCE TEST (SAT) PROCEDURE

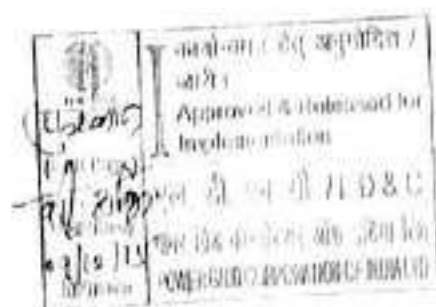
DOCUMENT NO : POWERGRID -URTDSM-A/GEN-1-600

ISSUE & DATE : Rev. D, 09.12.2015

P.O. No : CC-CS/363-NR1/SCADA-2161/3/G1/R/NOA-I/4857
CC-CS/363-WR1/SCADA-2162/3/G1/R/NOA-I/4859
Dated: 15/01/2014

PROJECT NAME : WAMS - PKG-I & II UNDER URTDSM

	NAME	SHORT NAME	SIGNATURE
Prepared	G.SIVASANKAR	GSS	
Checked	S. BALAJI	SB	
Approved	RITESH DHARAT	RB	



1 ORGANIZATION OF VALIDATION TASKS

1.1 SITE ACCEPTANCE SESSION

1.1.1 Definition

The Site Acceptance Test document deals with the acceptance of the panels at respective sites by customer. It aims at checking, in accordance with approved documents

1.1.2 Program

The different steps of the Site Acceptance test include:

- Physical verification of hardware as per approved drawings.
- Functional verification of hardware.
- Connectivity with Phasor Development Server (PDS)/Phasor Data Concentrator (PDC).

1.1.3 Reference documents

1. Approved BOQ of the substation.
2. Approved typical scheme referred in BOQ
3. Installation guide.
4. PMU database (DB) (System generated copy)



1.1.4 Results

Upon completion of SAT, the final acceptance meeting assesses the acceptance of the system by the customer.

The assessment of the acceptance is based on the test results on the Site Acceptance Tests. The purposes of the review are:

- To state about the conformity between the offered system and the approved documents.
- To verify that every tests has been performed successfully.
- To issue the acceptance certificate.

Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD.	Rev	D
Project	WAMS - PKG-I & II UNDER URTDSM	Date	9-11-2015
MDL Ref	POWERGRID-URTDSM-A/GEN-3-600	Page	1 of 20

2 Sub-station Details

1	Region	East Region					
2	Site ID	—					
3	Sub-Station	JITPL 400kv Switch Yard					
4	Utility	Power Generation					
5	Sub-Station Type	Kiosk type.					
6	Voltage level	400kv					
7	Feeder Qty	2.					
8	PMU Qty	1					
9	PMU Panel Location	Kiosk-1					
10	Reporting Destination						
11	PMU IP DETAILS	PMU Ref	Panel Ref	IP	PMU Ref	Panel Ref	IP
		PMU-1	K1179	172.19.21.162	PMU-8		
		PMU-2			PMU-9		
		PMU-3			PMU-10		
		PMU-4			PMU-11		
		PMU-5			PMU-12		
		PMU-6			PMU-13		
		PMU-7			PMU-14		
12	LAN SWITCH IP DETAILS	EFS Ref	Panel Ref	IP	EFS Ref	Panel Ref	IP
		EFS-1	K1179	172.19.21.161	EFS-4		
		EFS-2	MV2	172.19.21.161	EFS-9		
		EFS-3			EFS-10		
		EFS-4			EFS-11		
		EFS-5			EFS-12		
		EFS-6			EFS-13		
		EFS-7			EFS-14		


 Approved & Authorized for Implementation
 एल सी एल सी / LO&C
 उपरि उक्त उपकरणों/ओई उपकरणों के

Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD		
Project	W/M/S - PKG-I & II UNDER URTDSM		
MDL Ref	POWERGRID-URTDSM-A/CEN-3-600		
Rev	1		1
Date	9.12.2015		1
Page	4 of 10		1

3 SAT REQUIREMENTS

3.1 Scope of Tests

The Site Acceptance Test (SAT) for Phasor Measurement Unit (PMU) Panels with Global Positioning System (GPS) Clock and Ethernet LAN switch will be performed at the respective substation. The tests shall verify that the PMU functions are normal during the test.

The SAT tests have been divided into relevant sections in order to perform the tests systematically. Upon successful verification of the tests in each section, pertinent forms and test results shall be signed by the customer or his representative.

3.2 Hardware Environment

The following hardware equipment is to be made available during SAT and in line with approved document.

- Phasor Measurement Units.
- Ethernet Switches
- Time Synchronizing Equipment (GPS)

List of testing equipment's used

- Multimeter for AC/DC measurements
- Clamp on meter
- Laptop with connecting cable



3.3 Software Environment

The following software shall be used for conducting the tests.

- MiCOM S1 Agile (IED configuration software).
- PMU data analyzer.

This software shall be utilized for local configuration of Intelligent Electronic Devices (PMU).

Sub-station specific configuration files shall be prepared offline for all the PMU's and downloaded to the PMU for testing.

Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD.	Rev	0
Project	WAMS - PKG-I & II UNDER URTDSM	Date	9.12.2014
MDL Ref.	POWERGRID-URTDSM-AGEN-3-600	Page	5 of 20

4 SAT Tests

4.1 Document Verification

Purpose

This test verifies that the applicable approved drawings and documents for the equipment / system under test are referred during the Site Acceptance Test and in case of dispute contract specification shall prevail.

Setup

Ensure that the documentations are readily available for reference during SAT.

Procedure

Verify the drawings (printed) of all assembled equipment are present.

Expected results

All documents are available as per the below table

S.N.	Document Reference	Available / Not available
1	Site Survey Report	✓
2	Approved Bill of Quantity (BOQ)	✓
3	Sub-station specific panel drawings – General Arrangement, Legend, Schematic Drawing	✓
4	Typical Adaptation Drawing	✓
5	System generated PMU database	✓

Test Result:

☒ PASS ☐ FAIL ☐ NOT TESTED

9523040605
11/10/2018
Customer Representative

9500180223
11/10/18
Penikasangar
ALSTOM Representative

Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD.	Rev	0
Project	WAMS - PKG-I & II UNDER URTDSM	Date	9.12.2015
MDL Ref.	POWERGRID-URTDSM-A/GEN-3-600	Page	6 of 10

4.2 Equipment Verification

4.2.1 Check for BQ, Technical details, & Wiring as per approved typical scheme referred in BQ.

Purpose

To verify that all hardware equipment required are as per approved documents.

Setup

Ensure that the PMU panels are readily available in SAT location.

Procedure

1. Visually inspect the equipment in panel for cleanliness and ensure that they are free from damage / foreign material (dust, solid material etc.).
2. Inspect the yard wiring terminated in panel for correctness and ensure that they are free from insulation damage.
3. Ensure that all earth/ground connections are correctly bonded from the panels to the substation earth.
4. Check the connectivity of all the equipment's are as per approved schematic drawings.

Expected results

1. Equipment in the panel is free from damage, rust and dust.
2. Terminated yard wiring in the panel are insulated and free from damage.
3. Panel earth/ground is connected to sub station earth.
4. All Equipment connections are as per the approved schematic.

 16/10/18 Unit No. 1000000 1000000 1000000	Equipment & Material / Location 1000000 1000000 1000000
--	---



Test Result:

☒ PASS	☐ FAIL	☐ NOT TESTED
--	-------------------------------	-------------------------------------

950830605
 11/10/2018
 Customer Representative

950010018
 11/10/18
 ALSTOM Representative

Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD.	Rev	3
Project	WAMS PKG-I & II UNDER LRTDSM	Date	11/10/2018
MDL Ref	POWERGRID-LRTDSM-AIGEV-I-500	Page	1 of 30

4.2.2 Physical Verification of Equipment

Check the Bill of Material as per approved PMU Panel drawings and Approved 600

Expected results

All equipment is in line with the Panel drawings and Approved 600

S.NO.	Equipment	Ok / Not Ok
1	PMU ✓	Ok
2	GPS ✓	Ok
3	EFS ✓	Ok

**Test Result:**

☒ PASS	☐ FAIL	☐ NOT TESTED
--	-------------------------------	-------------------------------------

9500180228
11/10/2018
Customer Representative

9500180228
11/10/18
ALSTOM Representative

Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWER GRID CORPORATION OF INDIA LTD.	Rev	01
Project	WAMS - PKG-I & II UNDER URTDSM	Date	9.12.2018
MOL Ref.	POWERGRID-URTDSM-AGEN-3-600	Page	4 of 20

4.2.3 Power Checks

Purpose

To ensure that the equipment installed in the panels are operating when connected to the nominal specified power supply (Substation specific DC voltage)

Setup

Perform the steps below to confirm correct operation of all equipment connected to the power supplies.

Procedure

1. Verify that all equipment's (PMU, GPS, and EPS) in panels are powered at the nominal AC/DC supply voltage input (Substation specific DC voltage). If required, using a Digital Multimeter verify the presence of nominal supply voltage.
2. Verify that there are no abnormalities seen when the equipment are turned ON.
3. Verify that all equipment's are operating normally.

Expected Result

- 1) DC voltage is available for the panel as per substation specific drawing
- 2) All equipment is powered up.

**Test Result:**

☒ PASS	☐ FAIL	☐ NOT TESTED
--	-------------------------------	-------------------------------------

950830605
10/10/2018
Customer Representative

9500180228
11/10/18
ALSTOM Representative
Pema Sony. D.

Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD	Rev	0
Project	WAMS - PKG-I & II UNDER URTDSM	Date	11/10/2018
MDL Ref	POWERGRID-URTDSM-AIGEN-3-600	Page	3 of 20

5 Time Synchronization Equipment Tests

5.1 Time Synchronization Equipment Functional Check

The time synchronization test shall be performed by making GPS as the time reference for synchronizing PMU's as per schematic drawing

Setup

- Ensure that the test equipment is available in SAT location.

Procedure

1. Verify the antenna installation
2. Check how many satellites are visible to the GPS and ensure that the GPS is locked to the satellites before starting any test
3. Check that the PMU does not have any GPS related alarm
4. Disconnect IRIG-B and/or PPS input from PMU and observe the alarms on the PMU along with error bit
5. Reconnect the above and check all alarms reset
6. Check the GPS configuration for PPS / IRIG-B settings Using the laptop
For configuration details refer GPS setting screenshot below:
7. Repeat step-1 to 6 for all other PMU's available in the substation

Expected Result

- 1) No GPS related alarms displayed in PMU after installation
- 2) Error displayed in PMU upon disconnection of IRTG-B and/or GPS input to PMU

95830605
11/10/2018
D. P. S. Dham
T. P. L.

4/10/18
(Date stamp)
Approved & Recommended for
border completion
LD&C
11/10/18
per [signature]

9500180228
11/10/18
per [signature]

AD India Limited
STATION

Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD	Rev	1
Project	WAME PKG I & II UNDER URTDSM	Date	9/12/2011
MDL Ref	POWERGRID-URTDSM-AYGEN-3-600	Page	10 of 20

ALSTOM

GPS Setting

ALSTOM

RT430 GPS Submodule Clock

Model	RT430
Version	1.0
Manufacturer	ALSTOM
Serial Number	123456789
Part Number	123456789
Lot Number	123456789
Production Date	12/10/2018
Test Date	12/10/2018
Test Result	PASS
Test Location	Powergrid Corporation of India Ltd.
Test Operator	Powergrid Corporation of India Ltd.
Test Equipment	Powergrid Corporation of India Ltd.
Test Method	Powergrid Corporation of India Ltd.

ALSTOM	ALSTOM T&D India Limited
ALSTOM	ALSTOM T&D India Limited
ALSTOM	ALSTOM T&D India Limited
ALSTOM	ALSTOM T&D India Limited
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ALSTOM	ALSTOM T&D India Limited
ALSTOM	ALSTOM T&D India Limited

Test Result:

☒ PASS	☐ FAIL	☐ NOT TESTED
--	-------------------------------	-------------------------------------

9583040605.
S.P. Sathyan
11/10/2018
Customer Representative

9580180288
11/10/18
ALSTOM Representative
Panya Seng R.



Title	SIT: ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD.	Rev	0
Project	WAMS - PKG-I & II UNDER URTDSM	Date	4/12/2015
MDL Ref	POWERGRID-URTDSM-A/GEN-3-600	Page	11 of 20

6.3 PMU Digital Input Data Retrieval

This test is conducted to check the Digital Inputs are mapped as per the approved drawing / BOQ.

- 1) Verify that the changes of state of Digital Input points are updated in the PMU local display.
- 2) External signal to be connected to the DI circuit and status to be verified.

Expected Results

- 1) Digital inputs are configured as per approved drawings / BOQ
- 2) The status of the DI is high/low based on the external signal

Tested by	Approved by
Tested on	Approved for
Tested at	Tested by
Tested for	Tested on
Tested by	Tested on
Tested by	Tested on

Test Result:

☒ PASS ☐ FAIL ☐ NOT TESTED

Customer Representative

ALSTOM Representative

Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD	Rev	0
Project	WAMS - PKG-I & II UNDER URTDSM	Date	4.12.2015
MDL Ref.	POWERGRID-URTDSM-A/GEN-3-600	Page	13 of 20

7 Operation check for communication port, auto restoration & configuration settings of PMU.

7.1 PMU Communication Port Operation

Connect current & voltage signals from CT/PT circuit

1. Connect laptop containing PMU Data analyzer software through copper RJ45 port & monitor the analog phasors measurements. *At present C.T/P.T going being not done. GATEWAY will be connected during line start down operation.*
2. Repeat step 1 for all the PMU's available in the S/S.

Expected Result

- 1) Analog phasors are displayed in PMU Data analyzer

7.2 Check for PMU configuration settings

To download & check the configuration settings of PMU

1. Power up the PMU with nominal power supply (Substation specific DC voltage)
2. The Healthy LED should turn ON.
3. Set the IP address of PMU manually or using the front RS232 serial port.
4. Download configuration to the PMU
5. Modify some of the parameters manually and upload the configuration file to the laptop.
6. The changes done manually should be reflected on the configuration file in laptop.
7. Now change settings in configuration file in laptop and download to the PMU
8. The changes done in step 7 should be reflected in the PMU.
9. Enable/Disable some phasors in the PMU configuration
10. In the PMU data analyzer software only the relevant phasors should be displayed.
11. Verify that Phasors and DI names are as per station specific documents
12. Repeat step 1 to 11 for all the PMU's available in the S/S

Expected Result

- 1) Modified configuration parameters are displayed in laptop and displayed in PMU on downloading the same to PMU.
- 2) Phasors displayed in PMU Data Analyzer is as per the modified phasor list
- 3) Final configuration are ensured as per the station specific documents

9583040602.

9500180298

10/10/2018

11/10/2018

POWERGRID CORPORATION OF INDIA LTD.

Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD.	Rev	D
Project	WAMS - PKG-I & II UNDER URTDSM	Date	9.12.2017
MDL Ref	POWERGRID-URTDSM A/GEN-3-600	Page	14 of 20

7.3 Check for auto restoration of PMU on DC power recovery after its failure

To ensure that PMU starts communicating with PMU data analyzer on resumption of power supply and no erroneous command is generated during DC power failure & start up.

1. Set the PMU to UDP multicast spontaneous mode
2. In power on condition, test the PMU communication with PMU data analyzer
3. Monitor the analog phasors, and Digital status with PMU data analyzer
4. Switch OFF main DC supply and check that protocol test tool communication with PMU is stopped.
5. Switch the DC supply ON and check that the PMU has started up without any manual assistance and starts communication with the PMU data analyzer automatically
6. Check that no erroneous command is generated during DC power failure or DC power starts up.
7. Repeat step-1 to 6 for all the PMU's available in the S/S

Expected Result

- 1) Communication established after error resumption of DC fail



Test Result:

☒ PASS ☐ FAIL ☐ NOT TESTED

Customer Representative

ALSTOM Representative

Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD.	Rev	0
Project	WAMS - PKG-I & II UNDER URTDSM	Date	9/12/2015
MDL Ref	POWERGRID-URTDSM-AGEN-3-600	Page	15 of 20

6 Station Level Architecture

This provides the overview of the PMU connectivity to SDH equipment in the substation. Typical architecture for is shown here for reference.

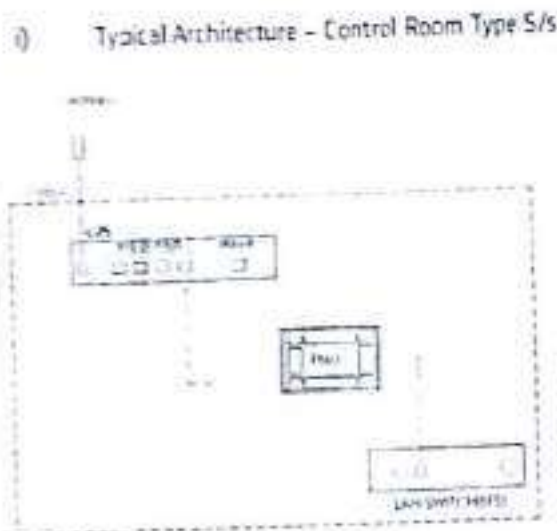
- i) Control Room Type S/s
- ✓ Bay Kiosk Type S/s

Procedure

- 1) Check the Sub-Station Architecture is as per as per Typical Schematic drawing referred in Approved BoQ
- 2) Typical architecture shown below for reference

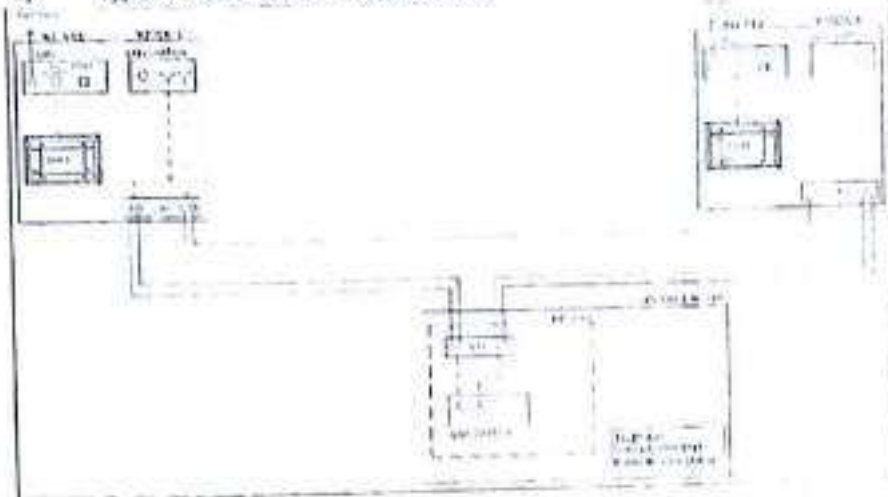
Expected Result

- 1) Connections established as per the Typical architecture referred in the Approved BoQ.



Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD.	Rev	D
Project	WAMS - PKG-I & II UNDER URTDSM	Date	9.12.2015
MDL Ref	POWERGRID-URTDSM-A/GEN-3-600	Page	10 of 20

ii) Typical Architecture - Kiosk Type 5/s



Test Result:

☒ PASS
 ☐ FAIL
 ☐ NOT TESTED

9583040605
 [Signature]
 Customer Representative

9500180880
 [Signature]
 11/10/18
 ALSTOM Representative
 Penra Sony R.

Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD.	Rev	0
Project	WAMS - PKG-I & II UNDER URTDSM	Date	9.12.2015
MDL Ref	POWERGRID-URTDSM-A/GEN-3-600	Page	17 of 20

9 Conclusion

Restore PMU for final setting and put PMU in service. Check for normal load current and voltage phasors values and record as per Annexure-01

 (Signature) (Name) Designation Telephone	Approved & Released for Implementation Date: 11.10.18 Signature: [Signature] Name: [Name] Designation: [Designation]
	Approved & Released for Implementation Date: 11.10.18 Signature: [Signature] Name: [Name] Designation: [Designation]



Test Result:

☒ PASS	☐ FAIL	☐ NOT TESTED
--	-------------------------------	-------------------------------------

9583060405
 11/10/2018
 Customer Representative

11.10.18
 9500180222
 Penya Sany R
 ALSTOM Representative

Title	SITE ACCEPTANCE TEST PROCEDURE		
Customer	POWERGRID CORPORATION OF INDIA LTD.	Rev	0
Project	WAMS - PKG-I & II UNDER URTDSM	Date	3.12.2015
MDL Ref	POWERGRID-URTDSM-A/GEN-3-500	Page	16 of 20

Annexure-01 - PMU Functional Test Report

GPS SI no:	R14303XCCP06A
IP Address:	192.168.0.254

Time Sync Check (HMI):

No of Satellites Visible	GPS locked	LED Status
11	locked	ON

GPS Clock Status check - ☒ OK ☐ NOT OK

PMU ID:	5629	Substation ID:	-
IP Address:	192.168.0.254	Substation Name:	JITPL 400KV Switchyard.

METERING CHECKS Logical PMU-1

Feeder name	Phase Description	Visible at PMU HMI	Visible at PDS/PDC
At Presently Non availability of shutdown CR/PT gnd. for not connected will be connected during shutdown operation of 0.9 shed have. 12/10/2018	A-ph Current (Ia)	/	/
	B-ph Current (Ib)		
	C-ph Current (Ic)		
	Positive Sequence current Magnitude		
	A-ph Voltage (Va)		
	B-ph Voltage (Vb)		
	C-ph Voltage (Vc)		
	Positive Sequence voltage Magnitude		
	MW		
	MVAR		
	Frequency		
	df/dt		

PMU measured values are getting updated in real time ☐ OK ☒ NOT OK

Title	: SITE ACCEPTANCE TEST PROCEDURE		
Customer	: POWERGRID CORPORATION OF INDIA LTD.	Rev	D
Project	: WAMS - PKG-I & II UNDER URTDSM	Date	9.12.2015
MDL Ref.	: POWERGRID-URTDSM-A/GEN-3-600	Page:	19 of 20

METERING CHECKS Logical PMU-2

Feeder name	Phase Description	Visible at PMU HMI	Visible at PDS/PDC
AMUL-1	A -ph Current (Ia)		
	B -ph Current (Ib)		
	C -ph Current (Ic)		
	Positive Sequence current Magnitude		
	A -ph Voltage (Va)		
	B -ph Voltage (Vb)		
	C -ph Voltage (Vc)		
	Positive Sequence voltage Magnitude		
	MW		
	MVAR		
	Frequency		
	dI/dt		
PMU measured values are getting updated in real time ☐ OK ☒ NOT OK			

Note:

Visibility of data at PDS/PDC is subject to availability of link between PMU and PDS/PDC at control center. The test stands completed with the data Visibility at PMU HMI in the absence of link between PMU and PDS/PDC at control center.



Title	: SITE ACCEPTANCE TEST PROCEDURE	Rev	D
Customer	: POWERGRID CORPORATION OF INDIA LTD.	Date	9.12.2015
Project	: WAMS - PKG-I & II UNDER URTDSM	Page:	20 of 20
MDL Ref.	: POWERGRID-URTDSM-A/GEN-3-600		



पावरग्रिड
POWERGRID

ANNEXURE B.2.11.2

पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड

(भारत सरकार का उद्यम)

POWER GRID CORPORATION OF INDIA LIMITED

(A Government of India Enterprise)

Ref: POWERGRID/ODP/ULDC/PMU/

Date: 04/09/2025

To,
HOD (Electrical)
Jindal India Power Ltd.
At/PO-Derang, Dist-Angul
Odisha-759117

Subject: Extension of connection of CT & PT wiring in PMU installed at JITPL-Regarding

Ref:

- i) E-mail correspondence on 30.06.2025
- ii) E-mail correspondence on 30.07.2025

Dear Sir,

As you aware that PMU has been installed by POWERGRID at JITPL under URTDSM package for reporting of real time data to ERLDC. PMU was installed at JITPL and SAT was signed on 12.10.2018 with remark "CT & PT connection will be done by JITPL during opportunity shutdown" (SAT report attached). The optical link from JITPL to Angul has been commissioned and the PMU is reporting to ERLDC since 07.03.2025 only with digital signal. Analog signals are not reporting due to non-extension of CT & PT inputs from JITPL end. Vide e-mail communication mentioned under reference-(i) &(ii), JITPL was informed to extend CT/PT connection to PMU for the reporting of analog data to ERLDC. In reply JITPL has informed that CT/PT connection will be extended during line shut-down. (Mail communication attached)

As commissioning and complete data reporting of JITPL PMU to ERLDC is a critical long pending issue, hence it is requested to extend CT/PT connection to PMU at the earliest.

Thanking You

Yours Faithfully

(K. K. Prusti)

Sr. GM (AM/ULDC)

CC: for kind information please

1. CGM (Odisha)
2. Member Secy, ERPC, Kolkata

**Re: Deployment of Engineer for Commissioning of PMU at JITPL (Angul)**

From Ajay Kumar Sau {अजय कुमार सौ} <ajaysau@powergrid.in>

Date Wed 7/30/2025 10:46 AM

To electrical orissa <electrical_orissa@Jindalgroup.com>

Cc headelectrical orissa <headelectrical.orissa@Jindalgroup.com>; Rohan Prakash {रोहन प्रकाश} <rohanp@powergrid.in>; Pranab Kumar Sahoo {प्रणब कुमार साहू} <pranab.sahoo@powergrid.in>; Ajaya Kumar Sahu {अजय कुमार साहू} <ajaya.sahu@powergrid.in>; K K Prusti {के.के.} <kkp@powergrid.in>

1 attachment (2 MB)

Jindal MOM (1).pdf;

Sir,
With reference to the trailing mail, it is once again requested to kindly confirm the connection of CT & PT wiring in PMU towards commissioning of the same and for applying trail operation certificate to ERLDC.

Regards
A K Sau
ULDC-BBSR
POWERGRID
Mob. No-7008755688

From: Ajay Kumar Sau {अजय कुमार सौ} <ajaysau@powergrid.in>

Sent: Monday, June 30, 2025 3:41 PM

To: electrical orissa <electrical_orissa@Jindalgroup.com>

Cc: headelectrical orissa <headelectrical.orissa@Jindalgroup.com>; Rohan Prakash {रोहन प्रकाश} <rohanp@powergrid.in>; Pranab Kumar Sahoo {प्रणब कुमार साहू} <pranab.sahoo@powergrid.in>; Ajaya Kumar Sahu {अजय कुमार साहू} <ajaya.sahu@powergrid.in>; K K Prusti {के.के.} <kkp@powergrid.in>

Subject: Re: Deployment of Engineer for Commissioning of PMU at JITPL (Angul)

Sir,
With reference to the trailing mail kindly confirm the connection of CT & PT wiring in PMU towards commissioning of the same and for applying trail operation certificate to ERLDC.

Regards
A K Sau
ULDC-BBSR
POWERGRID
Mob. No-7008755688

From: electrical orissa <electrical_orissa@Jindalgroup.com>

Sent: Monday, March 17, 2025 12:29 PM

To: Mamidi Prasad {मामिडी प्रसाद} <mamidi.prasad@powergrid.in>

Cc: headelectrical orissa <headelectrical.orissa@Jindalgroup.com>; Rohan Prakash {रोहन प्रकाश} <rohanp@powergrid.in>; Ajay Kumar Sau {अजय कुमार सौ} <ajaysau@powergrid.in>; Pranab Kumar Sahoo {प्रणब कुमार साहू} <pranab.sahoo@powergrid.in>; Somanath Nayak {एस. नायक} <snayak@powergrid.in>; electrical orissa <electrical_orissa@Jindalgroup.com>

Subject: RE: Deployment of Engineer for Commissioning of PMU at JITPL (Angul)

Dear Sir,

In refence with trailing mail, PMU and panel side(D.I) Breaker status and other signal already connected and remaining CT & PT wiring need to be connected at protection panels end during Line shutdown...Apart from that our SCADA Engineer last 5 days working on the Signal from IEC-101 to IEC-104 protocol .This is present status of IEC 104 communication protocol this FYI and noted.

Thanking You,



O.P.Sridharan

EMD

Jindal India Power Ltd,

M No 7077774083, 7978197349

Derang, Angul Dist, Odisha

Mail: electrical_orissa@jindalgroup.com



From: headelectrical orissa <headelectrical.orissa@Jindalgroup.com>
Sent: 13 March 2025 12:50
To: electrical orissa <electrical_orissa@Jindalgroup.com>
Subject: FW: Deployment of Engineer for Commissioning of PMU at JITPL (Angul)

From: Mamidi Prasad (मामिडी प्रसाद) <mamidi.prasad@powergrid.in>
Sent: 27 February 2025 17:48
To: headelectrical orissa <headelectrical.orissa@Jindalgroup.com>
Cc: Rohan Prakash (रोहन प्रकाश) <rohanp@powergrid.in>; Ajay Kumar Sau {अजय कुमार सौ} <ajaysau@powergrid.in>; Pranab Kumar Sahoo (प्रणब कुमार साहू) <pranab.sahoo@powergrid.in>; Somanath Nayak (एस. नायक) <snayak@powergrid.in>; electrical orissa <electrical_orissa@Jindalgroup.com>
Subject: Deployment of Engineer for Commissioning of PMU at JITPL (Angul)

Dear Sir,

It is known that erection of OPGW link of 400 kV Angul _ Jindal link and communication equipments completed and testing is under progress.

For providing the DEMO to ERLDC towards establishment of above link PMU data/SCDA data is to be provided through above link up to ERLDC.

PMU is already installed at your site. However due to non-availability of communication link ,the PMU is not commissioned.

Now M/s GE engineer will arrive tomorrow to JIPTL site to carry out the following activity as it is more than 6 years that PMU panels were installed.

- To access the conditions of Equipment.
- To Power-on the Equipment.
- To make the connection between PMU Panel to SDH Panel for data reporting to CC.

Name :- Ajit kumar sahou
 Aadhar no :- 2348-8419-7524

Please find attached herewith the SAT of JIPTL .As informed, due to non-availability of shutdown, the CT/PI connection were pending at the time of signing of the SAT. It is mentioned in the STA report that ,JIPTL official has agreed to complete the Interfacing of CT/PI during shutdown. Please intimate the status of the same.

In view of the above it is requested to allow the above GE engineer at your site for further completion of PMU Commissioning works.

अभिवादन/Regards

एम. प्रसाद /M.Prasad

उप महाप्रबंधक (ए.एम.), मुख्यालय - ओडिशा परियोजनाएँ

Deputy General Manager (AM), HQ, Odisha Projects

पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड

POWER GRID CORPORATION OF INDIA LTD

प्लॉट नं. 4 | यूनिट - 41 | नीलाद्रि विहार | चन्द्रशेखरपुर | भुवनेश्वर - 751 021 | ओडिशा | इंडिया

Plot no.4 | Unit-41 | Niladri Vihar | Chandrasekharpur | Bhubaneswar-751 021 | Odisha | INDIA

मोबाइल: +91 94379 62181 /8018780410 | ई-मेल: mamidi.prasad@powergrid.in

Mobile: +91 94379 62181/8018780410 | E-mail: mamidi.prasad@powergrid.in

<https://apps.powergrid.in/gridcon>

दावात्याग : यह ईमेल पावरग्रिड के दावात्याग नियम व शर्तों द्वारा शासित है जिसे <http://apps.powergrid.in/Disclaimer.htm> पर देखा जा सकता है। Disclaimer: This e-mail is governed by the Disclaimer Terms & Conditions of POWERGRID which may be viewed at <http://apps.powergrid.in/Disclaimer.htm>



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<http://www.jindalgroup.com/disclaimer.html>

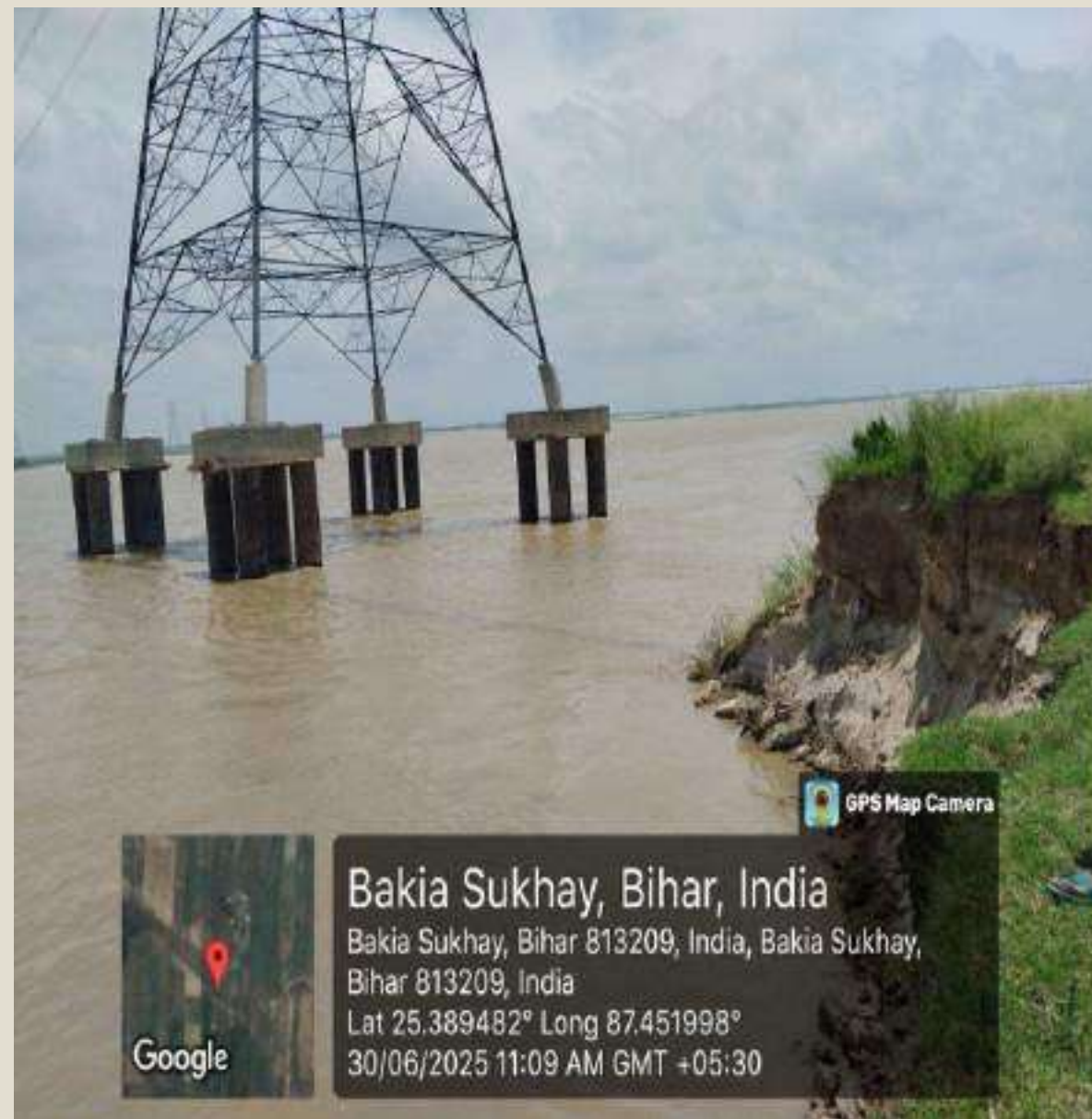
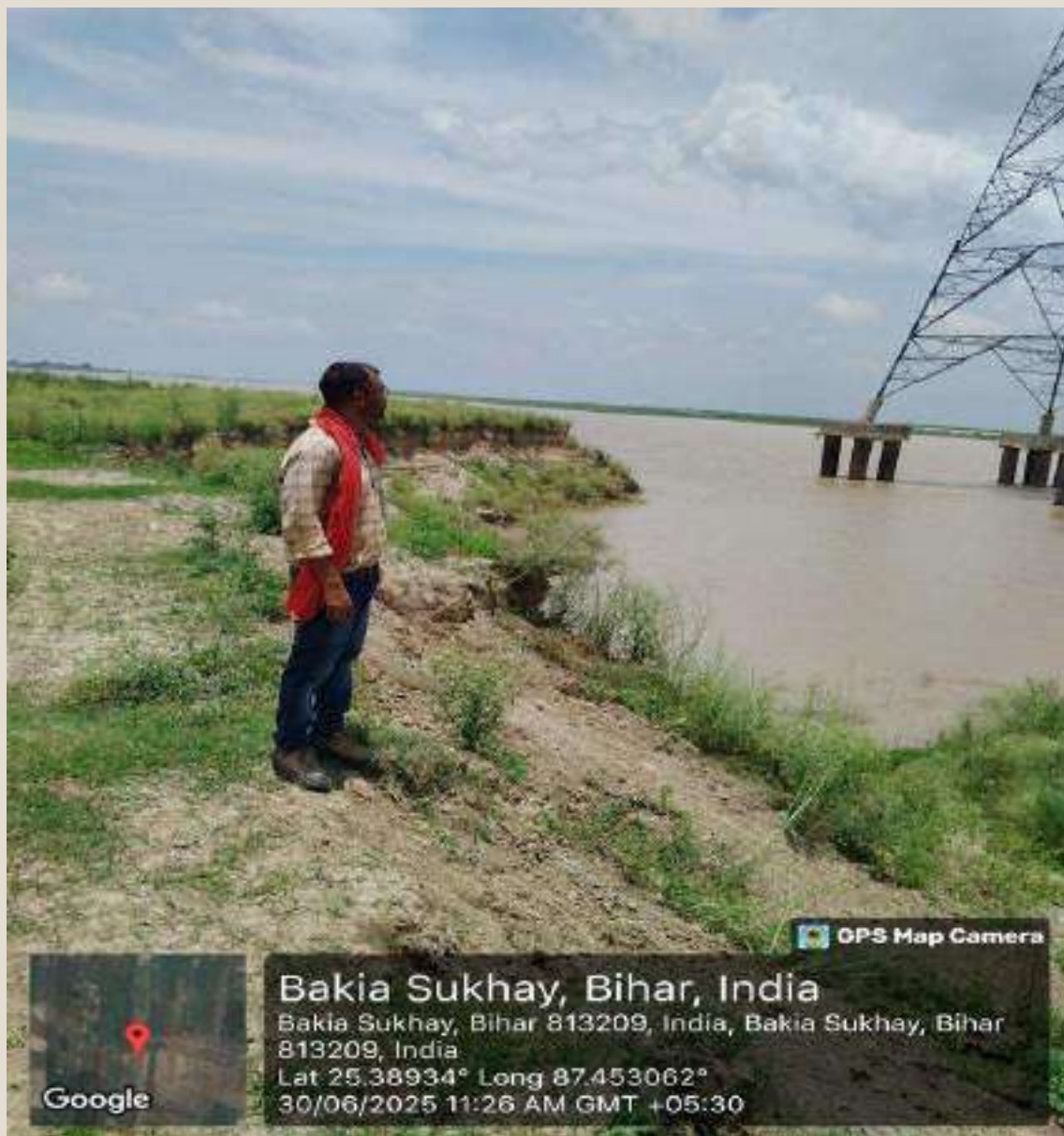
ANNEXURE B.2.14

**VULNERABLE LOCATION NO – 1100
(DD+0) OF 400KV D/C NEW PURNEA-
RAJARHAT LINE NEAR PIRPAINTI,
BHAGALPUR, BIHAR**

Tower No.1100, before monsson as on 30.06.2025

Sketch drawing of Tower no 1101 to 1100









After water
receded on
22.08.2025

Location 1101
was 310 mtr
from river bank





As on
06.09.2025

Location 1100
was 90 mtr
from river bank



GPS Map Camera

Bakia Sukhay, Bihar, India

9fp4+g58, Bakia Sukhay, Bihar 813209, India
Lat 25.386984° Long 87.45275°
06/09/2025 04:55 PM GMT +05:30



As on
13.09.2025

Location
1100 was 42
mtr from
river bank



As on
20.09.2025

De-stringing
of conductors
towards
location 1099
side
completed



THANKS

Sharing AMR System SEM data to SLDC, Utilities

Annexure B.2.21

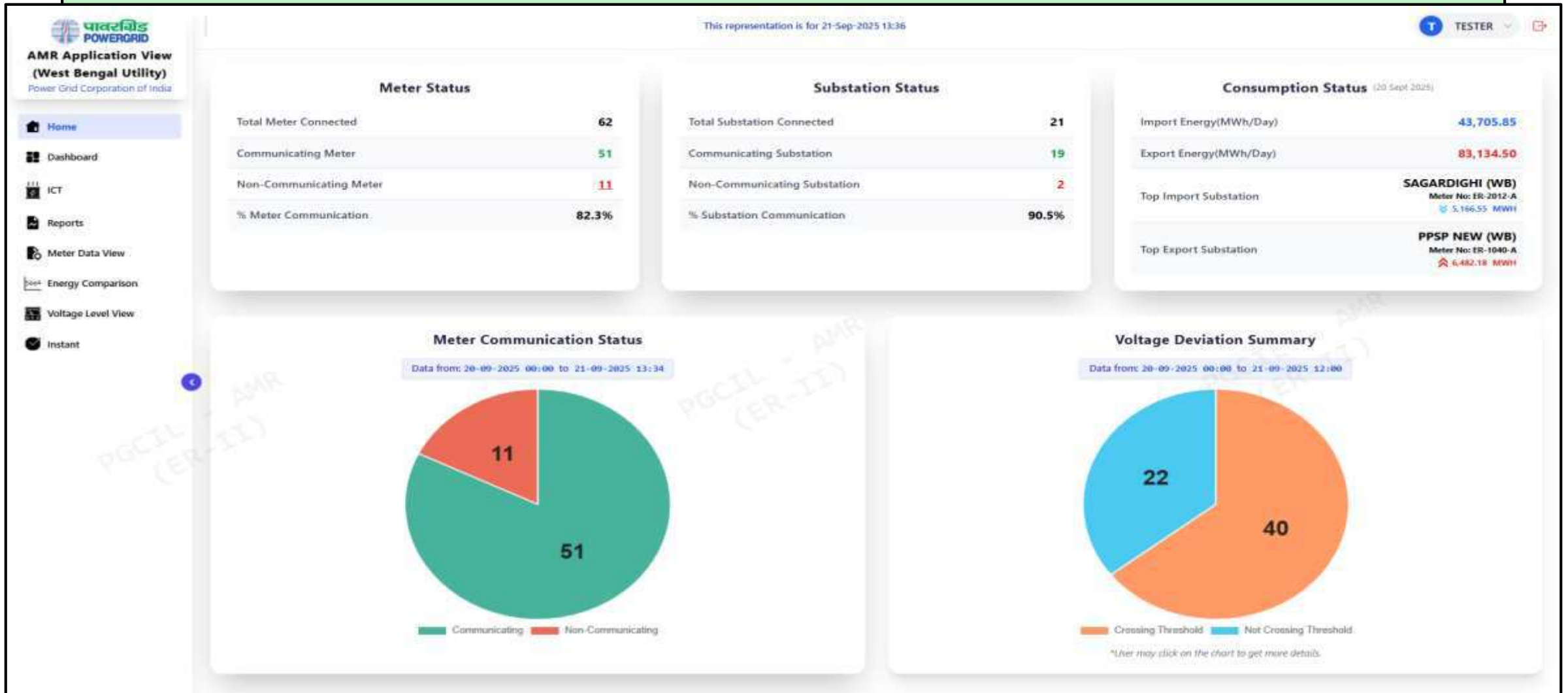
A screenshot of a login interface for PGCIL-AMR. At the top is the PGCIL logo, which consists of a blue circle with a stylized 'P' and the text 'पावरग्रिड POWERGRID' in green and blue. Below the logo is the text 'Login to PGCIL-AMR'. There are two input fields: 'Username' and 'Password'. Below these fields is a blue 'Login' button. At the bottom right of the login area is a link that says 'Forgot Password?'. The background of the slide shows a power line tower and a sunset.

**A Snap of the Application – for WBSLDC
(pilot project)**

Project Background vis-à-vis Application Development

- *Approved in 54th ERPC meeting. WBSLDC considered as pilot utility.*
- *New application development cycle is in progress.*
- *Tie-lines Feeders, opposite end feeders along with ICT Feeders are integrated.*
- *15 mins block-wise data, 01 min instant data (as per feasibility) are integrated.*
- *Dashboards, Reports, Meter Data-view modules are major features of the application.*
- *PGCIL, WBSLDC & TCS have several joint meetings during the development followed by demonstration.*

WBSLDC Application- Snapshots



WBSLDC Application- Snapshots

**AMR Application View**
(West Bengal Utility)
Power Grid Corporation of India

[Home](#)
[Dashboard](#)
[ICT](#)
[Reports](#)
[Meter Data View](#)
[Energy Comparison](#)
[Voltage Level View](#)
[Instant](#)

This representation is for 21-Sep-2025 13:33

TESTER

Substations List

Green: All meters of the substation are communicating Yellow: Partial meters of the substation are communicating Red: All meters of the substation are non-communicating

ALIPURDUAR (WB)

BIDHANNAGAR (WB)

BIRPARA (WB)

DALKHOLA (WB)

FALAKATA(WB)

GAZOLE (WB)

GOKARNA (WB)

JEERAT (WB)

KALIMPONG (WB)

KHARAGPUR (WB)

KLC BANTALA (WB)

KURSEONG (WB)

MALDA (WB)

MANIKCHAK (WB)

NBU (WB)

NEWTOWN (WB)

PPSP NEW (WB)

RAMMAM (WB)

SAGARDIGHI (WB)

SANTALDIH (WB)

SUBHASHGRAM (WB)

Substations List / SUBHASHGRAM (WB)

[Export](#) [Import](#) [Refresh](#) [Close](#)

Meter SNo.	Feeder Name	Meter Type	Substation Type	Meter Make	Energy Total (MWh)	Last LS Data	Last Midnight Data	Status
ER-1003-A	220 KV SUBHASHGRAM(WB)-SUBHASGRAM(PG)-1	STANDBY	STATE-TRANSMISSION	GENUS_15	2128.51	21-Sep-25 09:30 AM	21-Sep-25 12:00 AM	✓
NS-2942-A	220 KV SUBHASHGRAM(PG)-SUBHASGRAM(WBSETCL)-1	STANDBY	TRANSMISSION UTILITY	SECURE	2130.62	21-Sep-25 10:45 AM	21-Sep-25 12:00 AM	✓
ER-1127-A	220 KV SUBHASHGRAM(WB)-SUBHASGRAM(PG)-2	STANDBY	STATE-TRANSMISSION	GENUS_15	1836.38	21-Sep-25 09:30 AM	21-Sep-25 12:00 AM	✓
NS-2937-A	220 KV SUBHASHGRAM(PG)-SUBHASGRAM(WBSETCL)-2	STANDBY	TRANSMISSION UTILITY	SECURE	2451.58	21-Sep-25 10:45 AM	21-Sep-25 12:00 AM	✓

*Above meters are shown alternatively with opposite-end feeder details listed in yellow strips
*Energy column shows previous day's total consumption.

WBSLDC Application- Snapshots



WBSLDC Application- Snapshots

**पावरग्रिड
POWERGRID**

AMR Application View
(West Bengal Utility)
Power Grid Corporation of India

Home

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Energy Comparison

Voltage Level View

Instant

This representation is for 21-Sep-2025 14:04

T TESTER

Meter Data View / ER-1127-A / LS_15

GeneralLS_15Midnight

Meter SNo. ER-1127-AFeeder Name 220 KV SUBHASHGRAM(WB)-SUBHASHGRAM(PG)-2MF3200000Start 09/20/2025End 09/21/2025SubmitCTR: 1696/L PTR: 220/110

Export Excel

Time (LS Date)	Meter SNo.	Wh_net	Wh(I)	Wh(E)	kVAh	kVArh_Ig_kWh(I)	kVArh_Id_kWh(E)	kVArh_Ig_kWh(E)	kVArh_Id_kWh(I)	VR	VY	VB	IR	IY	IB
21-Sep-2025 09:30	ER-1127-A	-14.74	0	14.74	0	0	0.09	0.16	0	64.6	65.2	65	0.3	0.3	0.306
21-Sep-2025 09:15	ER-1127-A	-14.11	0	14.11	0	0	0.1	0.16	0	64.7	65.3	65.1	0.287	0.286	0.292
21-Sep-2025 09:00	ER-1127-A	-13.92	0	13.92	0	0	0.13	0.11	0	64.7	65.5	65.3	0.282	0.282	0.288
21-Sep-2025 08:45	ER-1127-A	-14.36	0	14.36	0	0	0.53	0.03	0	64.9	65.6	65.4	0.291	0.291	0.296
21-Sep-2025 08:30	ER-1127-A	-13.6	0	13.6	0	0	0.65	0	0	65.2	65.8	65.6	0.274	0.274	0.281
21-Sep-2025 08:15	ER-1127-A	-12.94	0	12.94	0	0	0.65	0	0	65.4	66	65.7	0.26	0.261	0.266
21-Sep-2025 08:00	ER-1127-A	-13.11	0	13.11	0	0	0.79	0	0	65.3	66.1	65.8	0.261	0.265	0.27
21-Sep-2025 07:45	ER-1127-A	-14.2	0	14.2	0	0	1.11	0	0	65.3	66	65.7	0.285	0.288	0.292
21-Sep-2025 07:30	ER-1127-A	-12.81	0	12.81	0	0	1	0	0	65.4	66	65.7	0.258	0.259	0.264
21-Sep-2025 07:15	ER-1127-A	-11.91	0	11.91	0	0	0.9	0	0	65.5	66.1	65.8	0.239	0.238	0.247
21-Sep-2025 07:00	ER-1127-A	-11.59	0	11.59	0	0	0.95	0	0	65.5	66.1	65.9	0.232	0.232	0.24
21-Sep-2025 06:45	ER-1127-A	-11.74	0	11.74	0	0	0.81	0	0	65.6	66.2	65.9	0.234	0.234	0.243

Prev12345...12Next

Showing 1 - 12 of 135

WBSLDC Application- Snapshots

**पावरग्रिड**
POWERGRID

AMR Application View
(West Bengal Utility)
Power Grid Corporation of India

Home

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Meter Data View

Energy Comparison

Voltage Level View

Instant

This representation is for 21-Sep-2025 21:37

TESTER

ICT Meters (765KV / 400KV / 220KV)

ALIPURDUAR

BINAGURI

BIRPARA

DURGAPUR

MALDA

MEDINIPUR

MEDINIPUR - ICT Meter Details

Export Import

SNo	Meter SNo	Feeder Name	Meter Type	Type	Make	Energy Total (MWh)	Last LS Data	Last MIDN Data	Status
1	NS-3301-A	765 KV SIDE OF MEDINIPUR(PG) 765/400KV 1500 MVA ICT-1	MAIN	TRANSMISSION UTILITY	SECURE	4087.72	21-Sep-25 06:30 PM	21-Sep-25 12:00 AM	✓
2	NS-3334-A	765 KV SIDE OF MEDINIPUR(PG) 765/400KV 1500 MVA ICT-2	MAIN	TRANSMISSION UTILITY	SECURE	4082.50	21-Sep-25 06:30 PM	21-Sep-25 12:00 AM	✓
3	NS-3298-A	400 KV SIDE OF MEDINIPUR(PG) 765/400KV 1500 MVA ICT-2	STANDBY	TRANSMISSION UTILITY	SECURE	-4083.42	21-Sep-25 06:30 PM	21-Sep-25 12:00 AM	✓
4	NS-3308-A	400 KV SIDE OF MEDINIPUR(PG) 765/400KV 1500 MVA ICT-1	STANDBY	TRANSMISSION UTILITY	SECURE	-2207.35	21-Sep-25 11:30 AM	21-Sep-25 12:00 AM	✓
5	ER-1686-A	765 KV MEDINIPUR(PG)-JEERAT(WB) LINE-1	STANDBY	TRANSMISSION UTILITY	GENUS_15	1166.43	21-Sep-25 02:30 AM	21-Sep-25 12:00 AM	✓
6	ER-1684-A	765 KV MEDINIPUR(PG)-JEERAT(WB) LINE-2	STANDBY	TRANSMISSION UTILITY	GENUS_15	0.00	06-Sep-25 10:30 PM	06-Sep-25 12:00 AM	✗

NEW JEERAT

RAJARHAT

WBSLDC Application- Snapshots

**AMR Application View**
(West Bengal Utility)
Power Grid Corporation of India

Home

Dashboard

ICT

Reports

Meter Data View

Energy Comparison

Voltage Level View

Instant

This representation is for 21-Sep-2025 13:39

TTESTER

Voltage Level View

09/14/202509/21/2025SubmitSearch Meter SNo./I

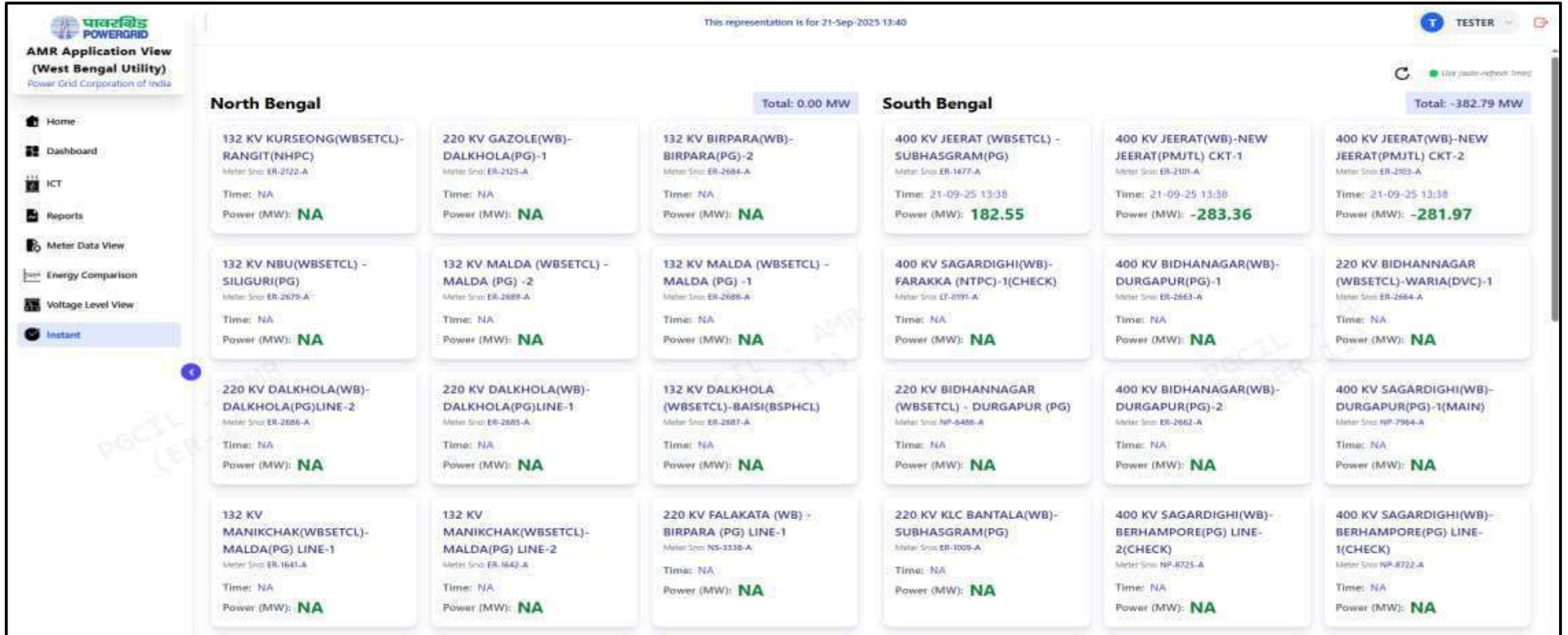
Calculations used:
Max Volt.(Primary)(KV) = $\sqrt{3} \times \text{MCC}(V_{ph}) \times V_{L2} \times \text{PTR}$
Permissible Voltage (KV) = $V_{max} \times 1.05$

S.No	All Substations	Meter SNo.	Feeder Name	Max Volt.(Primary) (KV)	Permissible Voltage (KV)	Difference (in KV)	Deviation (in %) ± 1	Occurrence Timestamp ± 1
1	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	137.59	135.96	1.63	1.20	21-09-25 12:00
2	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	137.80	135.96	1.84	1.35	21-09-25 11:45
3	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	137.80	135.96	1.84	1.35	21-09-25 11:30
4	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	137.80	135.96	1.84	1.35	21-09-25 11:15
5	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	138.01	135.96	2.05	1.50	21-09-25 11:00
6	BIRPARA (WB)	ER-2684-A	132 KV BIRPARA(WB)-BIRPARA(PG)-2	137.63	135.96	1.67	1.23	21-09-25 10:45
7	SAGARDIGHI (WB)	ER-1985-A	400 KV SAGARDIGHI(WB)- DURGAPUR(PG)-2(MAIN)	412.40	412.00	0.40	0.10	21-09-25 10:45
8	SAGARDIGHI (WB)	ER-2889-A	400 KV SAGARDIGHI(WB)- DURGAPUR(PG)-2(CHECK)	412.28	412.00	0.28	0.07	21-09-25 10:45
9	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	138.01	135.96	2.05	1.50	21-09-25 10:45
10	SAGARDIGHI (WB)	ER-2899-A	400 KV SAGARDIGHI(WB)-DURGAPUR(PG)-1(CHECK)	413.54	412.00	1.54	0.37	21-09-25 10:45
11	SAGARDIGHI (WB)	ER-1985-A	400 KV SAGARDIGHI(WB)- DURGAPUR(PG)-2(MAIN)	412.85	412.00	0.85	0.21	21-09-25 10:30
12	BIRPARA (WB)	ER-2684-A	132 KV BIRPARA(WB)-BIRPARA(PG)-2	137.82	135.96	1.86	1.37	21-09-25 10:30
13	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	138.01	135.96	2.05	1.50	21-09-25 10:30
14	SAGARDIGHI (WB)	ER-2889-A	400 KV SAGARDIGHI(WB)- DURGAPUR(PG)-2(CHECK)	412.72	412.00	0.72	0.17	21-09-25 10:30
15	SAGARDIGHI (WB)	ER-2899-A	400 KV SAGARDIGHI(WB)-DURGAPUR(PG)-1(CHECK)	414.04	412.00	2.04	0.50	21-09-25 10:30
16	SAGARDIGHI (WB)	ER-1985-A	400 KV SAGARDIGHI(WB)- DURGAPUR(PG)-2(MAIN)	413.16	412.00	1.16	0.28	21-09-25 10:15
17	BIRPARA (WB)	ER-2684-A	132 KV BIRPARA(WB)-BIRPARA(PG)-2	137.94	135.96	1.98	1.46	21-09-25 10:15
18	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	138.01	135.96	2.05	1.50	21-09-25 10:15

Showing 1 - 18 of 10288

Prev12345...572Next

WBSLDC Application- Snapshots



Visit to WBSLDC office- for application demo



THANK YOU

CENTRAL ELECTRICITY REGULATORY COMMISSION
New Delhi

Petition No. 309/TT/2023
alongwith I.A. No. 87/2023

Coram:

Shri Jishnu Barua, Chairperson
Shri Ramesh Babu V., Member
Shri Harish Dudani, Member

Date of Order: 29.05.2025

In the matter of:

Petition under Section 62 read with Section 79 of the Electricity Act, 2003 along with Regulations 111 to 113 of the Central Electricity Regulatory Commission (Conduct of Business) Regulations, 1999, for determination of transmission tariff of the deemed inter-State transmission lines owned by the West Bengal State Electricity Transmission Company Limited with the transmission system operated by the Central Transmission Utility of India Limited in terms of the Commission's order dated 5.9.2018 in Petition No. 7/Suo-Motu/2017, for inclusion in PoC transmission charges for the period FY 2014-15 to FY 2018-19 in accordance with the Central Electricity Regulatory Commission (Terms And Conditions of Tariff) Regulations, 2014.

And in the matter of:

West Bengal State Electricity Transmission Company Limited,
Vidyut Bhavan, Bidhannagar,
Block-DJ, Sector-II,
Kolkata-700091

...Petitioner

Vs.

- 1. Central Transmission Utility of India Limited,**
B-9, Qutab Institutional Area Road,
NRPC Colony, Block B,
Katwaria Sarai,
New Delhi-110019
- 2. Powergrid Corporation of India Limited,**
B-9, Qutab Institutional Area,
Katwaria Sarai, South Delhi,
Delhi-110016

...Respondents

Parties present : Ms. Molshree Bhatnagar, Advocate, WBSETCL
Shri Nimesh Jha, Advocate, WBSETCL
Ms. Shaide Das, Advocate, WBSETCL
Ms. Kavya Bharadwaj, CTUIL

approved in this order for the transmission assets shall be recovered in accordance with the applicable Sharing Regulations as per Regulation 43 of the 2014 Tariff Regulations and Regulation 57 of the 2019 Tariff Regulations. Further, the transmission charges allowed in this order shall be adjusted against the ARR approved by the State Commission.

63. In view of the above, we observe that once the transmission charges of non-ISTS lines are included in the ISTS pool, the availability of such lines needs to be verified by the respective RPCs, and the recovery of tariff should be linked to their availability. To facilitate this, the necessary mechanisms may be put in place by the RPCs. Accordingly, we direct that the YTC of such intra-State lines shall be included in the PoC Pool, subject to certification of each line's availability by the WRPC, in accordance with the provisions under the 2014 Tariff Regulations and the 2019 Tariff Regulations, as applicable.

64. We further direct the Petitioner to approach the State Electricity Regulatory Commission for the adjustment of such recovery against the ARR of the respective years of the Petitioner.

Filing Fee and Publication Expenses

65. The Petitioner has claimed reimbursement of the fee paid by it for filing the Petition and the publication expenses. The Petitioner shall be entitled to reimbursement of the filing fees and publication expenses in connection with the present Petition directly from the beneficiaries on a pro-rata basis in accordance with Regulation 52 of the 2014 Tariff Regulations.

Licence Fee and RLDC fees and Charges

66. The Petitioner has requested to allow the Petitioner to bill and recover License fee and RLDC fees and charges, separately from the Respondents. The Petitioner

**CENTRAL ELECTRICITY REGULATORY COMMISSION
New Delhi**

Petition No. 324/TT/2023

Coram:

**Shri Jishnu Barua, Chairperson
Shri Ramesh Babu V., Member
Shri Harish Dudani, Member**

Date of Order: 31st May, 2025

In the matter of:

Petition under Section 62 read with Section 79 of the Electricity Act, 2003 along with Regulations 111-113 of the Central Electricity Regulatory Commission (Conduct of Business) Regulations, 1999, for determination of the transmission tariff of the deemed inter-State transmission lines owned by West Bengal State Electricity Transmission Company Limited with the transmission system operated by Central Transmission Utility of India Limited, for inclusion in PoC transmission charges for the period FY 2019-20 to FY 2023-24 under the Central Electricity Regulatory Commission (Terms and Conditions of Tariff) Regulations, 2019.

And in the matter of:

**West Bengal State Electricity Transmission Company Limited,
Vidyut Bhavan, Bidhannagar,
Block- DJ, Sector- II,
Kolkata-700091.**

...Petitioner

Vs.

- 1. Central Transmission Utility of India Limited,
B-9, Qutab Institutional Area Road,
NRPC Colony, Block B,
Katwaria Sarai,
New Delhi-110019.**
- 2. Power Grid Corporation of India Limited,
B-9, Qutab Institutional Area
Katwaria Sarai, South Delhi,
Delhi-110016.**

...Respondents

Parties present : Ms. Molshree Bhatnagar, Advocate, WBSETCL
Shri Nimesh Jha, Advocate, WBSETCL
Ms. Shaida Das, Advocate, WBSETCL
Ms. Kavya Bharadwaj, CTUIL
Shri Ishrat Ali, CTUIL



transmission systems was governed by the provisions of the 2010 Sharing Regulations. However, with effect from 1.11.2020, the 2010 Sharing Regulations have been repealed, and the sharing of transmission charges is governed by the provisions of the 2020 Sharing Regulations. Accordingly, the transmission charges approved in this order for the transmission assets shall be recovered in accordance with the applicable Sharing Regulations as per Regulation 43 of the 2014 Tariff Regulations and Regulation 57(2) of the 2019 Tariff Regulations. Further, the transmission charges allowed in this order shall be adjusted against the ARR approved by the State Commission.

43. We observe that once the transmission charges of non-ISTS lines are included in the ISTS pool, the availability of such lines needs to be verified by the respective RPC, and recovery of tariff should be linked with its availability, for which necessary mechanisms may be put in place by the RPC. We direct that YTC of such intra-State lines shall be included in the PoC Pool based on the availability of each of the lines to be certified by the ERPC in terms of the provisions under the 2014 Tariff Regulations and the 2019 Tariff Regulations, as applicable.

44. We further direct the Petitioner to approach the WBERC for the adjustment of such recovery against the ARR of the respective years of the Petitioners.

45. To summarize, the Annual Fixed Charges (AFC) allowed in respect of the transmission asset for the 2019-24 tariff period in this order are as follows:

		(₹ in lakh)				
AFC	Line Length (Ckm)	2019-20	2020-21	2021-22	2022-23	2023-24
Asset-1 (Claimed)*	98.6	172.39	172.10	171.80	171.64	171.61
Asset-1A (Allowed)	94.726	215.51	216.98	218.47	220.34	222.90
Asset-1B (Allowed)	3.874	34.70	33.42	32.37	31.64	31.07



सेंट्रल ट्रांसमिशन यूटिलिटी ऑफ इंडिया लिमिटेड

(पवर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)

(भारत सरकार का उद्यम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.

(A wholly owned subsidiary of Power Grid Corporation of India Limited)

(A Government of India Enterprise)

Ref: CTU/BCD/AVC/01

Date: 26.08.2025

To,

As per distribution list

Sub: Availability Certification of deemed ISTS lines/natural ISTS lines of STUs.

Sir,

Central Transmission Utility of India Ltd., as CTU, has been mandated for Billing, Collection and Disbursement of ISTS charges as per provisions of CERC Sharing Regulations, 2020. Billing for ISTS charges includes transmission lines of ISTS Licensees and intra-state lines considered as deemed ISTS lines/natural ISTS lines of STUs. Hon'ble CERC is determining tariff for such lines of STUs as per provisions of CERC Tariff Regulations applicable from time to time and CERC Suo-moto order 15/SM/2012 dtd. 14.03.2012 & 07/SM/2017 dtd 12.05.2017.

CERC while approving tariff in following tariff orders observed that once transmission charges of non-ISTS lines are included in the ISTS pool, the availability of such lines needs to be verified by the respective RPC and recovery of tariff should be linked with their availability, for which necessary mechanisms may be put in place by the RPCs. CERC further directed that YTC of such intra-State lines shall be included in the PoC Pool based on the availability of each of the lines as certified by the RPCs in terms of the provisions under the 2014 Tariff Regulations and the 2019 Tariff Regulations, as applicable.

Sr. No.	STU	Pet. No.	Tariff Block	Order date	Concerned RPC
1	WBSETCL	324/TT/2023	2019-24	31.05.2025	ERPC
2	WBSETCL	309/TT/2023	2014-19	29.05.2025	ERPC
3	TG Transco (Telangana)	330/TT/2022	2019-24	01.08.2024	SRPC
4	RRVPL	112/TT/2017	2014-19	02.08.2024	NRPC
5	RRVPL	215/TT/2017	2014-19	02.08.2024	NRPC
6	RRVPL	212/TT/2022	2019-24	20.03.2024	NRPC
7	OPTCL	288/TT/2023	2014-19	26.02.2025	ERPC
8	MSSETCL	82/TT/2022	2014-19 & 2019-24	11.11.2024	WRPC
9	MPPTCL	329/TT/2022	2019-24	30.04.2024	WRPC
10	MPTCL	73/TT/2024	2014-19 & 2019-24	28.04.2025	WRPC
11	KPTCL	353/TT/2023	2019-24	07.05.2025	SRPC

Further, Clause 5 of Regulation 93 of Tariff Regulations, 2024 applicable for 2024-29 Tariff block also provides that Tariff of Non-ISTS Lines carrying Inter-State Power shall be approved based on provisions of these Regulations, and the fixed charges of such system shall be allowed based on the availability as certified by respective RPCs and shall be allowed to be recovered as per the mechanism specified in CERC (Sharing of Inter-State Transmission Charges and Losses), 2020.

In the absence of availability certificate of intra-state/natural ISTS lines of STUs, CTU is including transmission charges of intra-state/natural ISTS lines in ISTS pool on normative availability basis i.e. without any incentive/penalty, however, in light of above CERC orders and provisions of Tariff Regulations, 2024,

Office : 5th -10th Floor, Plot No. 16, Icon International Tower-I, Sector 32, Gurugram, Haryana- 122003

Registered Address : 2nd Floor, Plot No -2, Sector 29, Gurugram, Haryana-122001

कार्यालय : 5^{वीं} -10^{वीं} मंजिल, प्लॉट नंबर 16, इकॉन इंटरनेशनल टॉवर -1, सेक्टर 32, गुरुग्राम, हरियाणा- 122003

पंजीकृत कार्यालय : दूसरी मंजिल, प्लॉट नंबर -2, सेक्टर 29, गुरुग्राम, हरियाणा-122001

recovery of ISTS charges to STUs is subject to the availability of transmission system as certified by RPCs. Any under recovery /over recovery on account of certified transmission system availability needs to be adjusted in ISTS pool to comply the CERC directions/Regulations.

In view of the above, all STUs & DVC, whose intra-state lines/natural ISTS lines are included in ISTS system as per CERC Sharing Regulations, 2020, are requested to obtain availability certificates of their intra-state lines/natural ISTS lines from respective RPCs as directed by CERC in tariff orders/provisions of Tariff Regulations, 2024 and submit the same to CTU for necessary adjustment in ISTS pool.

Further, in some cases of intra-state/natural ISTS lines of STUs, where tariff has been approved by CERC for 2014-19 and 2019-24 tariff block and as per approved tariff, amount over recovered needs to be refunded in ISTS pool, therefore, STUs are requested to refund the over recovered amount in ISTS pool immediately.

STUs & DVC are also requested to file petition for determination of tariff for 2024-29 tariff block on priority basis and intimate the status of petition filing to CTU.

Thanking you,

Yours faithfully,



(V. C. Sekhar)
Sr. GM (Commercial)

Distribution list:

1. Rajasthan Rajya Vidyut Prasaran Nigam Ltd.
Vidyut Bhawan, Jan Path, Jyothi Nagar, Lal Koti, Jaipur,
Rajasthan 302005
2. Tamil Nadu Transmission Corporation Limited
NPKRR Manligai
144, Annasalai, Chennai 600002
Tamil Nadu, India
3. Madhya Pradesh Power Transmission Co. Ltd.
Block No.2, Shakti Bhawan, Rampur,
Jabalpur 482 008 (M.P.)
4. Chhattisgarh State Power Transmission Company Ltd
O/o The Chief Engineer (ETC), Energy Info Tech Centre
Daganiya, Raipur (CG) - 492013
5. Delhi Transco Limited
Shakti Souda, Kotla Marg,
New Delhi-110002
6. Karnataka Power Transmission Corporation Limited
Corporate Office, Kaveri Bhavan
K. G. Road, Bengaluru-560009
7. Himachal Pradesh Power Transmission Corporation Ltd
Himfed Bhawan, New ISBT Roa, Panjari (Below Old MLA Quarters),
Shimla -171005
8. Transmission Corporation Of Andhra Pradesh (APTRANSCO)
Vidyut Soudha, Gunadala, Eluru Rd.,
Vijayawada, Andhra Pradesh 520004
9. Damodar Valley Corporation
DVC HEADQUARTERS DVC Towers, VIP Road,
Kolkata-700054
10. Odisha Power Transmission Corporation Limited
9th Floor, TECH Tower Building, Saheed Nagar,
Bhubaneswar, Odisha-751007
11. Transmission Corporation of Telangana Limited (TG TRANSCO),
Vidyut Soudha, Khairatabad, Hyderabad-500082

12. Power Transmission Corporation of Uttarakhand Limited
Vidyut Bhawan, Near ISE Crossing,
Saharanpur Road, Majra, Dehradun-248002
13. Managing Director
UP Power Transmission Corporation Ltd.
7th Floor, Shakti Bhawan, 14-Ashok Marg,
Lucknow-226001 (Uttar Pradesh)
14. Chief Engineer (STU)
Maharashtra State Electricity Transmission Company Limited
18, Private Road, E Block BKC, Bandra Kurla Complex,
Bandra East, Mumbai - 400051, Maharashtra
15. Office of The Director(Transmission) ,
Meghalaya Power Transmission Corporation Limited
Short Round Road, Lum Jingshai,
Shillong, Pin Code-793001(Meghalaya)
16. Chief Engineer (TRAC)
KSEB Ltd.
Vidyuthi Bhavanam, Pattom Palace,
P.O- Thiruvanthapuram-695004 (Kerala)
17. Chief Engineer (Central Planning Deptt.),
West Bengal State Electricity Trans. Co. Ltd.
9th Floor, Vidyut Bhawan, Block-DJ, Sector-II, Bidhannagar,
Kolkata -700 091 (West Bengal)
18. Chief Engineer (SO & Commercial)
Haryana Vidyut Prasaran Nigam Limited (HVPNL)
Shakti Bhawan, Sector-6, Panchkula- 134109 (Haryana)
19. Chief General Manager (LAR)
Assam Electricity Grid Corporation Limited
Bijulee Bhavan, Paltan Bazar,
Guwahati- 781001 (Assam)



ଓଡ଼ିଶା ବିଦ୍ୟୁତ୍ ଶକ୍ତି ସଂଚାରଣ ନିଗମ ଲି.8. ODISHA POWER TRANSMISSION CORPORATION LIMITED

(A Government of Odisha Undertaking)

Regd. Office: Janpath: Bhubaneswar-751007.

Telephone: (0674) 2540051 (EPABX), Website: www.optcl.co.in

CIN: U4102OR2004SGC007553

No. RT&C-05/2023

330⁽⁴⁾

Date 12.09.2025

To,

The Member Secretary
Eastern Regional Power Committee
14, Golf Club Road,
Tollygunge, Kolkata-700033

Sub: Certification of non-ISTS lines of OPTCL carrying ISTS Power for FY 2024-2029.

Ref: 1) Your letter no ERPC/COM-I/NON-ISTS/2018/5246 dated 12.10.2018.

2) CERC order dated 26.02.2025 in Petition No. 288/TT/2023.

3) CTU letter no CTU/BCD/AVC/01 dated 26.08.2025.

Sir,

With reference to the subject cited above, we would like to bring to your kind attention that, upon our request based on the requirement of CERC, the following 9nos. Non-ISTS lines carrying ISTS power were certified by ERPC for the period 2014-2019.

Sl. No	Line Name	Voltage Level	Connecting Status	Ckt. Kms	Type Conductor	DOCO
1	Indravati-Indravati PG SC	400KV	Odisha, AP	3.970	Twin ACSR Moose	1999
2	Rengali-Keonjhar- SC	400KV	Odisha, WB	115.530	Twin ACSR Moose	1995
3	Keonjhar-Baripada-SC	400KV	Odisha, WB	104.243	Twin ACSR Moose	1995
4	Baripada-Kharagpur (Up Odisha Border)-SC	400KV	Odisha, WB	21.727	Twin ACSR Moose	1995
5	Jaynagar-PGCIL-DC	220KV	Odisha, CTU	15.460	ACSR Zebra	1990

6	Rengali–Rengali PGCIL-DC	220KV	Odisha, CTU	2.000	ACSR Zebra	1997
7	Balimela PH–Upper Sileru	220KV	Odisha, AP	24.760	ACSR Zebra	1982
8	Joda–JSPL SC	220KV	Odisha, JSEB	14.110	ACSR Zebra	1984
9	Joda–Kenduposi SC	132KV	Odisha	49.900	ACSR	1985

CTU vide letter dated 26.08.2025 has directed all STUs to file petitions for determination of tariff for the FY 2024–29 tariff block on a priority basis and to intimate the status of petition to CTU.

Regulation 93 of the CERC (Terms and Conditions of Tariff) Regulations, 2024 extracted below.

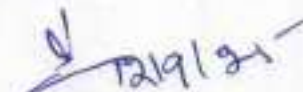
"Existing Intra-state transmission lines other than Natural ISTS lines, as certified by CEA based on the **recommendations** of the STU and **RPC**, shall be considered as ISTS systems, provided that these transmission lines are being used for evacuation and transfer of inter-State power on a regular basis, as identified by CTU in consultation with the concerned **RPC** and RLDC."

In compliance with the above provisions and direction of CTU dated 26.08.2025 for filing of tariff petitions for the 2024–29 tariff block, we are in the process of preparing our petition for determination of tariff of Non-ISTS transmission lines of OPTCL that carry ISTS power.

Accordingly, we request ERPC to kindly issue the required certification/ recommendations for above Non-ISTS lines carrying ISTS power for the period from 01.04.2024 to 31.03.2029, to facilitate the petition filing before the Hon'ble CERC.

Yours faithfully

Encl: as above


Sr. GM (RT&C)

CC

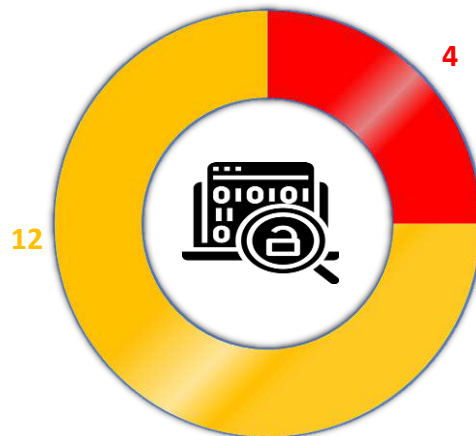
1. Sr.PS to CMD, OPTCL, Bhubaneswar for kind information of CMD.
2. PS to Director (O), OPTCL, Bhubaneswar for kind information of Director (O).
3. PS to Director (F), OPTCL, Bhubaneswar for kind information of Director (F).

ANNEXURE B.2.20

Annexure

IT Devices

OVERALL VULNERABILITIES



■ Critical

■ High

■ Medium

■ Low

■ Informational

IP Address / Hostname wise Vulnerability Count - Initial:

Sl. No.	IP Address/Hostname	Vulnerability Type					Total
		Critical	High	Medium	Low	Informational	
1	172.20.20.18	0	2	2	0	0	4
2	172.20.20.19	0	1	3	0	0	4
3	AMR-PC	0	1	5	0	0	6
4	172.20.20.15	0	0	1	0	0	1
5	172.20.50.1	0	0	1	0	0	1
6	172.20.20.5	0	0	0	0	0	0
7	172.20.20.6	0	0	0	0	0	0
8	172.20.20.20	0	0	0	0	0	0
9	172.20.20.22	0	0	0	0	0	0
10	172.20.20.36	0	0	0	0	0	0
11	DESKTOP-23NBTTO	0	0	0	0	0	0
Overall Findings		0	4	12	0	0	16

1. IP Address: 172.20.20.18 MAC: E4:62:C4:FE:5E:A6

Summary of findings:

Sl. No.	Vulnerability Name	Risk Level	CVSS Score
1	Apache Tomcat 9.0.0.M1 < 9.0.108	High	7.5
2	SSL Medium Strength Cipher Suites Supported (SWEET32)		7.5
3	TLS Version 1.1 Protocol Deprecated	Medium	6.5
4	TLS Version 1.0 Protocol Detection		6.5

Open port details:

Port	Service	Version
3389	tcp	ms-wbt-server
		Microsoft Terminal Services
8080	tcp	http
		Apache Tomcat (language: en)

2. IP Address: 172.20.20.19 MAC: E4:62:C4:FE:5E:A6

Summary of findings:

Sl. No.	Vulnerability Name	Risk Level	CVSS Score
1	SSL Medium Strength Cipher Suites Supported (SWEET32)	High	7.5
2	TLS Version 1.1 Protocol Deprecated	Medium	6.5
3	TLS Version 1.0 Protocol Detection		6.5
4	SMB Signing not required		5.3

Open port details:

Port	Service	Version
3389	tcp	ms-wbt-server
		Microsoft Terminal Services
445	tcp	microsoft-ds
		Microsoft-DS

3. IP Address: 172.20.20.37 HOSTNAME: AMR-PC MAC: B8:CA:3A:95:15:1D

Summary of findings:

Sl. No.	Vulnerability Name	Risk Level	CVSS Score
1	SSL Medium Strength Cipher Suites Supported (SWEET32)	High	7.5
2	TLS Version 1.1 Protocol Deprecated	Medium	6.5
3	TLS Version 1.0 Protocol Detection		6.5
4	HTTP TRACE / TRACK Methods Allowed		5.3
5	SMB Signing not required		5.3
6	Terminal Services Encryption Level is Medium or Low		4.3

Open port details:

Port		Service	Version
135	tcp	msrpc	Microsoft Windows RPC
137	udp	netbios-ns	NETBIOS Name Service
139	tcp	netbios-ssn	Microsoft Windows netbios-ssn
1801	tcp	msmq	Microsoft Message Queuing (MSMQ)
2103	tcp	msrpc	Microsoft Windows RPC
2105	tcp	msrpc	Microsoft Windows RPC
2107	tcp	msrpc	Microsoft Windows RPC
2869	tcp	http	Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
3389	tcp	ms-wbt-server	Microsoft Terminal Services
445	tcp	microsoft-ds	Microsoft-DS
7070	tcp	ssl/realserver	RealServer
5353	udp	mdns	Multicast DNS (MDNS)
80	tcp	http	Microsoft IIS httpd 10.0

4. IP Address: 172.20.20.15 MAC: E4:62:C4:FE:5E:80

Summary of findings:

Sl. No.	Vulnerability Name	Risk Level	CVSS Score
1	TLS Version 1.1 Protocol Deprecated	Medium	6.5

Open port details:

Port		Service	Version
443	tcp	ssl/http	Apache httpd

5. IP Address: 172.20.50.1

Summary of findings:

Sl. No.	Vulnerability Name	Risk Level	CVSS Score
1	SSH Terrapin Prefix Truncation Weakness (CVE-2023-48795)	Medium	5.9

Open port details:

Port		Service	Version
22	tcp	ssh	Cisco SSH 1.25 (protocol 2.0)
443	tcp	ssl/http	OpenResty web app server
123	udp	ntp	Network Time Protocol

80	tcp	http	OpenResty web app server
----	-----	------	--------------------------

6. IP Address: 172.20.20.5 MAC: 8C:84:42:2C:40:00

No vulnerability found

Open port details:

Port	Service	Version
22	tcp	ssh
443	tcp	ssl/http
80	tcp	http

7. IP Address: 172.20.20.6 MAC: 4C:D5:87:7C:5F:01

No vulnerability found

Open port details:

Port	Service	Version
22	tcp	ssh
443	tcp	ssl/http
80	tcp	http

8. IP Address: 172.20.20.20 MAC: 5C:BA:2C:45:8F:C8

No vulnerability found

Open port details:

Port	Service	Version
135	tcp	msrpc
137	udp	netbios-ns
139	tcp	netbios-ssn
445	tcp	microsoft-ds

9. IP Address: 172.20.20.22 MAC: 2C:EA:7F:6D:F0:FA

No vulnerability found

Open port details:

Port	Service	Version
135	tcp	msrpc
137	udp	netbios-ns
139	tcp	netbios-ssn
445	tcp	microsoft-ds

10. IP Address: 172.20.20.36 MAC: 6C:92:CF:06:0B:5E

No vulnerability found

Open port details:

Port		Service	Version
135	tcp	msrpc	Microsoft Windows RPC
137	udp	netbios-ns	NETBIOS Name Service
139	tcp	netbios-ssn	Microsoft Windows netbios-ssn
445	tcp	microsoft-ds	Microsoft-DS

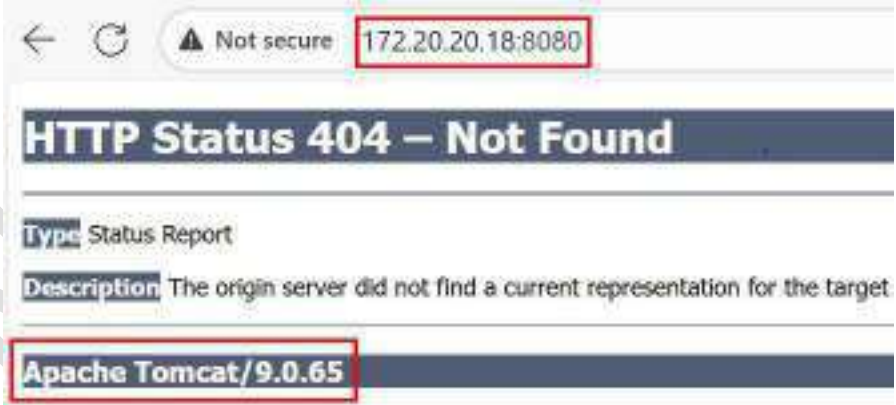
11. IP Address: 172.20.20.81 HOSTNAME: DESKTOP-23NBTTO MAC: 10:FF:E0:5D:1D:60

No vulnerability found

Open port details:

Port		Service	Version
135	tcp	msrpc	Microsoft Windows RPC

➤ **Detail Findings:**

1. Vulnerability Name: Apache Tomcat 9.0.0.M1 < 9.0.108 Vulnerability Rating: High		
CVSS: 7.5	CVSS:3.0/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H	CVE- 2025-48989
Affected System: 172.20.20.18:8080		
Vulnerability Description:	Apache Tomcat 9.0.65, released in May 2022, contains multiple known security vulnerabilities, including risks of remote code execution (RCE), privilege escalation, information disclosure, and denial-of-service (DoS). Several CVEs have been reported against this version, particularly affecting Tomcat's HTTP/2 request handling, session management, and request-parsing mechanisms. Since 9.0.65 lacks critical security patches introduced in later releases, it remains vulnerable to exploitation by attackers leveraging these weaknesses.	
Impact:	Running an outdated Apache Tomcat version such as 9.0.65 exposes applications and infrastructure to serious security risks, including unauthorized access, sensitive data exposure, server compromise, and potential service outages. Threat actors may exploit these vulnerabilities to execute arbitrary code, escalate privileges, disrupt operations, or bypass authentication controls. Such compromises can result in reputational harm, financial losses, and regulatory non-compliance with frameworks like OWASP, NIST, and PCI-DSS.	
Proof of Concept:		
Recommendation:	It is strongly recommended to upgrade Apache Tomcat to the latest stable release (9.0.108 or later) to ensure that all known vulnerabilities are patched and the platform is secure. In addition to upgrading, organizations should enforce hardened configurations by disabling or restricting the Manager application in production, limiting access to administrative interfaces, enforcing strong authentication and authorization, and applying modern security headers, thereby minimizing the attack surface and mitigating potential exploitation risks. Reference Link(s): https://tomcat.apache.org/download-90.cgi https://www.filehorse.com/download-apache-tomcat/96145/	

2. Vulnerability Name: SSL Medium Strength Cipher Suites Supported (SWEET32)

Vulnerability Rating: High

CVSS: 7.5 CVSS:3.0/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:N/A:N **CVE-**2016-2183

Affected Systems: [172.20.20.18:3389](#), [172.20.20.19:3389](#), [AMR-PC:3389](#)

Vulnerability Description: The server supports medium-strength cipher suites that use 64-bit block ciphers, such as 3DES (Triple DES), which are vulnerable to the SWEET32 attack. SWEET32 is a cryptographic vulnerability that allows attackers to perform collision attacks on 64-bit block ciphers, potentially leading to the decryption of encrypted communications. This weakness arises when a large amount of data is transmitted over a single session using a vulnerable cipher, allowing attackers to exploit statistical weaknesses in the encryption.

Impact: Supporting medium-strength cipher suites, especially those using 3DES, exposes encrypted sessions to plaintext recovery attacks, man-in-the-middle (MITM) attacks, and data leakage. Exploitation of the SWEET32 vulnerability can allow attackers to decrypt HTTPS traffic, steal sensitive user information, and compromise secure communications, resulting in data breaches, financial fraud, and regulatory non-compliance with security frameworks such as PCI-DSS, HIPAA, and NIST.

Proof of Concept:

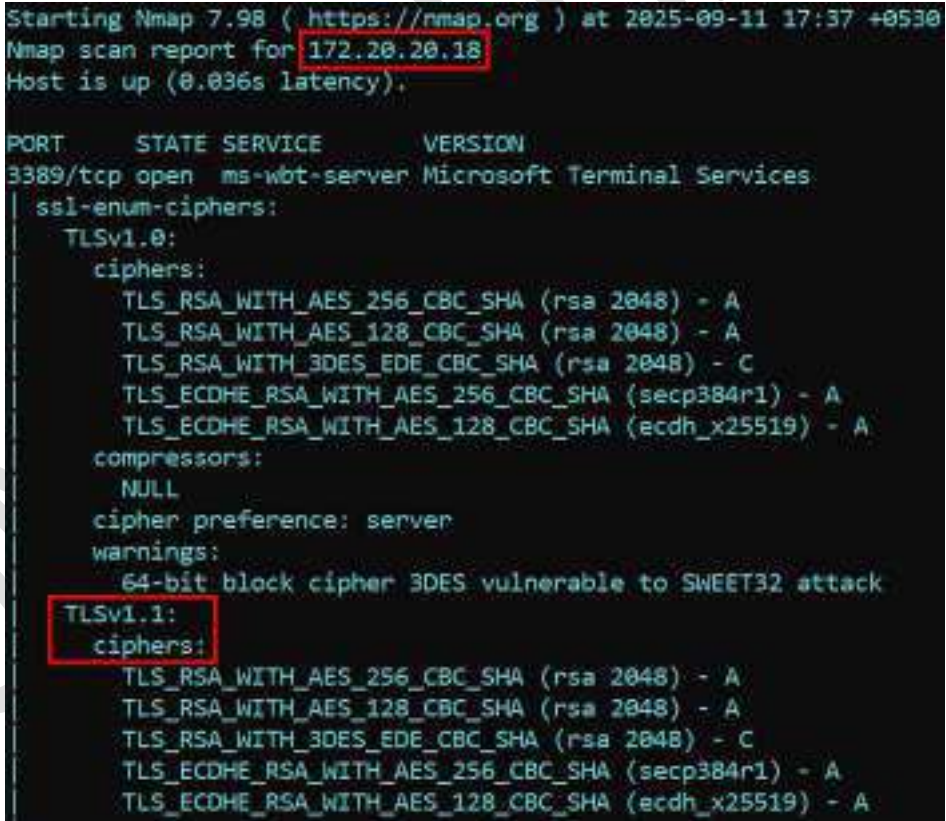
```
Starting Nmap 7.98 ( https://nmap.org ) at 2025-09-11 17:19 +0530
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS
Nmap scan report for 172.20.20.18
Host is up (0.036s latency).

PORT      STATE SERVICE          VERSION
3389/tcp  open  ms-wbt-server    Microsoft Terminal Services
ssl-enum-ciphers:
  TLSv1.0:
    ciphers:
      TLS_RSA_WITH_AES_256_CBC_SHA (rsa 2048) - A
      TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
      TLS_RSA_WITH_3DES_EDE_CBC_SHA (rsa 2048) - C
      TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA (secp384r1) - A
      TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
    compressors:
      NULL
    cipher preference: server
  warnings:
    64-bit block cipher 3DES vulnerable to SWEET32 attack
```

Recommendation: It is strongly recommended to disable all 64-bit block ciphers, including 3DES (TLS_RSA_WITH_3DES_EDE_CBC_SHA), and enforce the use of stronger cipher suites such as AES-128-GCM, AES-256-GCM. Additionally, update SSL/TLS configurations to support TLS 1.2 or TLS 1.3 exclusively, as these versions use modern cryptographic standards that mitigate SWEET32 and other related attacks. Regular security audits should be conducted to ensure compliance with best practices.

Reference Link(s):

<https://help.defense.com/en/articles/6302810-ssl-medium-strength-cipher-suite-supported-sweet32-windows>

3. Vulnerability Name: TLS Version 1.1 Protocol Deprecated Vulnerability Rating: Medium	
CVSS: 6.5	CVSS:3.0/AV:N/AC:H/PR:N/UI:N/S:U/C:H/I:L/A:N CWE- 327
Affected Systems: 172.20.20.15:443 , 172.20.20.18:3389 , 172.20.20.19:3389 , AMR-PC:3389	
Vulnerability Description:	The system was found to support TLS 1.1, which are outdated and insecure cryptographic protocols. These versions suffer from known vulnerabilities, including weak cipher suites, susceptibility to downgrade attacks (such as POODLE and BEAST), and lack of modern cryptographic features. Due to these weaknesses, TLS 1.0 have been officially deprecated by major security standards, including PCI-DSS, NIST SP 800-52, and the CA/Browser Forum, and have been discontinued in modern browsers and applications since 2020.
Impact:	Using TLS 1.1 increases the risk of man-in-the-middle (MITM) attacks, session hijacking, and data interception. Attackers can exploit these weak protocols to decrypt encrypted traffic, impersonate legitimate users, and compromise sensitive data.
Proof of Concept:	 <pre> Starting Nmap 7.98 (https://nmap.org) at 2025-09-11 17:37 +0530 Nmap scan report for 172.20.20.18 Host is up (0.036s latency). PORT STATE SERVICE VERSION 3389/tcp open ms-wbt-server Microsoft Terminal Services ssl-enum-ciphers: TLSv1.0: ciphers: TLS_RSA_WITH_AES_256_CBC_SHA (rsa 2048) - A TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A TLS_RSA_WITH_3DES_EDE_CBC_SHA (rsa 2048) - C TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA (secp384r1) - A TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A compressors: NULL cipher preference: server warnings: 64-bit block cipher 3DES vulnerable to SWEET32 attack TLSv1.1: ciphers: TLS_RSA_WITH_AES_256_CBC_SHA (rsa 2048) - A TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A TLS_RSA_WITH_3DES_EDE_CBC_SHA (rsa 2048) - C TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA (secp384r1) - A TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A </pre>
Recommendation:	It is strongly recommended to disable TLS 1.1 and enforce the use of TLS 1.2 and TLS 1.3, which offer stronger encryption and improved security features. Additionally, administrators should remove weak cipher suites, enable Perfect Forward Secrecy (PFS), and regularly audit SSL/TLS configurations to ensure compliance with industry security best practices. Reference Link(s): https://theseckmaster.com/blog/how-to-disable-tls-1-0-and-tls-1-1-on-windows-server https://techpress.net/disable-tls-1-0-and-tls-1-1-on-windows-10/

4. Vulnerability Name: TLS Version 1.0 Protocol Detection

Vulnerability Rating: Medium

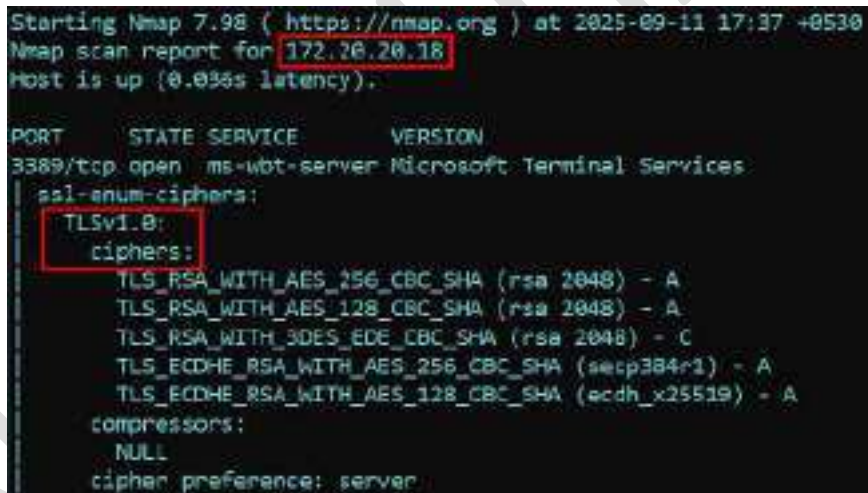
CVSS: 6.5 CVSS:3.0/AV:N/AC:H/PR:N/UI:N/S:U/C:H/I:L/A:N **CWE-**327

Affected Systems: [172.20.20.18:3389](#), [172.20.20.19:3389](#), [AMR-PC:3389](#)

Vulnerability Description: The system was found to support TLS 1.0, which are outdated and insecure cryptographic protocols. These versions suffer from known vulnerabilities, including weak cipher suites, susceptibility to downgrade attacks (such as POODLE and BEAST), and lack of modern cryptographic features. Due to these weaknesses, TLS 1.1 have been officially deprecated by major security standards, including PCI-DSS, NIST SP 800-52, and the CA/Browser Forum, and have been discontinued in modern browsers and applications since 2020.

Impact: Using TLS 1.0 increases the risk of man-in-the-middle (MITM) attacks, session hijacking, and data interception. Attackers can exploit these weak protocols to decrypt encrypted traffic, impersonate legitimate users, and compromise sensitive data.

Proof of Concept:



```
Starting Nmap 7.98 ( https://nmap.org ) at 2025-09-11 17:37 +0530
Nmap scan report for 172.20.20.18
Host is up (0.036s latency).

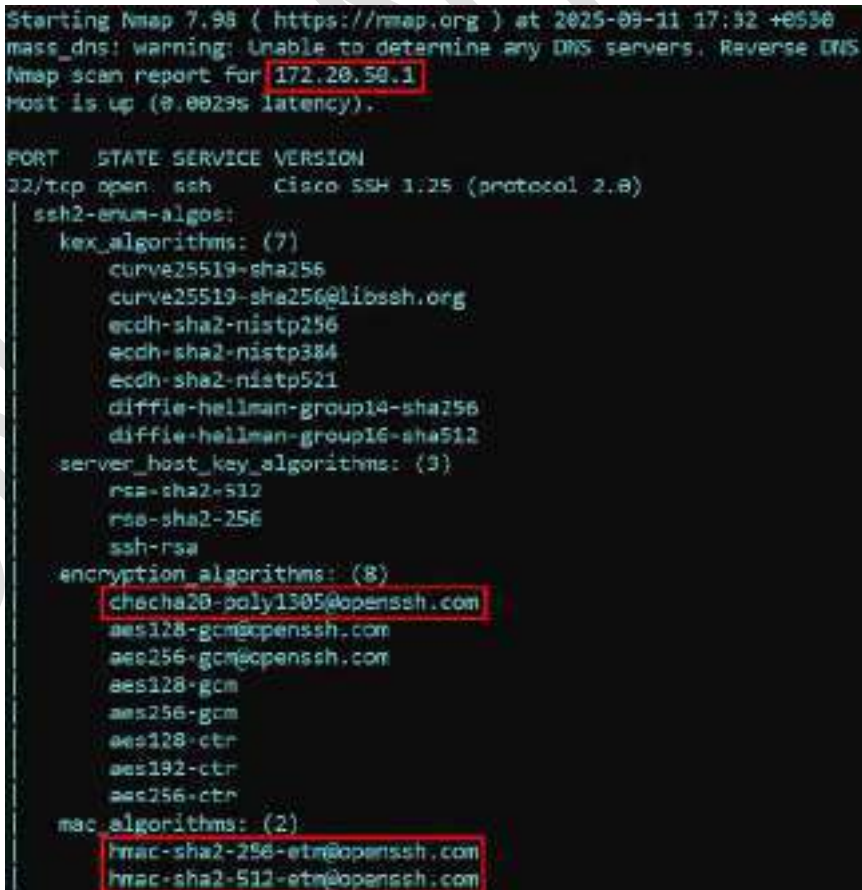
PORT      STATE SERVICE
3389/tcp  open  ms-wbt-server Microsoft Terminal Services

ssl-enum-ciphers:
  TLSv1.0:
    ciphers:
      TLS_RSA_WITH_AES_256_CBC_SHA (rsa 2048) - A
      TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
      TLS_RSA_WITH_3DES_EDE_CBC_SHA (rsa 2048) - C
      TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA (secp384r1) - A
      TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
    compressors:
      NULL
    cipher preference: server
```

Recommendation: It is strongly recommended to disable TLS 1.0 and enforce the use of TLS 1.2 and TLS 1.3, which offer stronger encryption and improved security features. Additionally, administrators should remove weak cipher suites, enable Perfect Forward Secrecy (PFS), and regularly audit SSL/TLS configurations to ensure compliance with industry security best practices.

Reference Link(s):

<https://thesecmaster.com/blog/how-to-disable-tls-1-0-and-tls-1-1-on-windows-server> <https://techpress.net/disable-tls-1-0-and-tls-1-1-on-windows-10/>

5. Vulnerability Name: SSH Terrapin Prefix Truncation Weakness (CVE-2023-48795) Vulnerability Rating: Medium	
CVSS: 5.9	CVSS:3.0/AV:N/AC:H/PR:N/UI:N/S:U/C:N/I:H/A:N CVE-2023-48795
Affected System:	172.20.50.1:22
Vulnerability Description:	The system is affected by the SSH Terrapin Prefix Truncation Weakness, a vulnerability that allows attackers to manipulate the length field of SSH packets, causing specific parts of the transmitted data to be truncated or altered. This issue primarily affects the SSH transport layer encryption, enabling an attacker to tamper with the integrity of SSH communications. As a result, an attacker could bypass authentication mechanisms, modify transmitted commands, or exploit other protocol weaknesses to gain unauthorized access.
Impact:	The Terrapin attack weakens the security guarantees of SSH authentication and session integrity, leading to man-in-the-middle (MITM) attacks, session hijacking, and potential remote code execution. Attackers exploiting this weakness may manipulate login credentials, alter command execution, or degrade SSH security to gain persistent access to sensitive systems.
Proof of Concept:	 <pre> Starting Nmap 7.98 (https://nmap.org) at 2025-09-11 17:32 +0530 mass_dns: warning: Unable to determine any DNS servers. Reverse DNS Nmap scan report for 172.20.50.1 Host is up (0.0029s latency). PORT STATE SERVICE VERSION 22/tcp open ssh Cisco SSH 1.25 (protocol 2.0) ssh2-enum-algos: key_algorithms: (7) curve25519-sha256 curve25519-sha256@libssh.org ecdh-sha2-nistp256 ecdh-sha2-nistp384 ecdh-sha2-nistp521 diffie-hellman-group14-sha256 diffie-hellman-group16-sha512 server_host_key_algorithms: (3) rsa-sha2-512 rsa-sha2-256 ssh-rsa encryption_algorithms: (8) chacha20-poly1305@openssh.com aes128-gcm@openssh.com aes256-gcm@openssh.com aes128-gcm aes256-gcm aes128-ctr aes192-ctr aes256-ctr mac_algorithms: (2) hmac-sha2-256-etm@openssh.com hmac-sha2-512-etm@openssh.com </pre>
Recommendation:	It is strongly recommended to update OpenSSH to the latest stable version, as security patches have been released to mitigate this weakness. Additionally, organizations should disable weak SSH algorithms, enforce key-based authentication, and use cryptographic integrity checks to prevent session tampering.

Regular security audits and monitoring of SSH logs for anomalies should be conducted to detect potential exploitation attempts.

Reference Link(s):

<https://medium.com/@haris29/ssh-prefix-truncation-vulnerability-terrapin-1343bee378fa>

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6. Vulnerability Name: HTTP TRACE / TRACK Methods Allowed

Vulnerability Rating: Medium

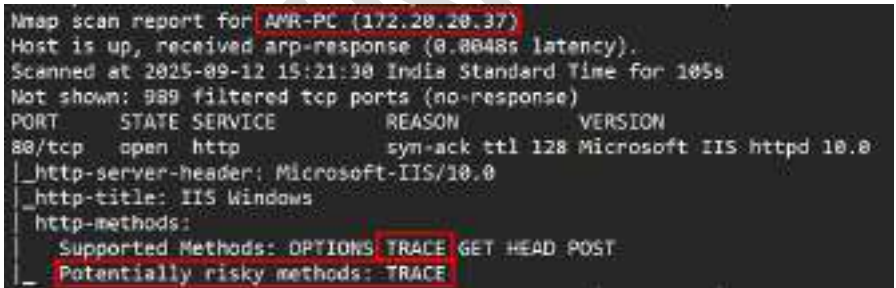
CVSS: 5.3 CVSS:3.0/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N **CWE-**16,200, **CVE-** 2010-0386

Affected System: 172.20.20.37:80

Vulnerability Description: The server is configured to allow HTTP TRACE and TRACK methods, which can be exploited in cross-site tracing (XST) attacks. These HTTP methods are intended for debugging purposes, but they echo back request headers, potentially exposing sensitive information such as cookies, session tokens, and authentication credentials. Attackers can leverage this behavior to bypass security restrictions, conduct session hijacking, and steal user data. Modern web servers and security standards strongly recommend disabling these methods to prevent exploitation.

Impact: Allowing HTTP TRACE and TRACK methods increases the risk of information disclosure, session hijacking, and cross-site scripting (XSS) attacks. Attackers can use XST to steal session cookies and impersonate legitimate users, leading to unauthorized access and potential data breaches. Additionally, enabling these methods may result in non-compliance with security frameworks such as PCI-DSS, HIPAA, and OWASP, which mandate the restriction of unnecessary HTTP methods to protect sensitive data.

Proof of Concept:



```

Nmap scan report for AMR-PC (172.20.20.37)
Host is up, received arp-response (0.8048s latency).
Scanned at 2025-09-12 15:21:30 India Standard Time for 105s
Not shown: 989 filtered tcp ports (no-response)
PORT      STATE SERVICE      REASON      VERSION
80/tcp    open  http        syn-ack ttl 128 Microsoft IIS httpd 10.0
|_ http-server-header: Microsoft-IIS/10.0
|_ http-title: IIS Windows
|_ http-methods:
|_   Supported Methods: OPTIONS TRACE GET HEAD POST
|_   Potentially risky methods: TRACE
  
```

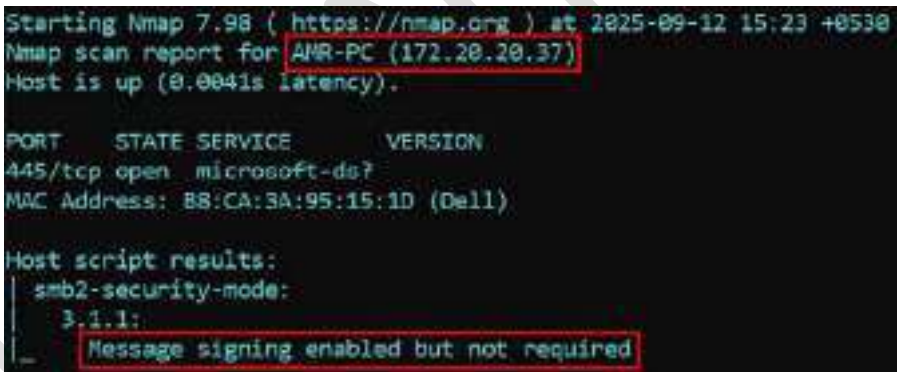
Recommendation: It is strongly recommended to disable HTTP TRACE and TRACK methods by configuring the web server appropriately. For Apache, this can be done by adding the following directive in the configuration file (httpd.conf or .htaccess):

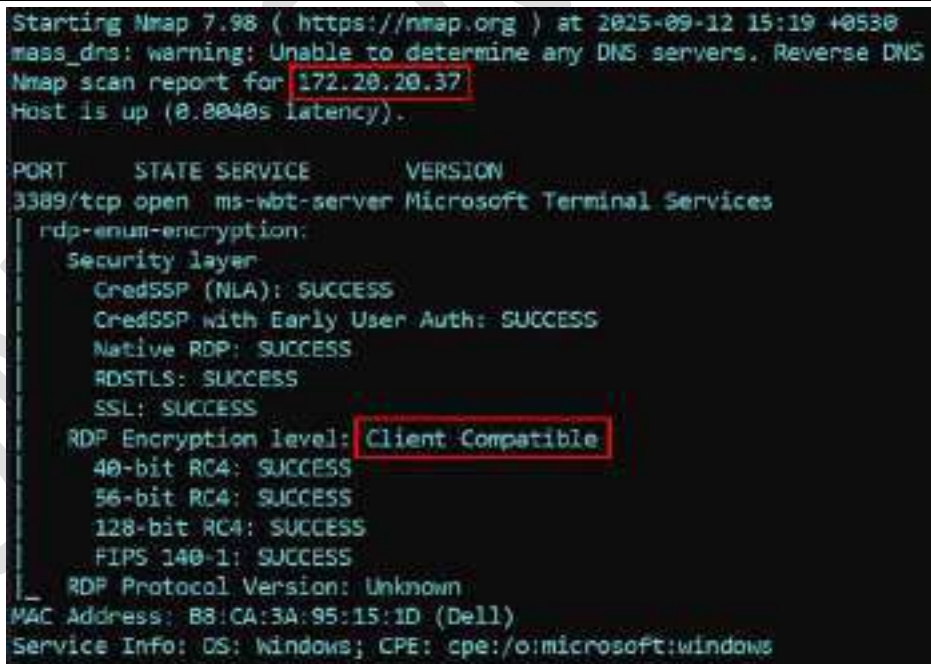
```

TraceEnable Off
if ($request_method ~* ""(TRACE|TRACK)"" ) {
    return 405;
}
  
```

Additionally, security controls such as Web Application Firewalls (WAFs), strict HTTP headers, and regular security audits should be implemented to mitigate further risks.

Reference Link(s):
<https://access.redhat.com/solutions/198813>
<https://stackoverflow.com/questions/10673306/disable-track-and-trace-in-apache>

7. Vulnerability Name: SMB Signing not required Vulnerability Rating: Medium	
CVSS: 5.3	CVSS:3.0/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:L/A:N
Affected Systems: 172.20.20.19:445 , AMR-PC:445	
Vulnerability Description:	The system is vulnerable because SMB (Server Message Block) signing is not required, allowing SMB communications to be transmitted without integrity verification. Without SMB signing, an attacker could intercept and tamper with SMB traffic, potentially altering data, stealing credentials, or impersonating legitimate systems. This exposes the organization to man-in-the-middle (MITM) attacks and unauthorized access.
Impact:	The lack of SMB signing significantly weakens the security of SMB communications. Attackers can exploit this weakness to perform MITM attacks, hijack sessions, or inject malicious commands into SMB traffic. This can lead to unauthorized access, data tampering, credential theft, or the introduction of malicious payloads, jeopardizing the integrity and confidentiality of network resources.
Proof of Concept:	 <pre> Starting Nmap 7.98 (https://nmap.org) at 2025-09-12 15:23 +0530 Nmap scan report for AMR-PC (172.20.20.37) Host is up (0.0041s latency). PORT STATE SERVICE VERSION 445/tcp open microsoft-ds? MAC Address: 88:CA:3A:95:15:1D (Dell) Host script results: _ smb2-security-mode: _ 3.1.1: _ Message signing enabled but not required </pre>
Recommendation:	It is highly recommended to enforce message signing in the host's configuration. On Windows, Configure SMB signing to be required on all systems to ensure message integrity. Open Group Policy Editor (gpedit.msc) or use domain-level Group Policy. Navigate to: Computer Configuration → Windows Settings → Security Settings → Local Policies → Security Options Enable the following policies: Microsoft network client: Digitally sign communications (if server agrees) – Set to Enabled. Microsoft network client: Digitally sign communications (always) – Set to Enabled. Microsoft network server: Digitally sign communications (if client agrees) – Set to Enabled. Microsoft network server: Digitally sign communications (always) – Set to Enabled. Reference Link(s): https://help.defense.com/en/articles/6714765-smb-signing-not-required-windows-vulnerability

8. Vulnerability Name: Terminal Services Encryption Level is Medium or Low Vulnerability Rating: Medium	
CVSS: 4.3 CVSS:2/CVSS2#AV:N/AC:M/Au:N/C:P/I:N/A:N	
Affected System: AMR-PC:3389	
Vulnerability Description:	When Windows Terminal Services (Remote Desktop Services) is configured with an encryption level set to Medium or Low, the communication between client and server is not fully secured. At these levels, session data, including credentials and other sensitive information, is either weakly encrypted or partially exposed, making it susceptible to interception and replay attacks. Attackers positioned on the same network can potentially capture or manipulate traffic, exploiting weak encryption to compromise the confidentiality and integrity of remote desktop sessions.
Impact:	Using medium or low encryption levels for Terminal Services significantly increases the risk of sensitive data exposure and unauthorized access. Adversaries could intercept session traffic, steal user credentials, or launch man-in-the-middle (MitM) attacks to gain control over active sessions. Successful exploitation could lead to compromised accounts, lateral movement within the environment, unauthorized access to critical systems, and overall weakening of the organization's security posture, resulting in potential data breaches, downtime, and compliance violations.
Proof of Concept:	 <pre> Starting Nmap 7.98 (https://nmap.org) at 2025-09-12 15:19 +0530 mass_dns: warning: Unable to determine any DNS servers. Reverse DNS Nmap scan report for 172.20.20.37 Host is up (0.8040s latency). PORT STATE SERVICE VERSION 3389/tcp open ms-wbt-server Microsoft Terminal Services rdp-enun-encryption: Security layer CredSSP (NLA): SUCCESS CredSSP with Early User Auth: SUCCESS Native RDP: SUCCESS RDSTLS: SUCCESS SSL: SUCCESS RDP Encryption level: Client Compatible 40-bit RC4: SUCCESS 56-bit RC4: SUCCESS 128-bit RC4: SUCCESS FIPS 140-1: SUCCESS _ RDP Protocol Version: Unknown MAC Address: B8:CA:3A:95:15:1D (Dell) Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows </pre>
Recommendation:	It is recommended to mitigate this vulnerability, by configuring Terminal Services (Remote Desktop Services) to enforce High or FIPS-compliant encryption levels. This can be achieved by modifying the Group Policy setting: Computer Configuration → Administrative Templates → Windows Components → Remote Desktop Services → Remote Desktop Session Host → Security → Set client connection encryption level, and setting it to High Level or FIPS Compliant. Additionally, ensure that Network Level Authentication (NLA) is enabled to further protect against unauthorized access and credential theft, and restrict RDP access to trusted networks using firewalls or VPNs. Reference Link(s):



Business With Wisdom
...Growth With Assurance

https://www.syxsense.com/syxsense-securityarticles/cis_benchmarks/syx-1033-12465.html

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Sharing AMR System SEM data to SLDC, Utilities

Annexure B.2.21

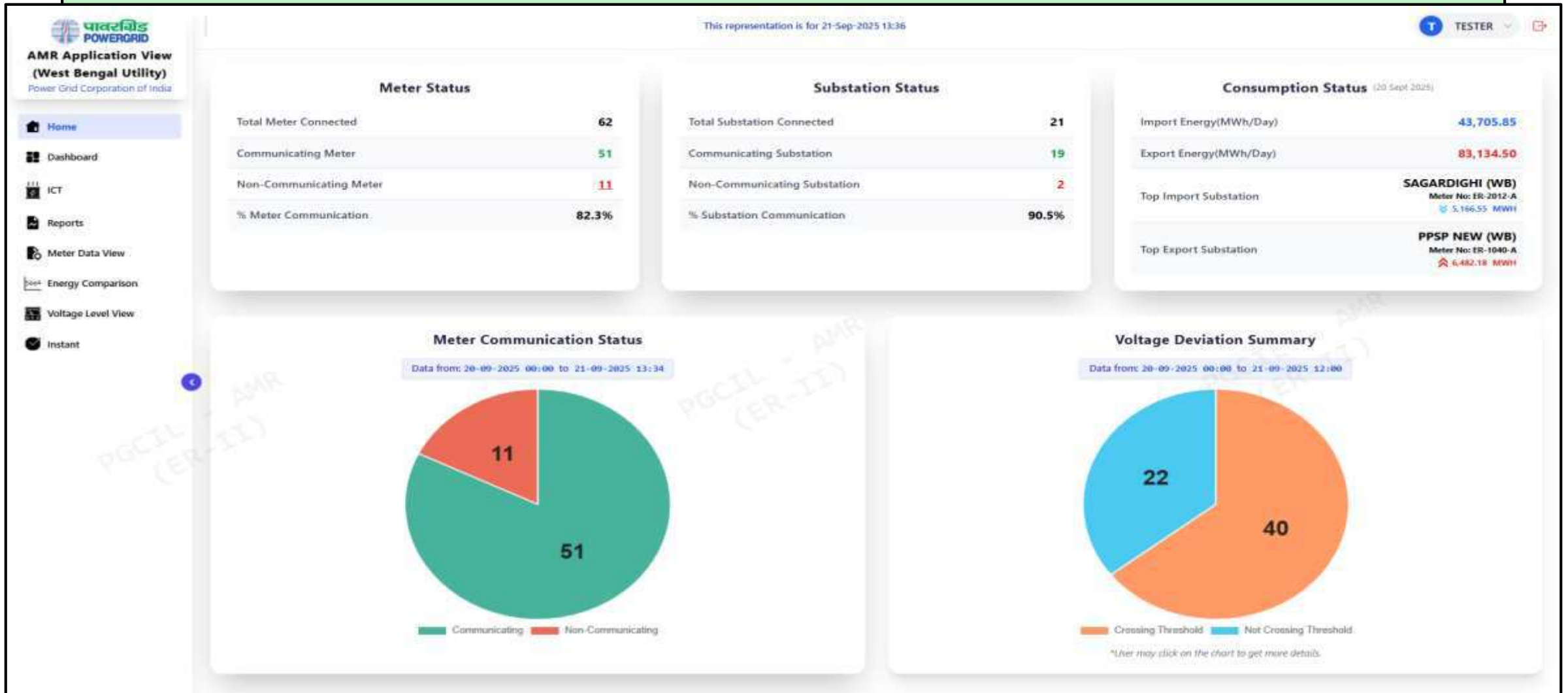
A screenshot of a login interface for PGCIL-AMR. At the top, there is a logo for 'पावरग्रिड POWERGRID' (Power Grid) with a stylized tower icon. Below the logo, the text 'Login to PGCIL-AMR' is displayed. The interface includes two input fields: 'Username' and 'Password'. Below these fields is a blue 'Login' button. At the bottom right, there is a link that says 'Forgot Password?'. The background of the interface is white, and it is overlaid on a larger background image of power lines and towers.

**A Snap of the Application – for WBSLDC
(pilot project)**

Project Background vis-à-vis Application Development

- *Approved in 54th ERPC meeting. WBSLDC considered as pilot utility.*
- *New application development cycle is in progress.*
- *Tie-lines Feeders, opposite end feeders along with ICT Feeders are integrated.*
- *15 mins block-wise data, 01 min instant data (as per feasibility) are integrated.*
- *Dashboards, Reports, Meter Data-view modules are major features of the application.*
- *PGCIL, WBSLDC & TCS have several joint meetings during the development followed by demonstration.*

WBSLDC Application- Snapshots



WBSLDC Application- Snapshots

**पारवर्तित
POWERGRID**

AMR Application View
(West Bengal Utility)
Power Grid Corporation of India

Home

Dashboard

ICT

Reports

Meter Data View

Energy Comparison

Voltage Level View

Instant

This representation is for 21-Sep-2025 13:33

TESTER

Substations List

Green: All meters of the substation are communicating Yellow: Partial meters of the substation are communicating Red: All meters of the substation are non-communicating

ALIPURDUAR (WB)

BIDHANNAGAR (WB)

BIRPARA (WB)

DALKHOLA (WB)

FALAKATA(WB)

GAZOLE (WB)

GOKARNA (WB)

JEERAT (WB)

KALIMPONG (WB)

KHARAGPUR (WB)

KLC BANTALA (WB)

KURSEONG (WB)

MALDA (WB)

MANIKCHAK (WB)

NBU (WB)

NEWTOWN (WB)

PPSP NEW (WB)

RAMMAM (WB)

SAGARDIGHI (WB)

SANTALDIH (WB)

SUBHASHGRAM (WB)

Substations List / SUBHASHGRAM (WB)

SUBHASHGRAM (WB) - Meter Details

Export Import

Meter SNo.	Feeder Name	Meter Type	Substation Type	Meter Make	Energy Total (MWh)	Last LS Data	Last Midnight Data	Status
ER-1003-A	220 KV SUBHASHGRAM(WB)-SUBHASGRAM(PG)-1	STANDBY	STATE-TRANSMISSION	GENUS_15	2128.51	21-Sep-25 09:30 AM	21-Sep-25 12:00 AM	✓
NS-2942-A	220 KV SUBHASHGRAM(PG)-SUBHASGRAM(WBSETCL)-1	STANDBY	TRANSMISSION UTILITY	SECURE	2130.62	21-Sep-25 10:45 AM	21-Sep-25 12:00 AM	✓
ER-1127-A	220 KV SUBHASHGRAM(WB)-SUBHASGRAM(PG)-2	STANDBY	STATE-TRANSMISSION	GENUS_15	1836.38	21-Sep-25 09:30 AM	21-Sep-25 12:00 AM	✓
NS-2937-A	220 KV SUBHASHGRAM(PG)-SUBHASGRAM(WBSETCL)-2	STANDBY	TRANSMISSION UTILITY	SECURE	2451.58	21-Sep-25 10:45 AM	21-Sep-25 12:00 AM	✓

*Above meters are shown alternatively with opposite-end feeder details listed in yellow strips
*Energy column shows previous day's total consumption.

WBSLDC Application- Snapshots



WBSLDC Application- Snapshots

**पावरग्रिड
POWERGRID**

AMR Application View
(West Bengal Utility)
Power Grid Corporation of India

Home

Dashboard

ICT

Reports

Meter Data View

Energy Comparison

Voltage Level View

Instant

This representation is for 21-Sep-2025 14:04

T TESTER

Meter Data View / ER-1127-A / LS_15

GeneralLS_15Midnight

Meter SNo. ER-1127-AFeeder Name 220 KV SUBHASHGRAM(WB)-SUBHASHGRAM(PG)-2MF3200000Start 09/20/2025End 09/21/2025SubmitCTR: 1696/L PTR: 220/110

Export Excel

Time (LS Date)	Meter SNo.	Wh_net	Wh(I)	Wh(E)	kVAh	kVArh_Ig_kWh(I)	kVArh_Id_kWh(E)	kVArh_Ig_kWh(E)	kVArh_Id_kWh(I)	VR	VY	VB	IR	IY	IB
21-Sep-2025 09:30	ER-1127-A	-14.74	0	14.74	0	0	0.09	0.16	0	64.6	65.2	65	0.3	0.3	0.306
21-Sep-2025 09:15	ER-1127-A	-14.11	0	14.11	0	0	0.1	0.16	0	64.7	65.3	65.1	0.287	0.286	0.292
21-Sep-2025 09:00	ER-1127-A	-13.92	0	13.92	0	0	0.13	0.11	0	64.7	65.5	65.3	0.282	0.282	0.288
21-Sep-2025 08:45	ER-1127-A	-14.36	0	14.36	0	0	0.53	0.03	0	64.9	65.6	65.4	0.291	0.291	0.296
21-Sep-2025 08:30	ER-1127-A	-13.6	0	13.6	0	0	0.65	0	0	65.2	65.8	65.6	0.274	0.274	0.281
21-Sep-2025 08:15	ER-1127-A	-12.94	0	12.94	0	0	0.65	0	0	65.4	66	65.7	0.26	0.261	0.266
21-Sep-2025 08:00	ER-1127-A	-13.11	0	13.11	0	0	0.79	0	0	65.3	66.1	65.8	0.261	0.265	0.27
21-Sep-2025 07:45	ER-1127-A	-14.2	0	14.2	0	0	1.11	0	0	65.3	66	65.7	0.285	0.288	0.292
21-Sep-2025 07:30	ER-1127-A	-12.81	0	12.81	0	0	1	0	0	65.4	66	65.7	0.258	0.259	0.264
21-Sep-2025 07:15	ER-1127-A	-11.91	0	11.91	0	0	0.9	0	0	65.5	66.1	65.8	0.239	0.238	0.247
21-Sep-2025 07:00	ER-1127-A	-11.59	0	11.59	0	0	0.95	0	0	65.5	66.1	65.9	0.232	0.232	0.24
21-Sep-2025 06:45	ER-1127-A	-11.74	0	11.74	0	0	0.81	0	0	65.6	66.2	65.9	0.234	0.234	0.243

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This representation is for 21-Sep-2025 21:37

T **TESTER**

ICT Meters (765KV / 400KV / 220KV)

ALIPURDUAR

BINAGURI

BIRPARA

DURGAPUR

MALDA

MEDINIPUR

MEDINIPUR - ICT Meter Details

Export

Import

SNo	Meter SNo	Feeder Name	Meter Type	Type	Make	Energy Total (MWh)	Last LS Data	Last MIDN Data	Status
1	NS-3301-A	765 KV SIDE OF MEDINIPUR(PG) 765/400KV 1500 MVA ICT-1	MAIN	TRANSMISSION UTILITY	SECURE	4087.72	21-Sep-25 06:30 PM	21-Sep-25 12:00 AM	✓
2	NS-3334-A	765 KV SIDE OF MEDINIPUR(PG) 765/400KV 1500 MVA ICT-2	MAIN	TRANSMISSION UTILITY	SECURE	4082.50	21-Sep-25 06:30 PM	21-Sep-25 12:00 AM	✓
3	NS-3298-A	400 KV SIDE OF MEDINIPUR(PG) 765/400KV 1500 MVA ICT-2	STANDBY	TRANSMISSION UTILITY	SECURE	-4083.42	21-Sep-25 06:30 PM	21-Sep-25 12:00 AM	✓
4	NS-3308-A	400 KV SIDE OF MEDINIPUR(PG) 765/400KV 1500 MVA ICT-1	STANDBY	TRANSMISSION UTILITY	SECURE	-2207.35	21-Sep-25 11:30 AM	21-Sep-25 12:00 AM	✓
5	ER-1686-A	765 KV MEDINIPUR(PG)-JEERAT(WB) LINE-1	STANDBY	TRANSMISSION UTILITY	GENUS_15	1166.43	21-Sep-25 02:30 AM	21-Sep-25 12:00 AM	✓
6	ER-1684-A	765 KV MEDINIPUR(PG)-JEERAT(WB) LINE-2	STANDBY	TRANSMISSION UTILITY	GENUS_15	0.00	06-Sep-25 10:30 PM	06-Sep-25 12:00 AM	✗

NEW JEERAT

RAJARHAT

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This representation is for 21-Sep-2025 13:39

T

TESTER

Voltage Level View

09/14/202509/21/2025SubmitSearch Meter SNo./I

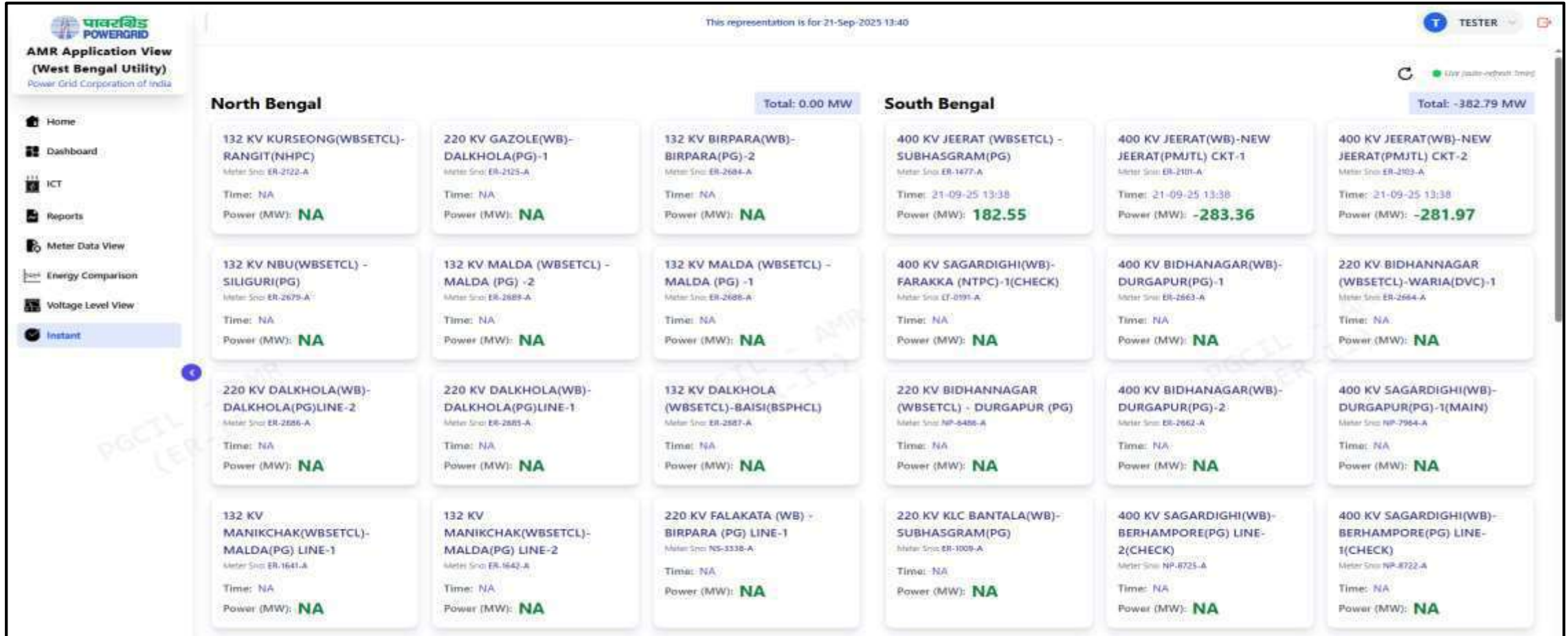
Calculations used:
Max Volt.(Primary)(KV) = $\sqrt{3} \times \text{MCC}(V_{ph}) \times V_{L2} \times \text{PTR}$
Permissible Voltage (KV) = $V_{max} \times 1.05$

S.No	All Substations	Meter SNo.	Feeder Name	Max Volt.(Primary) (KV)	Permissible Voltage (KV)	Difference (in KV)	Deviation (in %) ± 1	Occurrence Timestamp ± 1
1	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	137.59	135.96	1.63	1.20	21-09-25 12:00
2	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	137.80	135.96	1.84	1.35	21-09-25 11:45
3	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	137.80	135.96	1.84	1.35	21-09-25 11:30
4	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	137.80	135.96	1.84	1.35	21-09-25 11:15
5	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	138.01	135.96	2.05	1.50	21-09-25 11:00
6	BIRPARA (WB)	ER-2684-A	132 KV BIRPARA(WB)-BIRPARA(PG)-2	137.63	135.96	1.67	1.23	21-09-25 10:45
7	SAGARDIGHI (WB)	ER-1985-A	400 KV SAGARDIGHI(WB)- DURGAPUR(PG)-2(MAIN)	412.40	412.00	0.40	0.10	21-09-25 10:45
8	SAGARDIGHI (WB)	ER-2889-A	400 KV SAGARDIGHI(WB)- DURGAPUR(PG)-2(CHECK)	412.28	412.00	0.28	0.07	21-09-25 10:45
9	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	138.01	135.96	2.05	1.50	21-09-25 10:45
10	SAGARDIGHI (WB)	ER-2899-A	400 KV SAGARDIGHI(WB)-DURGAPUR(PG)-1(CHECK)	413.54	412.00	1.54	0.37	21-09-25 10:45
11	SAGARDIGHI (WB)	ER-1985-A	400 KV SAGARDIGHI(WB)- DURGAPUR(PG)-2(MAIN)	412.85	412.00	0.85	0.21	21-09-25 10:30
12	BIRPARA (WB)	ER-2684-A	132 KV BIRPARA(WB)-BIRPARA(PG)-2	137.82	135.96	1.86	1.37	21-09-25 10:30
13	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	138.01	135.96	2.05	1.50	21-09-25 10:30
14	SAGARDIGHI (WB)	ER-2889-A	400 KV SAGARDIGHI(WB)- DURGAPUR(PG)-2(CHECK)	412.72	412.00	0.72	0.17	21-09-25 10:30
15	SAGARDIGHI (WB)	ER-2899-A	400 KV SAGARDIGHI(WB)-DURGAPUR(PG)-1(CHECK)	414.04	412.00	2.04	0.50	21-09-25 10:30
16	SAGARDIGHI (WB)	ER-1985-A	400 KV SAGARDIGHI(WB)- DURGAPUR(PG)-2(MAIN)	413.16	412.00	1.16	0.28	21-09-25 10:15
17	BIRPARA (WB)	ER-2684-A	132 KV BIRPARA(WB)-BIRPARA(PG)-2	137.94	135.96	1.98	1.46	21-09-25 10:15
18	BIRPARA (WB)	ER-1879-A	132 KV BIRPARA (WB)-BIRPARA (PG)-1	138.01	135.96	2.05	1.50	21-09-25 10:15

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THANK YOU

Complete Protection Upgradation at Malda SS Under ADDCAP

ADDCAP Scope Details

- New control Room Building construction (02 Storied).
- Construction of 06 No's SPR for all 400/220/132 KV System.
- Replacement of all old pantograph/DBR Isolators in both 400 KV & 220 KV.
- New CRP system including distributed bus bar system commissioning at both 400 & 220 KV level.

Present Condition of Construction



SD Planning covering multiple elements

SL NO	Name of Element	From Date	To Date	No of Days outage	Concurrent outage	Associated works
01.	400 KV Malda-Purnea-I along with associated Line Reactor.	24.10.25	05.11.25	13		Dismantling & erection of new 89 L isolator & 89 T isolator of Purnea-I Bay. Complete old CRP system replacement along with distributed Bus Bar unit commissioning.
02.	400 KV TBC	24.10.25	05.11.25	13		Dismantling & erection of new 89 T isolator of TBC Bay. Complete old CRP system replacement along with distributed Bus Bar unit commissioning.

SD Planning covering multiple elements

SL NO	Name of Element	From Date	To Date	No of Days outage	Concurrent outage	Associated works
03.	400 KV Malda-Purnea-II along with associated Line Reactor.	06.11.25	15.11.25	10		Dismantling & erection of new 89 L isolator & 89 T isolator of Purnea-II Bay. Complete old CRP system replacement along with distributed Bus Bar unit commissioning.
04.	220 KV Malda-Gazole-I	06.11.25	15.11.25	10	220 KV TBC bay at Malda.	Dismantling & erection of new 89 T/89L isolator of Concerned Bay. Complete old CRP system replacement along with distributed Bus Bar unit commissioning.

SD Planning covering multiple elements

SL NO	Name of Element	From Date	To Date	No of Days outage	Concurrent outage	Associated works
05.	400 KV Malda-Farakka-II	16.11.25	23.11.25	08		Dismantling & erection of new 89 L isolator & 89 T isolator of subject Bay. Complete old CRP system replacement along with distributed Bus Bar unit commissioning.
06.	220 KV Malda-Gazole-II	16.11.25	23.11.25	08		Dismantling & erection of new 89 T/89L isolator of Concerned Bay. Complete old CRP system replacement along with distributed Bus Bar unit commissioning.

SD Planning covering multiple elements

SL NO	Name of Element	From Date	To Date	No of Days outage	Concurrent outage	Associated works
07.	400 KV Malda-Farakka-I	23.11.25	30.11.25	08		Dismantling & erection of new 89 L isolator & 89 T isolator of subject Bay. Complete old CRP system replacement along with distributed Bus Bar unit commissioning.
08.	500 MVA ICT-3 along with 400 KV & 220 KV Bays.	24.11.25	01.12.25	09		Dismantling & erection of new 89 T/89L isolator of Concerned Bay. Complete old CRP system replacement along with distributed Bus Bar unit commissioning.

SD Planning covering multiple elements

SL NO	Name of Element	From Date	To Date	No of Days outage	Concurrent outage	Associated works
09.	500 MVA ICT-5 along with 400 KV & 220 KV Bays.	02.12.25	07.12.25	06		Dismantling & erection of new 89 L isolator & 89 T isolator of subject Bay. Complete old CRP system replacement along with distributed Bus Bar unit commissioning.
10.	400 KV Bus Coupler	02.12.25	07.12.25	06	Complete 400 KV S/D at Malda for new Bus Bar commissioning.	Complete old CRP system replacement along with distributed Bus Bar unit commissioning.

SD Planning covering multiple elements

SL NO	Name of Element	From Date	To Date	No of Days outage	Associated works
11.	160 MVA 220/132 KV ICT-I	09.12.25	15.12.25	07	Dismantling & erection of new 89 L isolator & 89 T isolator of subject Bay. Complete old CRP system replacement along with distributed Bus Bar unit commissioning.
12.	160 MVA 220/132 KV ICT-II	16.12.25	21.12.25	06	
13.	160 MVA 220/132 KV ICT-IV	22.12.25	28.12.25	07	
14.	220 KV New Bus Bar Commissioning	04.01.25	04.01.25	01	Complete 220 KV Bus Outage at Malda for new distributed Bus Bar commissioning.

Bus Isolator Replacement shall be shared separately.